

5th Applied Active Inference Symposium (Part 1, Nov 12, 2025) ~ LIVE

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Greetings everyone. Welcome to the fifth applied active inference symposium. It's November 12th, 2025. Some brief introduction slides and then to Carl Fristen for a keynote address and discussion. So, welcome to the Active Inference Institute. We are a nonprofit volunteer-run organization dedicated to improving the rigor, accessibility, and applicability of the active inference framework. Since 2021, we have hosted an annual applied active inference symposium with Carl as the as the first presenter back in 2021, which serves as an endofear celebration and convening for our ecosystem. At the link symposium.activeinference.institute. You can look at all the previous programs and proceedings. We're here to highlight applied and translational work in active inference across domains and from groups around the world. Meanwhile, we surface opportunities and needs of our ecosystem, get feedback and facilitate connections among participants. That's you. You can join the institute discord, donate to support the work, stay tuned via newsletter and actively infer on about the program. There are three days of programs and the bulk of the proceedings will be live streamed to YouTube at the three links that are available at the program and you can watch it live and rewatch it at the same link and there may be some adjustment at the fine scale with time zones and pre-recorded and live and scheduling and all fun kinds of things but big picture there's going to be three live streams that will consist the bulk of the program. In the participant registration, we asked if people wanted their responses to be public and for those who did, we integrated their responses. So, you can read the raw responses to people's questions that people were asked like what would be interesting for them, what would be useful, how can active inference have a big impact in 2026. And we also used LLM to integrate the responses and do embeddings on the responses. All kinds of cool stuff. So you can check out that open source code. A big thank you to our team of co-organizers. Alexandra Mava, Andrew Pasha, Bradley Lucia, Christian Martin, Corey Slater, Daniel Friedman, Maria Garcia, Maria Louisa, Pablo FM, Sylvia Zang, and Zora Ramaned. It was a great and streamlined process with these co-

organizers. So absolutely was a team effort and I thank them all again. If there's some kind of session presenter or development that you'd like to see in next year's symposium, you would be welcome to provide that feedback or even to step up and be a co-organizer. It's a lot of fun. Email us if you're interested. Also, thank you to our sponsors. Our first sponsor is Navigating Uncertainty, a book bringing the science of active inference and the free energy principle to life through story, art, and human experience by Anna Pereira with a forward by Carl Fristen. And Anna will have a presentation later in the program. and also our sponsor for this symposium, Upcycle Club, an independent social initiative that actively supports a select group of noteworthy institutions and innovative projects dedicated to fostering research, education, and original thinking. So now without further ado to Carl's keynote and I thought how to introduce this keynote and on this slide I have Carl thinking of the 2012 paper the history of the future of the basian brain where there's a quote which was about the neuroimaging community but I believe it applies to the active inference community today as well and he wrote In short, the future of imaging neuroscience is to refine models of the brain to minimize free energy where the brain refines models of the world to minimize free energy. This endeavor itself minimizes free energy because our community is itself a self-organizing system. And so here from the past to the predictive future, we have generative AI and all these different domains of application for active inference. So hopefully a fittingly aesthetic introduction to what should be a very fascinating keynote followed by a discussion. So thank you Carl for joining and you can share your screen and looking forward to your keynote here. Well, thank you for that lovely introduction. Um, and just to say it's a great pleasure uh to kick off uh this symposium and I'd like also just to reiterate um my thanks on behalf of the the audience and the viewers to the co-organizers and particularly you Daniel for uh stewarding us all uh in this community direction. It is uh it is a great endeavor and I just want to again thank you for that. Um so I'm going to start um with a slightly retrospective feel. Um I'm mindful that now we're in almost the sort of second generation of people um applying or interested in the free energy principle and in particular active inference. So I'm going to stand back a bit and this presentation is going to have a certain retrospective global take on the foundations and the principles upon which active inference is based and in particular it's based upon a presentation given to the neuroai workshop at last year's Europe and also the um international active inference workshop uh earlier this year in Montreal. However, this presentation's got a twist because I know people always like to know what the latest thing is. So, at the end of this presentation, I'm hoping just to foreground um a domain of questions and

challenges that face artificial general intelligence research. um the kinds of um problem solving abstractions and reasoning that we uh that are characteristic of natural intelligence that uh now have become a goal or a target for certain sections of the machine learning uh community. So um what I'm going to do in order to sort of look at how one could deploy or apply active inference to understand formalize and simulate logical reasoning and abstraction. I'm going to refer back to a rather old paradigm now, the three ball paradigm, just to revisit that um and look at how we can use active inference in its most general sense to underpin the sample efficiency that we bring to the world when looking for evidence for the way that we understand the world. So, that's where we're going to start. Um, in fact, I'm going to cheat and um we'll get Jeffrey Hinton to actually start for us. Now, I'm not sure whether this is going to um be audible on the live stream. Uh, so Daniel, could if I could um um ask you uh I'm just going to play a few seconds and if you can hear it, then we'll just continue. And if you can't, I'll actually read out what Jeff what Jeff actually said. Um, so let me just start this. >> I would like to meet Cal Bolts. >> Was that audible? Excellent. Okay. So, let me just put this in context. Um, Jeff was being presented with the UCD Ulyses Medal uh a year or so ago. Um, and right at the end of his presentation and indeed right at the end of the question and answer um session from the audience, he was asked the following question. if you could go back in time and meet your hero, academic hero, who would you like to meet? And his response was, I'd like to meet Helm Holtz. So, let's just listen to his explanation for that. And I'd like to meet Helm Holtz believed in unconscious perceptual inference. He basically thought that when you're doing vision, you're doing lots of inference, but it's not kind of conscious deliberate inference. you're inferring from the distal from the proximal stimulus you're inferring what the state of the world is that gave rise to that and he was clearly right about that so that was one branch of Helms's work a different branch of Helms's work was um free energy there's something called Helmholtz's free energy and what Helmholtz didn't realize was that if you're trying to do inference in a complicated model with complicated lat variables a statistician would tell you to do inference by given the current parameters of your model, take the data and figure out the best way for your model to explain the data. And there might be several ways for your model to explain the data. But you have to figure out the probability of each of those ways in which your model could explain the data in order to adjust the parameter of your model. And what variational inference says is actually you don't have to get that probability right. You can approximate that probability and still guarantee you'll learn a better model. or rather than some band that'll improve. And so the two completely different bits of Helms's

career, the work on free energy and the work on perceptual inference were actually very closely related. And free energy is the key to being able to do perceptual inference tractably. And I'd love to tell Helms that lovely. Uh so I'm basically going to uh just try and unpack uh what Jeff was intimating there and do so with reference to three of the great um intellectual architects of the physics from my point of view at least um of artificial intelligence. uh clearly Helmholtz uh whose contributions come at many many different levels also a nod to Feman uh in the sense it wasn't quite Helmholtz free energy um that um underwrites variational inference the free energy in question um is actually due to uh Richard Feman but I also want to acknowledge um the sort of physics of measurement and observation and certainly the perspective uh afforded by Edwin gains in the context of things like the uh maximum entropy or caliber principles. Um so Helm Holtz, where do we start with Helm Holtz's contributions? Uh my favorite point is a slide which I've shown many many times. Um summarized by this beautiful quote, objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous system. So this is his notion of unconscious imperce. This is his notion that the brain is a constructive organ generating explanations in an inside out kind of way that provide an accurate and as we'll see simple account of sensory impressions on the nervous system. And clearly this is very closely related to the notions of uh perception as hypothesis testing by people like Richard Gregory. ideas that were taken with great effect and deployed in machine learning uh by people like Jeffrey Hinton and Peter Dine uh who actually built a Helmholtz machine um as a metaphor for the basian brain lending or borrowing from a probability theory in the sense of basian statistics and of course the work of Richard Fineman um on his work on the pathogal formulation of quantum electronamics. So what's this notion of um unconscious inference? Um one could argue it goes right back to the students of Plato. I'm looking it back to the 16th century. This um I think um beautifully illustrates the notion of unconscious inference or perhaps even conscious inference in this instance. Um, so this is a painting by a 16th century oil painter famed for painting still lifes that when viewed from another perspective give you a very different kind of visual impression. So if previously you saw a bowl of fruit but now you see a face, the point being made here is that you made that face. This is something that you brought to the table as an explanation, a hypothesis, a model, um a fantasy literally uh making the brain a fantastic organ, a purveyor of explanations and fantasies that best uh explain our sensory input. I think that perspective lends um a a a a certain um gravity to um Fineman's meme uh which we all uh quote um um regularly. um what I cannot create I do not understand

and you know that acquires I think a very concrete meaning when read in light of Helmholtz's notion of unconscious inference if you can't create it on the inside you have no understanding knowledge belief no real ability to recognize what it is so I'm here reading the creation again in this sort of constructive perspective on the brain in as a statistical organ creating explanations for its sensorium. Um how can one formalize that? Well, we can again appeal to Fman's free pathological formulation and free energy or use of variational calculus of variations to elaborate the right kind of objective functions that allow you to solve very difficult inference problems. um they are difficult because they entail marginalization or impossible integration problems but they can be rendered tractable and converted into optimization problems through these variational bounds and in particular the variational free energy as a bound on log evidence. So what do I mean by that? Well, um, if we just think carefully about the nature of things that could possibly, um, hold beliefs or, um, have beliefs or make inferences about other things. Then immediately we have to introduce a distinction between the thing that is observing, inferring, measuring and the thing that is observed, measured or we are making inferences about. And statistically uh this is the markov blanket due to Jude Pearl um and simply separates the world into the self or the observer and the non-self or the observed and the bidirectional exchange between these two systems here cast in terms of a markov blanket comprising sensory states and active states. The sensory states are mediating the effects of the world on the inside. Whereas the active states are reciprocally mediating the effects of the inside on the world through some kind of action autonomic reflexes or uh uh actually acting um with by say muscles or motor system. Um and this um this construction um this sort of partition separation disambiguation between the observed and the observer um enables one to apply Fman's path integral formulation to the dynamics of any system that yields to this kind of separation between self and non-self and I've just summarized this here in terms of uh the free energy principle where we can associate how or we can um describe and demonstrate that the dynamics the rate of change of my internal states so my neural activity and my active states together sometimes referred to as autonomous states why well because they don't depend upon the external states by construction by this separation of the self from the non-self um and if we can do that we can um now just write down uh the dynamics of those autonomous states I repeat my internal states and my active states as a gradient flow or a process that looks as if it is minimizing in the sense of optimization this variational function um of my sensory input and the beliefs about the outside that are parameterized by my internal states. And here's the familiar

expression or one uh way of writing down that expression. Um this has been used to great effect in terms of understanding uh things like predictive coding with reflexes which we'll uh cover briefly later. Um and sort of simple um from the point of view for example of um control theory sort of control as inference. And I I just wanted to sort of draw a bright line between two kinds of um application of active inference that rests upon the notion of beliefs about what I am doing. Um that takes us from control as inference into planning as inference where I now have in mind a probabilistic representation of what I am going to do uh and certainly what I'm going to do into the future. Um, interestingly, if one traces back the origins of that particular move, it just rests upon um considering big things possibly with a sort of deep or hierarchical structure on the inside where one now um where the active states no longer have direct access to the internal states deep inside um some uh internal structure. So if I just remove that, something really interesting happens. Um what happens is we can now regard the active states which were previously real variables that had direct access to the internal states as various causes of the sensory states. But if we're now reading the dynamics of my internal states as making inferences about the causes of my sensations, having beliefs about what is out there, then now I am licensed to describe my internal states as including beliefs about my own action. And this now um leads to the notion of planning as inference and introduces something called expected free energy. Why expected? Well, if I've got beliefs about my future and in particular the consequences of my actions in the future, those observations, those sensations, those data are now no longer real variables. They now become random variables because they haven't yet occurred. And now that means if I want to evaluate the free energy, I now have to take an expectation over the posterior predictive density over the sensations that I would cause in the future if I committed to this particular action. Um so you could say that this is a more if you like authentic kind of active inference in the sense that it could be applied to things like you and me. Um basically um using the path integral formulation to link us uh link the markoff blanket to full active inference where we have paths of least action that are determined by the free energy for my internal states my belief updating and for my action the expected free energy. So let's just visit those two sort of um aspects of free energy minimization sometimes cast I repeat as a path of least a principle of least action. Let's just look at the sort of the internal states minimizing variational free energy. Um and this is where the second foundational contribution of helmhotes comes into play. Um what we can do is if we now interpret the internal states as parameterizing some beliefs about the external states uh in this example parameterized in terms of a Gaussian distribution or Gaussian belief

um then because we know the expression for the dynamics of the internal states um we can use the help of decomposition to split the dynamics into two parts. And what I've done here is um associate those two parts for those people who are interesting that those two parts are effectively a dissipative flow that descends free energy gradients and a non-dissipative or conservative flow that um circulates in a certain fashion on the iso contours of uh of the free energy. I can associate this non-dissipative and dissipative part with the prediction and update of a Kalman filter which would be one very simple example of um um inference under a very simple uh generative model and uh the basis of control is inference things like linear quadratic control. But the story I want to tell here really is that we can associate if I knew the state of the world or had an expectation about the state of the world. I also know how it's going to change. I can predict how it's going to change the μ dot up here. But I also and that would be associated with the solenoidal the non-dissipative part. I you know I have not been informed by any sensory information at this stage. But when I do have access to sensory information, I can now evaluate the free energy gradient and update my expectation in the light of the sensory evidence and in particular um the uh prediction error that is effectively the free energy gradient. So what's a prediction error? Well, imagine we had this impression on our um sensory epithelia and I had in mind an internal state that represented the belief that this particular impression was caused by a howling dog. And if I had a generative model that could generate the consequences, the sensory consequences of a howling dog causing my sensory input, I could take the prediction, subtract it from the actual sensation and produce a prediction error, a mismatch. And then all this equation is saying, all the Helmholtz decomposition is saying is it will look as if this prediction error is driving my beliefs in order to minimize the prediction error where when the prediction error is um um minimized, I'm minimizing my free energy gradients and I found a free energy minima. So this um provides I think a really nice perspective on filtering on data evidence accumulation uh that is can be interpreted directly in terms of Helmholtz's uh decomposition. Um notice that all that this is saying is that you're minimizing your prediction error as a proxy for a variational free energy. You're never going to actually know uh what's out there. In this instance, it was actually a cat. Um so what about James? What's another way of looking at this sort of uh free energy minimizing um description of self-organization where the self is distinguished via Markov blanket from non-self. I put this up here um because I think that you know it does provide a really interesting uh dual perspective on the problem of inference or the description afforded by inference measurement and observation in general and I've just expressed the free energy now in terms of um a an energy

um and an entropy an expected energy or a potential uh and an entropy here And this um this entropy here um we can see if we want to minimize a free energy we need to maximize the entropy of our beliefs. And of course this is exactly what James was saying in terms of the measurements that we do in physics um but crucially under constraints. So what are the constraints? Well, they're just the constraints that are supplied by a generative model. Um for us you're affording an accurate explanation of some data um under a generative model uh where the generative model is usually expressed in terms of some likelihood and prior the probability of some outcomes or observations given the hidden states that are hidden behind the markoff blanket and some prior beliefs that characterize my model namely the probability of those latent or hidden states. I put this up because it there's a um there's often um a degree of confusion when talking about the um the thermodynamics of inference. Um so you know if I want to maximize um my uh the entropy of my beliefs why is it that often the free energy principle is cast as an account of or a description of things that sustain themselves by minimizing the dispersion or the dissipation of their say sensory states or observational states. Um the paradox is very easily resolved in this instance by noting that when people talk about entropy the you have to specify the particular probability density or distribution to which that entropy applies and in this particular instance um the maximum entropy uh principle applies to the entropy of the beliefs and this entails a minimum entropy production uh principle. um and that I think can be rendered very very um obvious in the sense if I now just take the time average of the free energy we can see immediately that as the path entropy goes down the uh goes uh in one direction um the entropy uh goes in the other direction. And so if I maximize my path integral by minimizing my free energy or my expected free energy then I'm implicitly minimizing the dispersion or dissipation of my uh exchanges with the environment through the outcomes or observations supplied by that environment. So maximum path entropy is basically equal to or equivalent to a minimum entropy production principle very much in the spirit of um the theory of dissipative structures. uh due to pri um another way of re um of um understanding or intuiting um the characteristic uh or the implications of free energy is just to re express it to somebody machine learning would express it u or possibly even um basic statistics. So I've just rearranged the terms and I've just arranged them um in terms of the accuracy, the expected probability of some outcomes um under some beliefs about what caused those outcomes, the latent states um minus a complexity. So what's the complexity? Well, it's just the divergence between my posterior beliefs, my um um the beliefs encoded by my internal states and prior beliefs. So that scores the divergence between what I believe now

a priori before seeing some data and after seeing some observations a posteriori it literally scores the degree to which I change my mind in order to provide an accurate account of the data. uh and of course that's just an expression of principles of indifference or Okam's uh uh principle but interestingly it also has thermodynamic uh correlates um by appeal to things like the Jazinski equality and Landau's principle you can now say that if I want to change my mind if I want to pay a complexity cost in accounting for my observations my measurement um in a way that provides an accurate account of that measurement or observation. Um then I'm going to have to spend energy. It costs some work, some thermodynamic work to erase bits and to move on some statistical manifold or to induce a divergence between my prior and my post area. Which means that in minimizing free energy equivalently in machine learning in maximizing the EV the negative free energy or the um the evidence lower bound I'm implicitly trying to minimize my computational cost and that um also impacts in a thermodynamic uh context that I'm going to do it as thermodynamically if as efficiently as possible. Um and of course this is just one aspect of biometric and or biotic self-organization um right through to uh kind of influence that Helm Holtz had in mind. So that perspective allows us to sort of look at the sort of what some people call merely reflexive active inference because we haven't as yet got to planning as inference um through a number of different lenses and I've just summarized these here in a slide which I'm sure many of you have seen before. Um so at the moment we're just talking about um the notion of describing ourselves as minimizing free energy or trying to maximize this um evidence bound or maximize the model evidence the evidence through our generative models through updating our beliefs um and noting that our actions should also minimize exactly this quantity. So our actions are just going to fulfill our predictions formally. Then um what does this mean or how could one interpret this? Well, we know that uh first of all the free energy is a bounded approximation to the log evidence um the log uh probability of some outcomes uh given me um as a um you know the self in question. Um and what that basically means is I can interpret this log probability as the kind of outcomes that are characteristic of the kind of thing that I am. So they have meaning for me. They characterize me. Uh you know I am um have a certain temperature. I have a certain status. I have all sorts of exchanges with the environment and all of those environments are characteristic or typical or define the kind of thing that I am. And from this that interpretation um of free energy um leads to things like reinforcement learning. If I was a control theorist, it could be cast in terms of optimal control theory. Or if I was an economist, it would be expected utility theory. If I was an information theorist, the negative of of this log marginal likelihood or log model

evidence is known as self-information. Um also known as surprisal, more simply surprise. And it just scores the improbability how uncharacteristic this exchange with the environment would be if I am me. Um and that reading of this quantity here uh the free energy in and of itself um can then um forge links with information theory the infamax principle the principle of maximum mutual information or equivalently minimum redundancy principles due to barlow and of course the free energy principle itself. That's good. If I now take the time expectation or average of self-information or surprise, I now get entropy. So now we come back to the minimum entropy production principle that James was considering from the point of view of maxent or max caliber uh principles. Um and what does that mean? Well, it just means that it will look as if my autonomous states, my sensory states and my control states or active states um are those that on average try to minimize the dispersion of my exchanges with the environment. Of course, this is the holy grail of self-organization as exemplified in synergetics uh due to Haken. And if I was a physiologist, it's just a statement of homeostasis that I am the kind of thing that keeps my essential variables within viable bounds and ensures that they they do not dissipate um uh beyond those uh characteristic values. The final interpretation which of course is the one we're familiar with is that this probability of some outcomes given me um can also be described as um model evidence if I associate me as a generative model of the world in which I am immersed. And this basically says that free energy minimizing is just maximizing model evidence um sometimes referred to as uh self-evidencing. And this is a provides a sort of formal account to the basian brain evidence accumulation and particular implementations such as predictive coding that rest upon the um the isomorphism between free energy gradients and uh precision weighted prediction error. So that's um from free energy to um self-evidencing. Um what about this sort of second kind of active inference where we now um instead of just picking those actions that minimize free um variational free energy. What if we now make inferences and now describe our internal states as holding beliefs about what we are doing uh and possibly what we are doing into the future. sometimes referred to as policies π_i here that now will minimize the expected free energy under some um I repeat posterior predictive density of the consequences the outcomes that ensue if I committed to a particular sequence of actions or combination of actions known as a policy and this is where we move from the free energy to the expected free energy and I've just written this out in full because you know even if you don't like equations I think there's a lovely iconic symmetry here um which is um um tells you a lot about ways in which you can understand the motivations for behavior um and the implicit um intentional aspects of the behaviors that I

infer myself committing to. Um so I've just written out the free energy in terms of a mixture of complexity and accuracy. um and um rewrite just moved terms around to uh foreground the fact that we can also express this as a KL bound or divergence um um or a band on the log evidence um and it's a band simply because the KL divergence between my beliefs about latent states and the true posterior can never be less than zero by construction. So if I now take the expectation of these quantities um under some consequences conditioned upon a policy what happens? Well the complexity becomes risk the inaccuracy becomes ambiguity and the divergence and log evidence become expected information gain and expected value. So just to unpack that just let us just sort of look at these terms and see how we can interpret them very much in the spirit of the uh the previous slide looking at merely um reflexive active inference. So let's assume that first of all I don't have any prior preferences. I don't have any probabilistic description of the characteristic outcomes um that I would expect given me or given my generative model. What are we left with? Well, we're left with another KL divergence, but this one is an interesting one um in the sense that it's um it is going to be maximized simply because now the outcomes are in the future and they now become random variables. Uh and what is it a um um an information gain or a pay divergence or um um an on average a mutual information? What is it between? Well, it's basically between the beliefs I would have about the states of affairs in the world out there in the future if I were able to see the consequences or observe the consequences of those states relative to if I did not. In other words, it's scoring the information that is obtained by acting in this way. It basically scores the epistemic value or the epistemic affordance. um it scores the degree to which I'm going to resolve my uncertainty about states of affairs in the future if I did this. Uh in the BA in the visual search literature, this is often referred to as basing surprise. um is also mathematically equivalent to maximizing mutual information between cause and consequence where cause are my hidden or latent states and the consequence are the observable outcomes or consequences of those states conditioned upon some policy or active intervention um on the world under my generative model. Let's take a particular kind of uncertainty off the table. Let's assume for simplicity that we can see every state of the world directly. So now my observations become the states of the world and the states of the world become the observations. So there's no ambiguity about what um about the states or the causes of the observed consequences. If I remove the ambiguity, I'm left with another KL divergence. This time we want to minimize it. And what are we minimizing? Well, effectively, we're just minimizing the divergence between what I believe will happen if I commit to this policy relative to my prior preferences that characterize the

outcomes that are characteristic of me. Um, and that basically um scores my policies in terms of how close can I keep myself to my prior preferences. Um in engineering this would be known as um path control. Uh you could also substitute model predictive control uh and some kinds of Bayesian RL um under this uh under this formalism. If you're an economist this would be um risk sensitive control. Let's now make a final move and remove all reducible uncertainty from the problem. So now there is no more information to be gained from this environment. I'm now completely familiar with this environment and I cannot resolve my uncertainty anymore. What am I left with? Well, I'm just left with this um uh quantity which we have previously associated with value. Um the expected value. Uh so from the point of view of economics, this would just be uh expected utility theory. the value I expect given my beliefs about the future under a particular policy. So notice what we've done here. We've actually derived a number of really important um characterizations or descriptions of motivated behavior or optimal behavior in some sort um simply by selectively removing different sources of uncertainty. But the most general case um is this decision-making under uncertainty that um entails a principled approach to information gathering or uncertainty reduction and I've tried to illustrate that here just by appealing to the dual aspects of Bayesian optimal action or decision-making um so it has two aspects um the first is nicely captured by the ability of the expected value to score the probability that I'm likely to to uh pursue this policy and of course that's exactly what you'll find if you read books on Bayesian decision theory. So um this would be if you like the part of the Bayesian optimality that rests upon uh optimal Bayesian decision theory. But there's another way of being Bayesian optimal and that's um often cast in terms of Bayesian optimal experimental design designing your experiment to elicit those data that resolve the greatest amount of uncertainty about your hypotheses u the uh pertaining to the way that those data were generated or caused. Um, and this goes right back to the 1950s with people like Dennis Lindley looking at the information theoretic criteria which of course adjusts the expected information gain or the expected epistemic affordance that we've just described um that underwrites exploratory behavior designing the right kind of experiments uh in order to resolve uncertainty so that you can become exploitative uh in accord with Bayesian optimal decision theory. Um together expected value and information gain just mean a negative expected free energy. Um um and you know one would could argue this is the essence of active inference beyond a merely reflexive active inference. Um what I want to do now is just sort of return to these three um three sort of perspectives uh or contributions uh to the free energy but from now a much more technical um

uh perspective or a formal perspective. And what I've done here is write out the free energy as an accuracy term and then all the different bits of complexity that are contributed by the various sets of unknowns in my generative model and every set usually brings along uh a mean field factor or partition of the unknowns the random variables of different kinds. Um so I picked out here the unknown states the random variables that are the latent or the hidden states behind the markoff blanket that change from moment to moment. Uh we don't know um we don't know our policies. Um this is another random variable that I'm making inferences over and then selecting the from the uh selecting the most likely policy. Um but also I have um model parameters. I have parameters of a model. Here I'm summarizing the parameters in terms of a likelihood tensor. um which we'll come back to later um um a um uh that has in terms of the deployment of the of the parameters within say a tensor, it also has a particular structure. So we not only do we have um beliefs about states of the affairs, the ways in which we're going to interrogate or act upon the world, but also we have beliefs about the contingencies, the laws um the parameterizations, the associations that underwrite the cause effect structure um that are installed in our generative models. Now the key observation here is that if I can be described as a free energy minimizing thing then there are a number of ways in which I can minimize my free energy. I can change my beliefs. I can change my mind about states of the world and we could call that perception um by changing Q through my internal states to minimize uh the uh the variational free energy or the negative free energy in this instance. um but I can also change my posterior beliefs about the parameters of my generative model. So we have a nice distinction now between sort of inference and learning and I'm using active learning in the sort of traditional sense um certainly promoted by people like David McKay in the in the 1980s and '90s. Um and the difference here is it's exactly the same maths. It's just it usually unfolds over progressively longer uh time scales. So I I have to do lots of evidence accumulation or information gathering to resolve uncertainty about states of the worlds and how they change. And then on the basis of those beliefs, I accumulate those beliefs and um assimilate that into um optimizing my beliefs about the parameters, say the connection weights in a uh in a neural network um or the dish counts in a uh in a tensor network. um um that would correspond to this slow process of free energy minimization. Um that basically is minimizing the complexity of my model of the world, maximizing its compressibility, maximizing mutual information. All of these being equivalent perspectives on the same imperative u to minimize to maximize the um the marginal likelihood or the model evidence. The final um thing though is that not only can we change our posterior beliefs encoded by the the the you

know our um internal states, we can also minimize the free energy by changing our prior beliefs. Now we're familiar with that in terms of um predicating our prior over what to do next, our policies on expected free energy, but we can also do the same with respect to a prize over the latent states and crucially the very model itself. And that's what I want to pursue in the final part um of the of this uh of this presentation. Um so what we're we're talking about now is that that there are lots of opportunities um to minimize free energy. Um and if we're all free energy minimizing things, it's likely that we are will show evidence of these kinds of minimization processes such that minimizing um changing our prior to minimize free energy could be regarded as a description of attentional selection. >> We're talking about now is that that there are lots of opportunities Sorry, continue. Carl, >> right, I'm talking to myself. Um, uh, and the pri um, sorry, policy selection would be um, you know, a neat description of, um, optimizing my prior beliefs about what I'm going to do next. Um and that leads to if with this sort of use of a selection word um appropriate to the kinds of unknowns that constitute my generator model leads to the notion that um we can also minimize free energy um through model selection uh through a process akin to structure learning. Um so I'm going to close with an example and then just show you um sort of the kind of things that we've we've been um certainly with colleagues at versus recently focusing on in terms of generalizing this notion of optimizing our priors over different aspects of the generative models. This is a paradigm which I'm sure most of you have seen before and if you haven't the purpose I'm of showing it again is just to enable you to think about the way that you abstract the way that you reason about the cause effect structure and in particular how do you test hypotheses about the mappings from states of play states of affairs in the world and some correct response response the rule that underlies um a reward or some feedback that you are correct. So what we're talking about here is sort of a a calculus of abstraction, a calculus of reasoning and a calculus of rule discovery where the rule can stand in for some kind of symmetry, some kind of equivariance, um some structure that explains when you are right, when you've got the right answer. Um, and this um this is a the three ball paradigm. Daniel, do you want to play this? Have you forgotten the rule? >> Go for it. >> Okay. Well, let's Yeah, if you haven't forgotten the rule, that's good because we can get through this more quickly, but I normally I would play this game with the audience uh just to get them into the spirit of introspecting. How do we solve these kinds of abstraction problems or rule discovery uh problems? Um, so the game normally is is presented as follows. You um you're going to be presented uh with three colored balls and you're going to choose or your game or your job is to choose the correct color based

upon the arrangement of the three large circles. Um, and the um location of the correct color depends upon the color of the upper circle, which in this instance is blue. And then what I'm going to do um is give you the correct answer um and just leave that on the screen so that you can remember past instances, past trials so that you can start accumulating evidence for against different hypotheses or rules that could be brought to the table to explain what's the correct color. So that's all you know. That's your prior beliefs. Uh and then we normally play with the audience a game. Uh and here there are um you know a series of games starting with uh the one that we saw previously and then a series of of examples. And what we're going to do or what I'm going to ask Daniel to do is to guess until he thinks he's discovered the rule and then tell us if he's got the right rule. And I'm interested in how sample efficient he is. I'm interested in how many exposures does he have to um entertain before he suddenly gains insight into the rule, before he has that aha moment. Um and indeed, if you're watching this um now or later, um you know, the the the just think about the uh the experience of this kind of rule discovery. You know, is it a gradual thing? Does it happen all of a sudden? Are you able to um mentalize it? Is it a propositional thing? Are you arguing in words? Or does it just come into pop into mind? These are all really interesting aspects of this how one would formulate or formalize um this kind of abstract reasoning or logical induction. Um right. So let's start. Daniel, what we start with this um this little game here, this epoch here. Uh what color do you think is the correct color? >> Green. Excellent. And for those of you um who might not have noticed, this is exactly the same as the example on the previous slide. So I'm assuming that Daniel was >> That was training data. That was training data. >> Ah, right. So what you what are you going to go for this time? >> Red. >> I'm sorry it was green. And of course there is not sufficient evidence at this stage for you to possibly guess any kind of rule. Um, how have another go. Let's go to this one. >> Blue, >> right? Well, think again. If if the correct color is is is on is in a particular location, um, can it be blue if there's no blue uh on the screen? >> I I'm open to some very creative rules. >> Okay, we'll go with red. >> Okay, then let's do that. Bye, Joe. You're correct, sir. And how about this one? >> Blue. >> You're going for blue. I'm afraid it was green. >> Okay, >> this is this is this is desperately unfair. So, when I normally do this with an audience, we give people about 20 or 30 seconds between each example so they can mull through alternative hypotheses and suddenly spot the one that's uh explained the data before. My uh at this stage I I no one in the audience will will have caught it. Uh D, let's give a bit more information. What do you think's the correct answer here? >> Well done. Do you have an idea as to what might be the rule? Always green unless otherwise

absent. That's close. It's not quite the rule. Uh but that's that's certainly close. Do you want to think about it just a little bit longer? This is relevant because um what when I'm going to simulate when I simulate this game um there is a process of offline introspection that requires you to spend some time rifling through different um hypotheses or explanations for this particular rule or cause effect structure. So, I'm deliberately witting to give Danielle some time to to spot any rules that he he might um bring to the table to make the next predictions. What are you going to predict for this one? You you've already got this one, so this is easy. >> Um, you've just told me the >> or green on top. Oh, it depends on the color of the upper circle. It doesn't say it doesn't depend on other things. Yeah, the all the information you need is in the the arrangement of of these of these three balls. And what you've told me so far is when um the uh and all all you the sort of the the cue if you like is the color of the central ball. So when you've already told it when the central ball is green uh the is green. >> Yeah, you've already told me that. And >> when the central ball is blue, it's also Yeah. Well, you don't know that. Oh, uh, well, that that was certainly one instance, but it could have been this one here. >> Fair enough. Perhaps have another choice. Ah, >> it's red. >> Okay, then red. Correct. >> Then green. >> Correct. >> Green. Green. >> Oh. Oh, you >> blue. If it's blue. blue. >> It can't It can't be green because there's no there's no green on the uh on on the I'm teasing you and it's very unfair to te I should uh the correct answer was in this instance blue. Um so the the solution now unfortunately we we don't have access to the audience but some of you um what normally happens is um after a couple of um trials following the point at which Daniel spotted the sort of first element of of the rule um um people suddenly get the insight with very very sparse information but so usually by about sort of uh uh between five and eight uh trials And the the correct answer was exactly as Daniel um realized even by the fifth one I think um that when the center is green the correct color is always green. When the center is red the correct color is on the left and when the center is blue the correct color is on the right. Um so discounting the first one you would not have known that until you now got to the 1 2 3 4 5 6 7. Um so this illustrates you know not a desperately complicated rule but certainly has a certain context sensitivity. Um and the question I want to ask now is why is it that Daniel partially was able to do this um um or was able to infer this on the basis of um well certainly part of the rule after 1 2 3 4 5 um given um he hadn't even seen a blue a Q apart from the training uh trial um before. So in principle this is quite remarkable that you he was able to make some inference to have a hypothesis in the absence of any definitive data which is um which you know is the kind of abstraction and our remarkable efficiency to discover latent structure in the

world even before we've actually seen it. Um but let's just see if we can actually um get a base optimal agent to um match that kind of performance. Um there's something which I have to implement in the simulations uh which was not made explicit in the task instructions. Um that the simulated agent has to actually look at each of the location from a start location and thereby has to choose where to look as well as whether to select a particular color um prior to it. uh um um uh after it's decided what color to choose. Um so there are two kinds of actions in this particular um example. It's where am I going to look and what am I going to choose? What would I need? What latent states would I need to generate um the outcomes? Well, the outcomes are basically what color am I seeing? Um and where am I looking? um and what kind of feedback am I getting? Is it sort of have I yet to make a choice or if I have made a choice is it right or is it uh is it wrong? So to generate these what where feedback outcomes I need to um I need to know what the latent states of the world are and the latent states are basically a sort of context that establishes the rules. So for example, if the correct color is on the right um then uh well what is the correct color? say it's red. Um, and if I know that the correct color is on the on the right and I know that the correct color is red and I know where I am looking and I know whether I have chosen or not, I can generate all of these things. And how do I generate these things? Well, I I do so through a likelihood mapping technically a likelihood tensor that goes from these um latent states and probabilistically maps to the outcomes. So what I've done here is just show the rule that we used in the previous example in the form of an unfolded um likelihood tensor um which is fairly simple to read. All it's saying is that you know if um the rule uh the current rule context is that the correct color is on the left then there's a lawful mapping between what I am seeing um and what the correct color is uh provided I am looking at the left. But if I'm looking at the center then I'll just see red. And hence that was the rule that whenever the center was red I would should look to the left. When I look to the left I can now identify the correct color because of this isomorphic mapping between the two. If I'm looking anywhere looking to right that doesn't count. Um so what we're doing is basically visualizing the the nature of a rule in terms of these sparse isomorphisms these sparse mappings in a likelihood um mapping um between cause and consequence. And of course you didn't know that you did know that if um the the color um of the uh center square so I'm looking at the center there's a unique color to one of three rules because there are three colors but you don't know what the rule is. So you started off with this uh and you were hoping to gather enough uh information or evidence to eventually disclose this structured um likelihood mapping. So we could actually simulate that and this is what's

um been done here. I'm just showing um the results of simulating a base optimal agent um with those ignorant prior as naive prior um and looking at its performance um over 32 games each game comprising sort of uh four sorry four moves five exposures or or uh to the same stimuli. Um we don't need to go into any detail here what's going on. This just shows you the the policies about where to look and whether to select something um that are selected given all the cues um um the central cues at the top here. Um what we what we will notice though is that after about um um 13 uh or 14 um observations, the agent is continually correct, makes no mistakes. Um which is um roughly the same as as Daniel's performance even though he half got the rule. Um at that point the free energy reduces on average and the confidence namely the uh the precision or the negentropy of the belief distribution over what to do next what policies to engage in uh suddenly um remains quite high. So there seems to be a sort of a reasonably abrupt change um after about the 15th um uh the 15th exposure uh to a low free energy confident correct kind of behavior. Um and um the question is um you know what what underlines what underwrites this this qualitative um uh sorry quantitative uh change in in behavior. Um what we can get an idea just by contrasting the belief updating um before and after learning sufficient amount of contingencies to have an insight into the rule. Um and what I'm showing here is just the simulated belief updating and image format but also the simulated local field potentials or if you do electrophysiology uh event related potentials um associated with looking at the different cues before and after acquiring some knowledge through parameter learning. So updating my beliefs about the likelihood mapping um uh denoted by a and the thing I just wanted to highlight it's really interesting that you know when I've learned what the meaning of these instructional clues are when I have some understanding of the cause effect structure then that meaning is actually reflected in belief updating. So suddenly certain cues um suddenly evoke belief updating and simulated neural responses in those representations of the context. So suddenly I can recognize I'm in one context simply because I and it's me and its implications simply because I've now learned the implications. I've learned the rules. Um but one has to ask well why look anywhere? There's no reward in this paradigm. um you know what why would you um what what what drives what motivates um scanning around in the first place? Well, of course, we know what the answer to that is. It's it's basically um the expected information gain but in this instance not over the normal kind that is associated with basin surprise namely resolving uncertainty about states of the world but using exactly the same construct but resolving uncertainty about the actual contingencies the rules the parameters of my generative model. And in some literature's this distinction um between

information gain over states and information gain over parameters is known as novelty and salience. And all we're saying is that generalizing expected information gain to cover the parameters of a generative model um equips you with a kind of curiosity that can be described in terms of what happens if I do that uh what will happen. So it's not what is behind this door but can I open this door? um and we are motivated to be curious in this kind of way. So our intentional behavior is now motivated by this um by responding to epistemic affordances in the environment that allows us to slowly unear the contingencies and the parameterizations um in our generative model. Um so is that a sufficient explanation of what's been going on? Well, not quite. remember that Daniel was able to um partially discover the rule after um six um uh present 1 2 3 4 5 five presentations. How did he do that? Because he'd never seen some of the information necessary to be confident that this was a rule in play. Yeah. At this point, I would you know, I'm tempted to ask Daniel, but I think I think that would be unfair. Um so I I asked the audience I said think about how on earth what did what did you know if you had discovered that rule before the uh the full 32 exposures how did you or how what what did what did you know that the moral didn't know and of course the normal response um which is completely correct is well I knew there was a rule I knew there was some kind of symmetry there was some kind of constraint some lawful relation ationship um which constrained my hypothesis space um which meant that I could actually rifle through different explanations that all had a rule-like uh structure um I may have done that serially or I may have done that well I don't know how I could because it's likely that I uh I was doing this unconsciously um but from the point of view of um how one might simulate the fact that you knew knew that there was a rule. How could one install that prior into this kind of active inference? And the idea is really just to focus again on optimizing the prior optimizing the prior via a process of structure learning. And basically what we're saying is that if this agent was equipped with an a priority set of rules, hypotheses um uh laws um that are going to be embodied or represented by a whole range a whole set of different kind of likelihood mappings or uh likelihood tenses here different models. Then um this puts constraints on the belief updating and it enables the agent to accumulate evidence for each of these models. And when the agent is sufficiently um confident that the one model is the best explanation what's going on, she can then commit to that particular model. So this is a process known as basin model selection. Um and we can formalize that just by looking at the sort of the maths of basin model selection. Uh and the picture that emerges is that you know we spend a lot of time um gathering information foraging the the environment in a way that allows us to build up associations and

contingencies and correlations um as parameters in our generative model. And then the idea now is that we're going to go back and find a simpler model that has a greater evidence that still complies with the particular model by removing certain parameters and scoring the whether the uh the free energy bound on the model evidence goes up or down. If I remove something from my model, if I reduce my model in some way uh and the evidence goes up, that means that that model was just contributing the complexity, not the accuracy. So I'm fully licensed to remove it and to reduce that model. And furthermore, if I have a space of hypothesis, all of which are reduced forms of some full model, I can go through evaluating the reduced free energy or the reduced evidence and score all my models in terms of how uh how much um how likely they are in relation to each other. And if there's a winning one, I can pick out that and I can uh use this bas model reduction in order to um very quickly and efficiently says no, this model is the best explanation for the data I have seen. So far, of course, this is very reminiscent of what we're doing in sleep um in terms of synaptic regression just by removing um redundant parameters crucially um minimizing free energy but in the absence of any sensory data. So there's no accuracy at all. So the free energy just reduces to complexity. I minimize my complexity just by removing um removing any um parameters um that have a substantive change in the free energy and that's relatively trivial to um evaluate um under these discrete state space models or likelihood tensors. So what we're saying is basically we go around with our eyes open accumulating all sorts of potentially um redundant associations and then we we just think about things. We close our eyes. we introspect offline um just to u make sure that we and just retain those uh parameters that are necessary to explain the data that we've just seen so that we can now maximize our model evidence and then open our eyes again uh and start uh start start again. And when we implement that in this agent um we can massively increase the um the sample efficiency um through this period of periodic introspection or having little naps between each of the trials which is why I was trying to give Daniel a break but didn't actually allow him a proper break uh to think about things um and um increase so increase the efficiency with which we can build our beliefs about the structure and the parameterization of our generative model. So here's the true rule. This is what um Daniel started with after about 12 trials. This is probably what he was representing if he didn't know that there was a rule there. If he'd had time or indeed did have time to a certain extent um to now look at this uh through the lens of all possible rules, he could have identified this particular combination of subtenses or uh this rule here after the 12th observation. And indeed um when we simulate this um um we can show a marked reduction in the discovery time and

the associated um uh performance that can be scored in terms of the average confidence um and the uh free energy bound on model evidence. Uh this is probably the most interesting uh kind of result. It basically shows how long it was until the model had a na moment. how long it was until one of its hypotheses, one of its rules um became substantially more likely than any other to the extent that they could then confidently commit to that particular rule. Um and in some instances it can it can actually with the right kind of information. Um it can actually be uh very very uh you know be done after just three exposures. Um but you have to have the right kind of information. Uh this is the the top line is the first one. Um showing that uh to be to be confident about the best drill you you should have been about sort of eight uh observations which is roughly what um uh which is roughly what gen people do uh generally um um uh however notice the blue squares are when people jump to the wrong conclusion. So there is a possibility of prematurely committing to a particular model on the basis of what you've seen so far, but um um it's not going to be the best explanation for what you're going to see in the future. So that occasionally you get this sort of um jumping to conclusion effect, but on average for this particular game with random ordering of stimuli, we're looking at u you know the the aha moment emerging after about five to six um exposures which is uh exactly the same that um Daniel um evinced when uh getting the partial rule. The final the final thing I just wanted to um um speak to is the last generalization um from expected information gain over parameters to expected information gain over the model in of itself. So this is new. This has never been written up or um or described anywhere. But it it it's a natural extension that rests upon um basin model reduction and takes us now into the world of true epistemic foraging as a scientist would do when she is gathering data for or against disambiguate between a null and altered hypothesis but generalized to potentially hundreds of hypotheses. So now we're addressing the question um let's assume we are going to do some basis uh reduction. What's the best way to gather data that will disambiguate, maximize the information gain, maximally resolve the uncertainty about this hypothesis space? So I can remove one set of hypotheses successively and winnow down my hypothesis space until the last man is standing and I can commit to this particular explanation. And I would submit that this is if you like um active reasoning. It's it's sort of actively having in mind explanations for your world and then designing experiments in a way that is the most efficient in terms of resolving uncertainty about your hypothesis. And that just basically requires um adding an extra term to the expected free energy. Um that just scores the difference between my posterior over models uh with the outcomes relative to without the outcomes. And when we do that um we get sort of

you know very interesting uh behaviors um and you can actually score or plot the posterior probability over different models or hypothesis. Here this is the same paradigm but made slightly different. So we didn't tell the agent which of the um locations of the colored ball uh colored balls signified um which the um the criterion ball that was the contained the correct color. Uh so um in this in the game we just played that was a center ball but the this agent didn't know that. So it it has to acquire a lot more information before it can commit to a particular hypothesis. That hypothesis um well the correct um the correct rule is the one that we just um um entertained which the center is read that the left state is the correct choice and so forth. Um that's rule number 65. that there are 81 other rules that all have to be um eliminated before we before we can get to the correct rule. And this is a process of elimination here starting off with rather uninformative posterior beliefs about the particular rule or model in play. And when we um are confident by applying uh Okam's razor um to this distribution of posteriors of hypotheses or beliefs or models um we can suddenly commit to this model and then we can identify the true likelihood mapping um for subsequently and basically come very confident um in terms of what we're what we're doing uh and perform perfectly um under the constraints in the game uh with a minimum divergence between the posterior the true uh true likelihoods. Um just for those people who are interested um it's interesting to think about well how much does this um expected information gain over models contribute to the expected information gain over other things and how do they interact with each other and and just to speak to that it's um what we've done here is to simulate that game again um but using all the different kinds of um components of the expected free energy the expected information gainer over models states um parameters and states and the expected value which in this instance we can be ignored. Um and then remove the various sources of or kinds of expected information gained to see what would happen. And interestingly what happens if we remove this um this scientific or um um information gain over hypothesis or models. um we can still do quite a lot lot of um resolving uncertainty about um models just by um resolving uncertainty about the parameters that constitute those models. But we have real difficulties when it comes to two equally plausible explanations because um this agent is actually deliberately sampling the world in order to maximally disambiguate between the two uh the two last hypotheses. And when we remove everything we we we we still there is information there but it can take a very long time before we actually commit to any particular hypothesis. So in summary um uncertainty can be you know thought of in terms of expected surprise namely entropy um and we can think of different sources of uncertainty and their resolution in the following way that

latent states parameters and structure all um offer different sources of uncertainty that can be resolved through free energy minimizing processes of inference learning and selection of different kinds uh leading to um uh one way of reading artice artificial curiosity in terms of succumbing to or finding affordances that can be described as salience, novelty and uh knowledge. Um and just to reiterate the importance of sleep. Um in the words of Einstein, the merely reflexive act of inference is just basically appealing to the best explanations for the world. Put very simply, everything should be made as simple as possible but not simpler. Um when we move to um playing as inference and the importance of expected information gain that is constrained by expected value. Uh the important thing is to not stop is to not to stop questioning. Curiosity has its own reason for existing. And on this analysis, you probably invert that that if you exist, you have to be curious in the special condition that you don't have direct access to your own actions upon the world. And with that, it just remains for me to thank those people whose ideas I've been talking about and of course to thank you for your attention. Thank you very much. >> Thank you, Carl. Awesome. All right. Great. I will now pass to Michael Garfield who will facilitate this next section. There will be some roundt discussion and thank you Michael. Take it away. Yeah, wonderful. Uh that was a bracing way to wake up immediately jump into your lecture and and uh begin one's day. Um, I guess the first question I have for you and then I I think this touches directly on Hungju's work and uh I would love to solicit input from other people on this as well, but Carl, let's let's have you go first has to do with the surprise of advanced meditators or psychonauts in and you know non-ordinary states of consciousness and like the the the loss of the distinction between self and other and um there's a what Eric Davis calls a kind of a metabolic ontology that um there's a threshold beyond which the uh these like diatic distinctions no longer uh seem to pertain. pain and you end up with like a kind of co-imcation of self and environment or a uh like an n-dimensional figuration of of these relationships. So, I'm I'm I'm curious to hear folks on this panel wax on uh phenomenology of non-ordinary self- other relationships in the in light of this framing that you just laid out for us. I'll start that because I'm the oldest. Um and I I get the impression it's very early in the morning for you. So So I'm assuming you're in California, which of course >> Santa Fe. Yeah, >> Santa Fe. Okay, that's still that's still about sort of 11 hours I thought 8 hours. Uh so it must be very early in the morning. My apologies for that. For me, it's getting getting into mid-after afternoon. uh in relation to your question and I'll try and set up the the the other uh panelists for this um a very sort of uh technical view of this. I guess what you're talking about is sort of non-dual states the the distinction between self and others

certainly the the narrative self or indeed the sort of semantic self just disappears. Um I think that that again I'm answering you know if I had to simulate this how would I simulate what kind of generative model would I have to put in play in order to plausibly simulate this kind of phenomenology and the first thing that I would have to do is to actually write down a generative model that actually had a distinction between self and non-self um and um or self in different uh in different contexts. So uh that would be quite a sophisticated generative model, very expressive generative model that naturally now have discovered the rule or the hypothesis that it is actually an entity and that many things can be explained by this particular entity that is me being in one of a small number of states sort of carving subjective selfhood into into um some plausible set of states that can now very efficiently recognize I'm in this state and therefore should attend to this and should attend to that and then what I would do to dissolve um that confident commitment I am in this state and and implicitly I am me um would be to look at those situations under which there's a loss of um precision over the self other or self in other contexts um distribution how would I do that well operationally that would require um decreasing this deep precision at um via increasing the precision of lower level processing. Um what would that look like? Well, it would look like I was attending to something that was um um closer to the sensorium. Um so you you you you you could sort of say this was some aspect of meditation for example that you know I am deliberately attending to assigning much greater precision to the um to you know to the sensorium um possibly a very limited part of the sensorium and that implicitly dissolves any conviction it doesn't I don't need to explain the world now in terms of I cause this and I cause that there is no need for any highness uh uh you know it it is not necessary in this particular instance simply because I've been able to and of course we aren't generally able to do this but if you become a skilled practitioner doing this some people can um you know redeploy their precision their entropy their their um their attention however you want to articulate this away from this very high level self kind of model of minimal selfhood at least that would be my That's how I would approach this. >> Oops. Now I'm unmuted. All right. Sure. Thanks. Um, thank you. Thank you for inviting me in this panel talk is really great honor. um before. So um well I would say from you know from a more I would say phenomenological perspective phenomenologistish perspective I would say um so it is also important to distinguish uh intentional intentional uh quality versus intentional the entity the was it the um so if say conscious conscious uh conscious qualia or qualia the qualia uh comes to to me like if we if I am observing a red apple then red apple can be also the qualia but but in the but in terms of perspective or the orientation subjective

orientation that if I would like like um be be um open to um approach I would say open to approach to the redness of the of the apple or am I close to like am I deliberately uh thinking that red apple is something bad or something something um harmful to to um my individual. Um so so those that kind of uh that kind of was it uh um type of I would say type of consciousness is well the questions are referred in general as intentional quality and um I would say much the easier easier way to describe this would be um having certain perspective toward some some kind of matter some kind of um conscious matter. So um I would say also modeling not just only the red apple the redness of the apple and the roundness of apple but also what uh what what exact perspective does the um an individual is oriented oriented toward um having having which which perspective is very important I would say when we try to model the consciousness and um in terms of in terms of how so how can the active inference framework can uh can play in in terms of modeling this perspective I would say um let's start from uh let's start from the very uh very basics of let's say um free energy principle the the physics principle um that the individual uh individual turns out to have certain certain kind of veilance or certain kind kind of feeling or orientation towards the uh toward the certain OB objects and that also might be uh might be self-organized within the within the um agent uh within the agent or conscious being. Um so what I would suggest would be um if we can if we can deliver this perspective system as something um something in in the active emperence model uh something that um makes makes the agent to infer or um so as a as a different inference layer um or so Sorry. Um as a as a inference different inference layer different dimension of inference layer that is uh that can be uh uh described in as a um a certain form of information I would say. Um so that would be one uh one meaningful path that we could model the perspective structure or the consciousness certain parts of consciousness. I would say Adam and Alex. I don't know if either of you want to Alex seems to be getting the the ball from Adam here. Uh well, my computer crashed in the middle of this, so I think I'm not 100% sure what we're talking about. Hi, Carl. Uh so uh I'm not exactly sure what we're discussing. I think I got a little bit of what Hungju was saying, but uh maybe remind me of the prompt. Yeah, the the the prompt has to do with uh like this like the self as a uh something that is is not necessarily persistent in you know it's an output of inference and there there are you know uh many well doumented states in which self other relationships are not uh in which we have a different syntax you know where like non-dual consciousness being one of them how do we you know how do we think about all of that within an active inference framework >> so are you saying the output of inference is oneself just to make sure I got it >> yeah it was a Yeah.

It's like that the idea of like a like there being a a self other relationship. >> Uhhuh. >> Is it >> with other as in like other entities outside of like outside blanket? Right. >> Okay. >> Well, this might not be the most appropriate answer, but I I tend to think of it as maybe a different direction. So, this is something Carl didn't also mention in his talk. Uh, but he and I have worked on this concept. So another corollary of the free energy principle is mortal computation. Um and you and I have written at length about that. Uh the audience can just type Google and it'll come up. And um so then that's a little bit about talking about embodiment and inactivism and embeddedness and extended theory of mind. And then Carl and added a fifth E to the five four E cognitive theory. So it's 5E cognitive theory elementary or basil cognition in reference to our friend Michael 11. Um, and so like these this like a whole framework for discussing like sentience. Uh, I I tend to avoid discussions of consciousness a little bit. Carl is far more braver and more equipped to address these things. So I usually stay away from that. I like to look at like a low level. But when you say the self, um, I think at least I haven't thought of active inference is outputting something. It's rather active inference is like a a reference to like the dynamics and mechanics of something that is embodied. So you need to talk about like a substrate and so we need to talk about well what platform are we building on like is it a robot and that is the self. So we have kind of this like entity and again it could be like nested markoff blankets and we could talk about lots of jargon but the idea is that you have some kind of object and it's uh its primary directive is it doesn't want to die or rather it is trying to preserve its non-equilibrium steady state its nest um because it doesn't want to disintegrate in its heat bath. So the whole point is it like moves through uh the universe and trying to collect resources. It does the epistemic foraging that active inference is all about uh because again that's critical to maintaining its nest. That's it's prime directive. Um and so like the self the identity you have to like touch on the notion of embodiment and you need to have like a body to talk about and so active inference is performed within the context of that body again can be described as a marov blanket and so that is the self I mean of course it can be destroyed it can disappear uh if of course it fails to maintain its nez um but the idea is like that's how I think of the self and then you can use the idea that that's the immortal computer I underneath all of this is the FEP is saying we're all mortal computers and we live in a collective of mortal computers and so that's the other right so there's other mortal computers so I think I kind of touch on that and that's kind of the way I think of it rather the self is kind of like inherent to the fact we have to start talking about an entity a substrate something physical and we also need an environment of course because activism is critical so you need some way

to describe the body it's interactions. It's thermodynamic coupling to some niche. Uh and then of course we need other other bodies that also interact with that niche and interact with that body. Uh and then of course you get embeddedness which is then how you get society and culture. And then we can start working our way up the the big hierarchies that Carl often likes to talk about. Not to speak about you when you're here but as we work up to bigger and bigger self-organizing systems. So that's my perspective. I might be misunderstanding, but that's how I see self is it's kind of like, well, you got to be able to talk about now a platform, a substrate, a body, whatever it is. It could be E.coli. Uh, and that could be fine. That's a a wonderful example of the most element, one of the more simpler mortal computers. So, it rather than an output. So, I don't know if maybe I misunderstood, but that's how I think of it. And so, I kind of like to touch on that is sentience. I don't know about the consciousness part. I very careful with that. So, >> Sure. Sure. Sure. I I I I think that uh you know just to qualify that uh a lot of meditative traditions see uh like laughter as like a an observation of surprise at realizing that you one is not other than uh figures into this that I I'm Adam, I want to hear you speak to this. the part about laughter in particular or >> well no just broadly >> um well what just came to mind with the part about um laughter also would be um some of the work Carl's done with um artificial curiosity and insight and some of um Schmidt Hoover's uh uh framings of uh humor in terms of like rapid compression of um uh sudden uh I guess a sudden arrival at an increasingly elegant models. But um in terms of I guess what's come up is um I was uh wondering about um in the process of um self-evidencing whether the most robust self-evidencing involves um having a certain porousness to the self. Uh there's um uh some work that's been uh called under uh different names such as uh psychotic mark psychotic marov blankets or like you can have marov blankets that that aren't well tuned enough that might be like overly rigid and um they don't um uh adjust to the multiscale competences they ought to be a part of um and the co-construct niches well and they can become like um pathological or um maybe like parasitical or like or cancerous in their in their nature. So any but the uh uh guess the thing I'm thinking of is um uh the the importance of maintaining metastability uh or or for um ongoing I guess um uh chaotic uh itinerency or ability to adapt or evolve like uh if if in the in the process of uh preserving the self in a given form The best way to do that would be have a grip that's not too tight or not too loose um on the self. So those are some things that came to mind. I don't know if that's made any sense whatsoever, but I I put some uh uh some of the thoughts in the chat and some papers. Um uh there's a interesting one by um my friend um Goku um um how higher goals are construct and collapse under stress where he'll actually

talk about how systems um uh uh when they they're put under allic load they can basically um their clarical structure can unravel a bit as like the deeper um nodes or predictions the ones related to things like self and world um uh potentially become overloaded and destabilized and this having a certain adaptive function. So those are things that came to mind when uh from from your prompt. But >> all right. Yeah. I had another question on uh this this Carl's talking about sleep and scaling. You know, I was I was uh just watching a a stream of octopus documentaries talking about the the development of the octopus brain and how rapidly these animals have to grow and learn and and uh how therefore so much of their life is spent in sleep. And uh I'm trying to like think about that in terms of um like what we have going on with uh AI which you know by contrast seems to uh you know we we don't we don't have uh at least in any way that we think about it like a like a dreaming like a sleeper dreaming for the models that that we're put that we're using in AI systems. Although I I have seen some research sort of simulated dreaming in in an an effort to like uh create geospatial world models in in AI. Um, so I'm curious about like the relationship between these things and and uh how uh energy inefficient our AI systems are and uh the possibility that like human input might be able to function as a sort of proxy of of dreaming or sleep in AI systems. Um, that's a big kind of wide ambit question, but I'll kick it off there that like, you know, how how do we how do we let the machines sleep and dream? Um, you know, what what might that look like technically? Uh, what role does uh? >> Yeah, I'll just I'll stop there, Carl. >> No, that's a great question. I' I've been making notes about sort of key words that people have brought to the table which I think sort of deserve deserve a little bit of a comment. That's just picking up on um that particular question. Um it is interesting that um from the point of view of people like Jeffrey Hinton um building the Helmhill's machine um sleep played a really important part um and if you go back and Google the weak the wake sleep algorithm um you'll see the first forerunners of the kind of mortal computation that Alex was talking about. And interestingly, what this basically entails is that you're um you're basically building machine that can learn to infer. And the way that you do it is you expose it to data um and then you learn the parameters of um a neural network that maps to posterior beliefs. And then you let it sleep. You take away the data and you let it have a little little period to sleep. And during the sleep you actually generate from the learned representations ficted data dreaming data and that allows you now to learn the parameters of the mapping from the latent representations of the latent causes to the actual content or the outcome. So this is very very similar to um in fact almost isomeorphic to the original introduction the free energy principle that was cast really in terms of learning to

infer um but um the ultimate objective is an inference machine exactly along the lines of the helm machine or or um you know the way that I introduced the sort of helm hulks in perspective. So I think your notion that you know sleep um is is important is absolutely right. It was there at the foundation of all um certainly the the useful um um engineering endeavors in machine learning and artificial intelligence research. It's lost its way a little bit with uh current transformer architectures and um um uh DRL um um however I think it's coming back again in one sort of um muted sense. If you just look at the um what reasoning means in the generative AI literature, reasoning means trying to suppress hallucinations which you could call a kind of dreaming um but it's all mind reading that the user could actually see but trying to resolve um um mistakes or you know false predictions or you know implausible predictions um by um recursively um um you know in the background um generating its prompts and or multiple prompts and then selecting. So reasoning as read in the generative AI literature which has nothing to do with reasoning in the vernacular or cognitive science or indeed in what I was talking about uh or indeed in terms of probabistic um uh induction uh you know logical induction of the kind that Josh Tenbar might be uh you know might be interested in. Uh but reasoning in this particular popular literature in machine learning um large language model literature does actually uh come back to this notion of just taking you know stopping worrying about the content and then just turning things over in your mind which has some similarity to the kind of um um um basic model comparison and uh uh and and reduction that um was um in my presentation. So I think that yeah it's a really important thing to think about and you know again I come back to the fact that you're you should be asleep now if you're in California. So I'm sorry about that sleep's really important. Uh but can I just comment upon some of the really just very quickly on on some words that were came up dur during during the panel discussion which I think um would be really usefully formalized within an active inference uh setting. So I I'm thinking about you know um initially we're talking about veilance. Uh I think veence is really important. Um so the most technical and simplest explanation of veilance I think would be come under the rubric of a effective inference and in this very simple view veilance just is the precision or it's the negative uncertainty. So things are negatively veanced when you're uncertain specifically when you're uncertain about what to do and that's sometimes associated with anxiety um sometimes in a biological context uh stress uh and chron and chronic stress. So just knowing not knowing what to do because you've got a very low precision belief belief over your policies uh can be read as and of course you have to recognize that and now you have if you like a generative model that can recognize

oh I am uh in a state of imprecision uh pertaining to my policies and therefore I must feel bad. um uh and you can now um sort of unpack the implications and the virtual and or possibly vicious circles in terms of interacting with the world. Um that um that I think um is one part of the story. It's not emotional inference though and I'm now speaking to Alex's focus on mortal computation. Of course, as soon as you talk about mortal computation and things like you and me, you now have to move to intraceptive inference that the actual body matters. So the vastness the the biggest sensory organ I have is my body. Uh and I have to make sense of my body. Uh and you know I have to infer the state of my body. And of course as soon as you do that you've got some sort of minimal kind of selfmodeling inferring my state of self because I have to insta infer it could be completely pre subconscious or subpersonal you know in the sense of homeostasis. But when we move to allostasis then I think that the lines get a bit you know the lines are not so distinct and you could argue that that intraception in the spirit of mortal computation has to be an aspect of emotional inference and of course only you can own an emotion. So I think that's important. Um I've also got down here self versus other. Um you know I think that's a really important observation. So there's a whole other argument uh which actually came under Adam's nod to co-construction of our niches. Um so one thing if we co-construct a niche together is that we our world now becomes populated by other things like us which suddenly calls upon generative models to disambiguate did I cause that or did you cause that because we are co-constructing and that may be if you like a really simple-minded explanation as to why we need a notion of self simply to disambiguate between self or the consequences of the action of self versus other and of course that's a you know a deep issue when it comes to um attachment theory or mergence of theory of mind, you know, in a developmental context for the baby to recognize that it's a self and it may actually have to first recognize that the mother is an autonomous self and then recognize perhaps I am similar to mom and therefore perhaps I'm a thing as well and suddenly you get this this this sort of distinction between self and other. So I just wanted to really endorse that that point uh I think that Michael was making that you know the the other hood uh is probably more important for selfhood than anything else. You don't need it uh if you didn't actually co-construct your niches. Um the dimension um the the dimensionality in terms of quality of experience and quailia um that you'll be searching for in um active inference schemes formally I would read as the order. So first order inference is basically inferring the content of what's going on. Second order inference is inferring the uncertainty or the precision of that. So it's not about the content but it's the precision you're inferring. As a psychologist you would read that as

attention for example. Um and I think that that the um some people would argue to have a qualitative experience you have to have some um agentic control over the attention to the redness of the apple. um which just means you've now got to have a sort of deep generative model of a particular kind where um the depth is not now in terms of the sort of composition of content but the uh the depth in terms of parameterizing the quality of your beliefs about the content. Sometimes you can describe this as sort of metacognitive aspect. So I think that that that that would be a really interesting line of inquiry to explain and people do explain this um you know the distinction say between something that's normally transparent and opaque uh being defined by the ability to mentally act upon that representation on the inside through affording it greater or lesser precision but of course to do that you have to have a representation that I want to afford this to. So again it it induces a dependency on on self-representation and there are all sorts of um ways that you you know one can impact that in terms of I repeat phenomenal transparency and mental action. So I just wanted to say all of all of those perspectives I think speak to really exciting opportunities to formalize them in terms of relative straightforward generative models but perhaps not too straightforward because they do involve sort of higher order aspects of of of of inference that uh that we are particularly skilled at skilled at solving. So then I want to add to Carl if you don't mind I want to extend because Carl brought up really interesting the Helmholtz machine and the wake sleep algorithm. Uh we also you were to Michael's point about generative models like again that's been around the dreaming concept like Boltzman machines also used to be called by Terry Sejnowski and Jeff uh harmoniums uh the idea with that learning like wake sleep got generalized to contrastive divergence. Another way that dreaming kind of looked like was you have what Carl was mentioning like data dependent inference and then you had uh independent or let's say decoupled kind of like exploring trajectories that are confabulations as Jeff would call them and then you kind of like push them away uh from what's reality which is data driven. So that's kind of been I just kind of wanted to highlight that sleep has kind of been at least through the view of like oh we have negative and positive data and then today another kind of branch of research that I think dreaming has sort of kind of lived uh is in that contrastive paradigm we see a lot of self-supervised learning in the neuroai machine learning communities um and again that's kind of like pushing away outofdistribution data from indistribution data uh and again I Carl and I have also done some of that too with like meta representational predictive coding. Um so I kind of and I think the dreaming another way that it kind of lives on is in the concept of catastrophic forgetting. Uh because no matter how much

transformers are heralded as wonderful and the solution to everything uh they still forget. Deep neural networks have not really address this issue. Uh and so uh you can see even like predictive coding literature from my groups like my own and others where you use like the generative model that's being learned in this neurocircuitry. You sample from it and then you kind of use that as a refresh mechanism to try to improve your longer term memory. And that could be viewed as like a dreaming state or little uh let's say daydreaming phases. Uh and that goes back to also classical literature like Robert French and those that tried to work on catastrophic forgetting like the 80s and the '9s. Uh in another place too I just want to highlight is let's not forget that active inference which shares a lot of connections to reinforcement learning we have experience replay which is again another way where we kind of replay a lot of data in a buffer um and trying to kind of uh improve the learning mechanism of a policy uh so we that kind of mimics episodic memory kind of sort of so that's another place where I think it lives so I just wanted to highlight that and then to you asked Michael very brief briefly you snuck in there about like energy efficient AI and how energy efficient it is. Obviously it is completely opposite of what we want. Transformers are again Carl and I have argued this they have carbon footprints of literally cities uh to run a you know chat GPT and LLM. So I think like to get to that energy efficiency that we want and I think also Carl highlighted in his uh mention of interoceptive inference uh I think the answers again we go back to the idea of a body uh you need an embodiment system they need a mortal substrate uh and then I think that kind of begins to answer where dreaming uh and this consolidation of information that again Carl mentioned in his talk which is the structural selection. I think you need a body because then it kind of motivates things like rest. Uh because again if your body is fatiguing and you need to kind of go into a state of like oh I need to repair because you guys have mentioned we've mentioned homeostasis you guys have mentioned allostasis we should not me forget autopsysis uh the idea that we also need to repair and construct and deal with damage and so I think that that's where you need a motivation for the dream state and the rest state. And I think that that's where the body sort of answers it because again that comes from the interoceptive inference. I need to understand my own internal states and then I am driven to preserve them and again go back to our directive. So I think that's where dreaming the future of dreaming comes into play because I think with a body and its own limitations and its own needs and requirements and then how you interact with the niche to get those resources uh motivates things like rest repair uh and again so we and not to keep saying the same thing but Carl and I highlighted all the levels right you have to start with like homeostasis or homeostasis

allostasis on top of that you need to you need to describe autopsis you kind of need to go all the way. You have to explain all the levels to build like a complete mortal computer that you know is again an embodiment of the free energy principle. And I think then that starts to motivate things like why you would dream. Uh and so I think that's kind of where the future might lie even though we see it embedded like you know Carl and I pointed out some examples uh where dreaming is kind of used as a mechanism to repair learning uh or to for example learn to contrast confabulations from real world data. So it's just like a little thing I wanted to make a comment or long comment on. And then if I'd I'll just add quickly I know we don't have too much a question for Carl. just wanted to ask him based on the talk and he mentions like a bunch of these different things about active inference and I love the formulation but something that he knows that I'm obsessed with is the role and Carl you mentioned neuroai neurobiological learning and I know from your group and a lot of those that work on it back propagation of errors is used often to train the deep neural network forms of active inference or you do things like probabilistic graphical models that are discreet which are kind of hard to scale up to very big systems. What do you think is the role of getting the learning rules right like Hebian plasticity and you know because again predictive coding has local learning rules uh I often see like my group tries to show you how you can do active inference with that but I don't see a lot of that what do you think is the role and the importance of getting that right that neuratomical description like synaptic plasticity correct and not using like the chain rule of calculus which is convenient but isn't really what our brains are doing I mean the neuroscience still has not confirm that we are just a chain rule system back propagating along the same synapses that we do forward transmission. So curious to know your thoughts on the role of that if you don't mind commenting if we have some time Daniel do we have time now? Yeah, go for it, Carl. And then any last comments from others? >> Right. I very Yeah, really excellent point. Excellent. I think at two levels just that sort of um the importance of mortal computation for the way that our tech industry is developing and the impact upon our lived world. I think that's a really important uh consideration which speaks to you know quite substantive issues uh in the way that we deal with big tech and the direction of travel in machine learning research and investment and building power stations in Virginia for example. Um uh so you know much of the motivation for biometric or neuromorphic uh mortal computation um inherits from that and from the the the physics of information uh sometimes cast in terms of say thermodynamic computing if you like but also the importance of neuromorphic. So that's I think a really important issue beyond academia. Um the second

issue is a specific question about the particular algorithms that are used and when they diverge from a neuromorphic or um um biometric um principle or when they diverge from the principles of least action um evidenced in biological and real world self-organization you know that's trouble ahead um and one example of that is back propagation of errors and of course the whole point of the freeji principle is it is a local free is you're an extensive property which means if you minimize it here and just here you're minimizing it everywhere if you're repeating that local um update road. So in answer to your specific question about the update rules I think they're vitally important um is there a right one or a wrong one? I according to the complete class theorem no um there isn't. What you would say is what is the implicit generative model that explains this particular functional form for for the uh um the um belief update uh rule for the parameters or the learning rule in machine learning talk. So there will be a generative model implicit in the functional form of your learning rule. Um and that has to have the form exactly of that hamph decomposition. So there has to be a sort of a solenoidal term. There has to be, you know, an update an update term and and a prediction term in it. Um there's also going to be some sort of volatility terms. There's going to be a decay term in it. Um so um but the key point is it has to be local because we know that the paths of least action um are defined by action which is a time integral of an energy potential and that enab potential is an extensive function. So it has to be local. So you know from the start that back propagation of errors is not the right way to do it. Um that would be my brief answer to to your question. Thanks Carl. Appreciate it. >> All the panelists now have to to say something very wise in their concluding comments. I think that's that's the pressure upon you now. >> Yeah. any short closing comments and then we'll go to the next session. >> Well then I'll just knock myself out quickly. Uh thank you Carl for again answering my question. I mean I obviously have argued for like a better part of a decade that back prop is not the right way to go. So uh and uh I think again through the passive lease action that kind of motivates why we should care about like local biological plasticity as kind of our guiding way of designing for example predictive coding active inference systems. So uh maybe to try to say something wise in the spirit of what you were just saying to the audience I think that uh yeah we should probably if you are interested in this like idea of neuroi Carl and I call it biomimetic intelligence uh but they're all kind of the same thing I think then we should be pu making a bigger push uh towards biological plausible forms of learning and rules uh and I think that that's kind of a key idea and then just remember that when you're constructing these As Carl has aptly pointed out, there is like an implicit generative model. Uh what you might not be

able to do with an implicit generative model, I'll just briefly comment, is you might not be easily able to sample from it. Sometimes it's better to have the explicit model. Uh but again, uh it's kind of like a trade-off in what kind of scientific question you want to answer and what type of model you want to build. So, uh, yeah, that's my minor comment, but I do think that there should be a lot a bigger push, uh, towards, uh, biologically plausible credit assignment learning, uh, in support of things like an active inference framework. And so maybe that's a call to the community to the active inference symposium. I know my students who will be presenting later uh, these coming days and today, they know that role, but I think more people need to do it. So, we need a bigger push for that, but maybe that's over like the next 10 years or so. So, thanks for that. Does anyone else >> You'll have to hand the baton by calling upon the >> Okay. Okay. Then I'll give it to uh Michael since it looked like you uh opened up your mic. Your turn. Say something wise in in the spirit. >> I have no wisdom here. I'll just say that uh this has been a a delight. I think you know it's it has prompted me to think about the the way that uh like why is it that the production of v like deep fake video has become so prominent for both open AI and for meta in their in you know public facing products and that it may that that like the insane uh finance like economics of these organizations is actually like leveraging humans trying to like uh dissipate out into like all of the possible fantasy worlds that we can imagine. and with these tools and and that that's somehow going to like hold this thing up. I mean, I'm sitt sitting here thinking about the AI bubble that we're in right now. And uh whether whether it's a bubble that's going to pop or whether whether it's actually like a a cell membrane now being held together by all of this this sort of uh collective fantasy. I'll leave it there. >> All right. Thank you everyone. We will be back soon. >> Yeah. Thank you. Good talking to you, Carl. All right, welcome back. This next session is going to be facilitated by Ian Tennant. So Ian to you and you have an hour and 15 minutes. So thank you. Go for it. Wait, I don't hear your audio. Yeah. All right. Apologies for that delay there. all right. The next video that is going to be played, sorry for the little bit of scheduling changes. This is Bradley Alisia, neurosimulation of spatial intelligence. It's going to be a 63minute recorded video. Hello, my name is Bradley Alisa and I'll be presenting today on the interdisciplinary underpinnings of the neurosimulation of spatial intelligence. Uh this work is in conjunction with the orthogonal research and education lab and particularly our cybernetic interest group. So an overview of where we'll be going today. Uh this is I'm going to sketch out a route to the neuro simulation of spatial intelligence. So we haven't done anything uh in detail but this is just a route to getting there. Uh the first and and a lot of these different topics are kind of

related and so I have this nested diagram. The first thing we'll talk about is vertebrate neuro anatomy. So thinking about the origins and correlates of spatial cognition and vertebrates. The next is ecological psychology and biocybernetics. So thinking about the brain as this sort of ecological ecologically influenced structure uh but also thinking about it as a circuit in terms of how biocyberneticists would think about the brain. The next is uh path integration and taxis. So we'll talk a little bit about path integration or root finding and then taxis which are movements towards different stimuli and movements among gradients and how those generate intelligent behavior. The fourth area is neuroethology. So we'll talk a little bit about neuroethology and how the brain interacts with sensory modalities to produce uh spatial cognition. And then finally we'll talk about embodiment in GS slam. So embodiment of course is very important for any sort of model of intelligence as well as this notion of GS slam which engineers use to build control systems that are spatially embedded in embodied models. So when we talk about neurosimulation we're talking about simulating the conditions of behavioral outputs. And so the neuro component refers to development, embodiment, sensory world, amplification and evolution. Development often is associated with generativity, amplification with something called good regulation and evolution is roughly corresponding to shared derived traits. The simulation part then replicates components of a system under plausible conditions. So to take all the aspects of the neuro and then simulate it in different ways. So developmental neurosimulation which is something our group has worked on is simulating the construction and expansion of the embodied nervous system. It allows us to ask what could happen in terms of morphology and behavior. From our work we might aim to model something like simple connetos self assembled and an embodied context. So our first stop is to consider the vertebrate neuroanatomical view. So spatial navigation in vertebrates is typically thought of as a learning memory and representational phenomena based on a process of consolidating episodic memory. And so in mammals specifically, but in other vertebrates as well, spatially explicit behaviors have been tied to place in grid cells in the inter rhinal cortex, which is mammal specific and their analogous structures in other vertebrates, but which provide a geometric representation or an abstract map of allocentric space that space outside of the body frame. So if you go back to 2014, the Nobel Prize in physiology and medicine was awarded for the brain's navigational place and grid cell system. And this was awarded to two groups. Uh one headed by Dr. John M. O'Keefe and the other headed by doctors Mabbrit and Edward I Moser for their discoveries of nerve cells in the brain that enable a sense of place and navigation. So you see the difference between place cells and grid cells. Grid cells set up

sort of a coordinate system where you can have these periodic episodes take place and then place cells represent specific discrete objects in the environment. These are the two papers that we'll be talking about here. The hippocampus is a cognitive map which is actually a book and then the Moser's grid cell and cortical representation. So the cognitive map is a system which generates an absolute reference of extrinsic space and it's located in the hippocampus. This is from John O'Keefe's work and so it's kind of thinking about spatial cognition being located in a cognitive map. This approach was based on years of neuroanatomical investigation. So before the late '70s there were a lot of lesions studies, behavioral experiments and neuroimaging that had been done that had basically been confirmed by Ke's team's work. So this work hypothesized place and place cells. So place cells are information about the environment and place cells are displacement and exploration of the environment. So you have two different cell types and they're working together in the same network to predict spatial locations and drive spatial behavior. Now of course we can look across uh we can look across vertebrates to see how diverse the hippocampus is. So the hippocampus is actually a homologous structure across uh vertebrates and you see this in humans of course uh which is in D down at the upper right hand side of the slide. You also see it in opossums uh and that's of course also a mammal. Then tu lizards and petrels uh petrels which is a fish. And you see those they look a little bit more they look a little bit different than the hippocampus that you see in the human opossum cat an analogous function. And so we can predict that there's a basal capability for hippocampal spatial processing. And this is like sort of the acquisition of it in the hip in the last common ancestor that had a hippocampus. And so we'll talk about two more papers. So the reason I showed you the foghorn is because we're going to be talking about the zebra fish to see and then talking about maze studies which you can uh investigate from rodents which is this paper that's cited here to fishes. One thing we can learn from the zebra fish larvae is that place cell analogues are found in the fish teleost. And this is showing how place cells come online in larval development. Place cells in the fish and teleosts are enriched in telencephalic development which happens during this larval period and it forms a spatial population code in the adult. So here we have these place cells that are sort of forming uh in the uh early you know it we see like learning in the teleost during this larval development and they go on to form the spatial population code which we see we also have these maze learning paradigms. So one of the things we want to work think about is we want to think about embodiment and how organisms with bodies kind of explore their space. So it's not just with the brain but space. And so there are these maze learning paradigms where

learning utilizes place information and builds a representation of space. And we'll talk later about how space itself is represented through a sensory information. Um but you know then we have to think about how spatial navigation and learning is is is acquired and is it generally learned this way. So we might actually use these maze learning paradigms to train an agent or to train an animal and that gives us interesting results. But in a naturalistic context, these mazes may be just kind of an artificial thing. And so the learning that we acquire might be an artifact of that. So what do you do if you do not have a hippocampus? So a whole lot of non-vertebrate animals where single cells conduct spatial navigation but they don't have a hippocampus and they don't have that kind of a learning structure. So what other anatomical mechanisms and functions are involved in spatial navigation? If we're interested in the simulation part and even in the neuro part we want to know if there are other mechanisms that are plausible. So this brings us to the ecological psychology and biocybernetics view. And in this part of the talk, we're going to look at this paper common and specific activation supporting optic flow processing and navigation in neuroimaging studies. And so we find that functional segregation between optic flow processing which is in the dorsal pathway which sometimes is called the what visual pathway is separate from spatial navigation which occurs in the ventral visual pathway or the where pathway. So we have optic flow processing and spatial navigation happening in the visual system and those are two they're basically the visual system is providing information of those two types of information to the hippocampus or to other spatial processing centers. And so uh one thing from the optic flow literature that's interesting is this idea of flow parsing mechanisms. and flow parsing mechanisms or have been identified that extract object motion from middle temporal units in primates. So there's this spatial uh processing mechanism that is uh used to extract object motion from visual information and this allows us to parse naturalistic flows. Uh spatially tuned units identify both object borders and relative or egocentric motion. So you have a lot of at least you have at least two mechanisms within the visual system that allow us to sort of parse information relevant to spatial cognition. And then of course we also have optic flow and allocentric navigation. And this is also another thing that happens to visual information to extract the spatially explicit information that can drive spatial navigation. And so if we look at a primate egocentric model, remember this is just in primates. We have different types of cells. Now we have our place cells of course, but we also have our head direction cells and we have a representation of speed which in so we have our speed cells of course. Then we have head direction cells and speed cells and those form this sort of network. And then you have

vestibular and optic flow information. So the optic flow and vestibular information is integrated by head direction cells. So you see those two inputs to head direction cells and head direction and speed cells iteratively update place cells. So place cells sit at the top of this hierarchy. Head direction and speed feed into it these cells their states and determine the uh state of the place cell. And then vestibular and optic flow information are going into head direction which completes the network. And so we're getting sensory information and ultimately it's driving the behavior of place cells. If we think a little bit about these head direction cells, they're very interesting. And so maybe we can follow more up on that. So we have two papers here. A bio inspired 3D neural compass based on head direction cells. This is an engineering take on this uh where we're looking at this model of of bat head direction cells. And then this more biological paper where they look at three-dimensional head direction coding in the bat brain and it's based on this model of the bat. So the bat uses echolocation and vision to sort of tune their spatial navigation. They have this flight room setup where the bat is flying around the room. They're taking pictures of the bat and they're looking at their brain. They're looking at the um HD cells, their tuning width and their firing rate, and they're looking at it for different behaviors. They're looking at the tuning width in percent. And then they're looking at the peak firing rate in hertz. So this shows that there's this response of head direction cells for different types of behaviors. And in flight, there are different types of neural responses. And so if we put this into a phylogenetic perspective, we have primates, corrupt and rodents. These are all mammals, but they're of course have different ways in which they treat optic flow. So in primates, we have optic flow contributing to a view representation. And this results in visual specializations and primates, but it has this view. This is why I showed the image of that network with the uh place cells and the visual information. So that's in primates. But in corrupt optic flow contributes to a view representation as well. But this is due to visual and auditory specialization. So they have this view representation and and corruptor is a sister out group of primates. So that makes sense. But they have this extra set of specializations. And then rodents being kind of distant from corruptor and primates optic flow contributes to a place representation where somata sensory and old factory specializations dominate. And so you can see that there's this relationship between how information like optic flow goes into the brain and depending on how the brain is adapted to different types of environments, different types of lifestyles, spatial cognition can occur in different ways but kind of rely on the same substrate. And so this is just a little uh at the lower right there's just this little visualization of how in flies visual motor transformations um basically stabilize flight through this optic flow

mechanism. So this is actually quite a bit different than the mammalian last common ancestor and this is quite different than the last common ancestor for mammals. But you see that uh you see that flies are also using uh optic flow to stabilize their movements and then that translates into how they can navigate through space. O'Keefe and Nadel actually had a lot to say about Gibson. And they quote is, "Spatial cognition is a set of transformations acting on sensory input and variance in the form of things like ambient optic arrays are extracted from multiple sets of alternative viewpoints or experiences. And so this paper below on the ambient optic array and eye perception kind of shows what this looks like. So basically uh you have all of these possible sensory transformations from the sensory world. We have this physical set of transformations and this visual set of transformations and spatial cognition is basically acting upon these transformations finding specific pathways forward. So you have all the sensory information which gives you a lot of possible places to explore. brain and spatial cognition filter out and leave you with a small set of alternatives that you can then explore. So sensory transformations can disambiguate objective or allocentric and subject ego subjective egocentric movement. These trajectories are the consequence of the sensory environment but also the way the agent moves through the sensory environment and it can kind of put things into different frames. And so of course then observers or agents being observers move to different locations relative to their target and they have to be able to track that target as they move even though the perspective changes and the object changes its kind of size and shape in the allocentric space. So this little set of pipes here is, you know, if you have this agent here and it's moving in sort of an arc, it's going to take different viewpoints of this set of pipes and it's going to look at it in different ways and it's going to get different information, but it can always tell that it's looking at an arc. Furthermore, it can remember all those positions and it can use those to navigate through these pipes knowing relative information, relative allocentric information about the object in the environment. And so this paper by Rushton and Warren is moving observers relative retinal motion and the detection of object movement talks a little bit about this in terms of your agent moving over time at a certain angle and then having these points of reference where their uh frame of reference shifts and then they have to use their memory system to sort of align everything later when they go back to navigate it. And of course we can look at very small connetos to think about different types of plausible visual pathways. So what I showed you in the last slide was this image of all these plausible visual pathways where we kind of the brain filters out all of the implausible ones and leaves us with the most likely visual pathways. But in a in an anoid conneto which is very small um it's less

than 100 neurons uh we get this we can do these experiments where we can look at all the plausible visual pathways available to the anoid and so we can look at this in terms of a two pixel system which is this sensory motor photo receptor uh that can spatially resolve light stimuli and then the information is sent to the aector on each side and it drives behavior. here. And then over time you get these different nerves that are ipsolateral and you know they cross the midline and then you have these uh more evolved eyes you know with lenses and eye spots and things like that. And then so you build up from that simple model to these more complex models and you can get these more complex behaviors like mutual inhibition, bilateral divergence, signal delays and all this. And so the reason I'm showing you this is to show you that uh spatial perception and spatial cognition can be relatively easy. You don't need to have a representational system. You don't even need to have that many cells. And but this is also it's a hypothesized precursor to visual system. So if this looks familiar this is because it is the sort of biocybernetic approach and it's similar to Brightenberg's approach to biocybernetics with vehicles and building vehicles that uh approximate small connetos nervous systems things like that. We can also look at spatially distributed light sources which require sensory arrays internal pathways and viewpoints on the light sources. So things now you get embodiment involved and body location and posture become important. And we actually had an interesting discussion at one of our Saturday morning neurosim meetings on Brightenberg's neuroanatomy and synthetic biology work. Bradenberg actually did a lot of neuroanatomy before he proposed the vehicles work and he built these he investigated fly connetos quite extensively looking at their neuroanatomy looking at their uh their output behavior and this is kind of what led him to the vehicles Oh, So you know there's a lot of kind of uh space between fly connetos and braenberg vehicles. It's really interesting biocybernetics and I think has a lot of relevance to this topic. So go back to our July 19th, 2025 meeting for more. And we have this new paper out called a brief history of cybernetic imperatives. This is on the SI archive. This is uh thinking about how we model plausible movements. So so far we've talked a lot about uh vision and visual information and then spatial cognition as sort of a neural circuit or a set of neural mechanisms. And so that's all well and good, but then you have this behavior and this behavioral output is related to that processing and how much information specifically how much regulation goes into those movements. So this again is kind of thinking about simplistic movements that result from maybe like a small agent like a small conneto or a vehicle. And so we have these different types of movements going from random and brownie in motion to levy flights where you kind of move around randomly but visit specific locations. And then you

have ballistic movements which are just kind of like um sort of moving in a certain direction and then moving again in a certain direction which is moving directionally but uncontrolled. And then four you have first order prediction and dynamics. So once you enter prediction in the mix, you bring in feedback, you bring in this sort of these dynamics that are smoother. You know, you can explore areas that are sort of controlled in a curved manner. And then finally, five, you have higher order prediction and dynamics where you can smooth out your curve and you can actually make these very intentional, very controlled movements. Maybe you're trying to uh reach a prey or something like that. And there are a lot of different uh behaviors in insects which are very interesting in terms of prey capture as bats as well where they kind of uh move towards their prey. They kind of move past the prey and then back towards the prey. And that's all based on their ability to get sensory information to update their spatial map and then capture the prey successfully. So this sort of typology of behaviors is something we're kind of working on as well and we want to fit this into this model of spatial intelligence. So next is the path integration and taxis view and I say photo chemotaxis that we want to talk about here. So photoaxis are light signals. Chemotaxis are chemical signals. Thermotaxis are uh thermal signals and thigotaxis are touch signals. And taxis refers to when you have an environmental signal in that form and then it forms a gradient in the environment and the agent or the organism will follow that gradient from where they are to the highest concentration of that gradient. So the toxic movements are organized not by some great regulatory mechanism but just kind of following the gradient to the next highest concentration. So this kind of relates to path integration. So I'm going to talk a little bit about path integration which is sort of um wayfinding or u homing if you are familiar with like ants or something like that where or insects where they use path integration quite extensively and it's kind of remembering space and remembering sort of the map of the space around you and then kind of going from a home location knowing how space is transformed and then moving back to the original home location. So you explore space a while and then you can always find your way back home. And so that's path integration. It has an interesting relationship with toxic behavior and we'll get into what that is. So we have these two papers, possible uses of path integration and animal navigation and then principles of insect path integration. And so this is path integration in ants. So you can see it's a very complex set of behaviors. It almost looks like our second phase of behavior is the uh levy flights where they kind of move in a certain direction but they make these leaps and uh this is kind of you know what what happens in ants where they leave home. They they forage in the environment and then they have to return to the nest.

And so they're using their ability uh they're using pheromones. They're using their ability to navigate and remember space and remember how it transforms when they turn their bodies and then make it back home. So path integration is the basis for cognitive mapping. Path integration will happen in other organisms as well, including mammals. So this isn't just a thing that you find in insects, but insects make great use of this. Uh so this is the basis for cognitive mapping. there's a limited use of landmarks and resetting processes. So they use landmarks but not exclusively. They also reset their uh their calibration tool. So usually they measure footfall or something like that and they use that as a heuristic. Uh they also reset their footfall information sometimes. So this there are all sorts of mechanisms involved in this. So the information needed to compute a trajectory involves a limited approximation of distance when the target is close. But when the uh target is farther away, you need to have a little bit different strategy. Sometimes you can use the direction to goal which is fixing and resetting your position and then matching behavior or having a coordinate system on top of a map and having that coordinate system be the guide to getting back to home base. And so there these different ways in which information can compute a trajectory and perform this path integration. Now there's some very interesting connections with taxis because like with path integration, taxis are cases where the organism goes out explores their environment. Maybe they're foraging or maybe they're doing something else trying to find some delicious uh meal of some chemical chemicals or something like that and then return home or return to some other place. And so instead of sort of these other mechanisms that we think about, we actually have these gradients that one can follow. You follow the gradient to its source and you then find your reward. So there's this very clear signal, environmental signal of where to go. But nevertheless, you have to it's not a straight line. It's not a straight shot to the source. There's a lot of exploration and there are mechanisms you need to use to compare two different places where you visited so you know where the greater concentration of this thing is. So we can actually build this interesting table which is to generate a sort of a contingency table for any one position as you're tracking or um whatever it is towards the source. So there are three different behaviors. You can go towards the source which is positive. You can partially go towards the source which is angular or you can be negative and move away from the source. And so this is all based on local information comparing two different places revisited and moving in that direction. And as you noticed with path integration, the path is wavy and it's curvy and and it doesn't it isn't a straight line. And then we can use different types of information in sensory information. So we can use light, photo, we can use chemistry, chemo,

we can use touch, thigma, temperature, thermo, hardness, duro and electricity or electro. So these are all known taxes. And sometimes you can use more than one taxis. You could use photoaxis and chemotaxis or photoaxis and thermotaxis and integrate them in a way that helps you to improve your performance. So for each of these then you can say if you're at a photoaxis you can move towards the photoaxis or towards the thermoaxis or you might move towards the photoaxis and partially towards the thermmoexus and this is all based on local information. So over time it'll average out to this path towards the source but it does produce some interesting dynamics especially when you have multiple modalities. And so that relates to path integration of course because the source might often be some sort of you know uh food source and then being able to return home from there is also a function of uh sort of following the outbound signal or maybe following sort of the decreasing gradient. These are in mechanisms that don't involve the type of hippocampal spatial cognition we talked about earlier but is nonetheless part of the spatial intelligence. We can also look at agents and we did this interesting work with agents where we were doing this sort of dendritic computing with agents. So we had these Bradenberg vehicles with antennas and the antennas had these different sensory uh sensors on them and they take in environmental information in this spatial array. So you have sort of this beam form of spatial information going from left to right and it comes into this uh antenna or this array and then it encounters these transformation points before it goes in the nervous system. So the way dendritic computing works is you have sort of the analogy of dendrites in the brain where you have these uh these treelike structures where environmental information hits the treelike structure and all of these sort of uh parts of the array interact with one another or you know maybe the ones to the left take in more information the ones to the right given the sort of the direction of the source of the information and then that influences how it's processed by maybe like the logic gate or some sort of transformational process before it gets into the into the nervous system. And so this is a interesting sort of paradigm because you can do a lot of interesting things with spatial processing. You can have environmental information that is directional. It could be far to your left or far to your right or it could be right in front of you and that information is processed differently even though it's the same source. So if you think of taxes, of course your taxes are going to be different. Your information is going to be maybe refined or maybe it'll be obscured by this kind of paradigm. Um but you know it's it's going to influence how things happen. And of course this is sort of how you know if we have if we care about embodiment, we're going to have this sort of array of sensors on the outside of the organism. It could be a visual system. It

could be other sensors. And so we need to consider how those are processed spatially and how that influences behavior. This is another example of sort of linking phenotype to the sort of processing where the surface perimeter is very curved and you have these these uh dendritic trees or these arrays that uh sort of conform to the shape of the the body of the agent. You have what we call sensory beam forming. This is a an idea from engineering where you take the different orientations of sensory information and process it based on its spatial location or its spatial orientation to the sensor and then to the organism as a whole. So this is very different from the way we kind of usually think about spatial processing as sort of like this undifferentiated signal that then gets processed in the brain. This is actually what's processed in these placement of the sensors on the body. We also have sensory modality specific gradients. This is from another paper I published. So in this case each sensory modality forms an input vector and these are doesn't necessarily need to be an embodied agent but this is like uh where we have different sensory modalities and the agent is sensing the modalities and the modalities form a gradient in the environment. So each to the agent. A directed vector describes the direction of exploration along the sensory gradient for each vector. So each of these vectors describes a certain modality. It describes a direction in which it's it's located and the direction of exploration that's expected. And so you can have these different vectors um that are aligned or they're misaligned. And so sometimes visual information is in a different place in the environment than auditory information or touch information. And those all have to get integrated. Uh either you know sometimes you'll have an angular displacement, sometimes they're all coming from the same direction. And so that's easy to determine the the sort of the direction of behavior, expected behavior. And so this fits into this model of openloop intelligence augmentation where we can have this model of intelligence as an openloop amplifier. to something like what W or Ashb predicted in uh his cybernetics work but instead of having uh this sort of model of the environment and sort of like an intelligence or an internal model we have an intelligent agent with a world model and then we have this uh formulation of intelligence which is sort of the uh sensory information that's incoming feed forward behavior the behavior that's generated And then the selection mechanism on that uh sensory pro sensory information. So sensory processing is incoming sensory information with selection that determines your initial sort of intelligence and then there's some model of the world that you refer to to refine that. And so thinking about spatial navigation in a mathematical framework, we're interested in spatial navigation as this continuous set of dynamics. So we can use a labbrian to describe this. We can draw this from the equations of

motion. So we're interested in spatial navigation, not necessarily as an agent that's visiting a number of locations and looking at the pretty uh squiggly lines that it generates, but we really want to know what the equations of motion are because the equations of motion can tell us something about the behavior of the agent and in turn what they're doing to process information to produce this trajectory. So this is a kinetic description and navigation where the segments of motion or output of place in grids rep representations for example where other types of spatial representations are put into the environment as behavior and this is an active process. So as representation changes so too do the equations of motion. This is a sketch of maybe what this looks like in terms of the uh taxis. So we have an agent Bradenberg vehicle. It's integrating taxis from multiple sensory sources and produces this grandian description or this trajectory that goes through space. You start near sort of a thermal thermal gradient. So you get these thermal taxis and then you leave the thermal gradient and you go to the chemical gradient and you venture through the chemical gradient. You come back out to the uh pH gradient and then you venture through that. come back to the sigma taxic gradient and then out. So you you explore these different gradients as you encounter them. Sometimes the gradients overlap but it basically describes sort of the spatial behavior in terms of these guiding cues and then the internal mechanisms. You know that's something in this model it's a black box but we can think back to some of the potential mechanisms that we might use to produce this behavior. So I want to go back to sort of the internal aspect of intelligence and talk about the neuroethological view. Neuroethology being sort of the study of the brain in naturalistic contexts. So again we revisit this idea of spatial congruence. We saw that back a few slides back that we had those uh sensory vectors that were maybe they were aligned and maybe there was some displacement. In this case, you can see kind of what that looks like. They train these cats to respond to signals that are both congruent, meaning that a visual signal and an auditory signal are coming from the same place, and signals that are in congruent. So, if the auditory signal and the visual system or visual signal are coming from different places and then the food reward is where you get the food, right? So, you know, it has if the signal is coming from the same place, the cat goes to that food tray, gets the reward. If the signal is coming from two different places, the cat has to sort of figure out where the food reward is. Is it the auditory signal or the vis uh visual signal and they have to disambiguate the two signals to get the food reward. So, disongruous sensory cues are feel spatial biases. Not all locations or orientations are equivalent. So it has to sort of sort through even though it has all these orientations and rotations spatial rotations and spatial location information stored in its internal

model of the environment. It still has to disambiguate a lot of information to make decisions to make the right decisions without error. We you know we have this model of coincidence detection spatial coincidence detection is rooted in the superior colliculus. You can see a model of superior colliculus at the lower right. And so these experiments are very famous for revealing yet another place where spatial processing happens. Uh related to this is the chemoaffinity hypothesis. This is another famous experiment where it's more about spatial framing. So the idea is that you have well in the Spir uh experiments on this he took a frog and he cut the eye out and he rotated at 180° . So when the eye is in its normal position, the frog can shoot out its tongue at like a fly and capture pretty quickly and do this pretty accurately. But when he moves the eye 180° , he rotates it in its socket. The tongue goes in the opposite direction of the fly almost consistently. And so the idea there is that you have this spatial frame that's based on the eye and the way it develops. that spatial frame is put into place and then when you rotate the eye you're basically shifting that spatial frame 180° off the spatial representation of the optic tectum is established in development and it stays like that throughout adulthood. When you shift the sensory array the corresponding reference frame is shift. So again we have these two cases of spatial incongruity. one where you have two senses that are uh decoupled, the other where you decouple the sensory array from the target in the environment. And so that leads us to this neurosimulated Morris water maze. And so this is interesting because you have visual and auditory sources and they're decoupled from each other. Not only are they decoupled, but you have multiple sources around the arena. The idea is these vehicles will travel and try to find the corresponding visual and auditory sources. So they there's kind of like this uh this are a great this spatial gradient of visual information and auditory information. And in some places it's mixing in some places it's independent. And so these vehicles are kind of exploring their space trying to fix on some source and exploring its space. And so this is kind of a continuous maze. If we think back to the mazes that we used for uh you know hippocampal learning, this is a continuous maze and this is something that will you know maybe tell us something about how we can process spatial incongruous spatial information in very simple nervous systems. And one more detail about this is that sometimes these sensors are all located sort of in the same direction. you have on this vehicle here, you have three sensors all pointed forward, but on this vehicle, but on this vehicle, you have sensors that are pointed in all different directions. You basically mutated them so that they're on different sides of the body. And so that affects processing. Sometimes that leads to a periodic orbit, sometimes it leads to something else. And we don't know, we haven't run these

experiments, but this is just kind of a thought experiment uh where, you know, we're kind of thinking about varying the sensor uh location, varying the sensory source location, and then having this naturalistic behavior where we can observe gradient following and behave output behaviors that are spatially interesting. So if we have this spatially incoherent framing plus spatially in in congruent multiensory information it can lead to release interesting dynamics. So then the next two papers are the brain stem integrator for self-location and memory and positional homeostasis and zebra fish and topographic maps of visual space in the human cerebellum. So this starts with a a quote from this Vaness paper from 2019. That's the second one. Topographic maps of visual space in the human cerebellum. And this is uh as follows. Retinotopic mapping under fixation precludes a distinction between purely retinopic or eye-centered, craniotopic or head-centered, spatiootopic or world center and other spatial encoding schemes. This is in line with what we've been talking about. We've been talking about different types of sensory centered uh processing, head centered processing, head direction cells, etc. And then this aloscentric view of the world where things are spatiial. We have this uh retinopic mapping that in humans at least or in mammals is precludes these distinctions. And so this actually points to the cerebellum is another site of spatial processing. The cerebellum encodes multiple polar coordinate maps of visual information that provides information to the aectors. So now we're not talking about the sensors, we're talking about the aectors. So now this is the outbound information and the cerebellum encodes these maps for that outbound information to be transformed. Okay. So now we're dealing with embodiment. We're dealing with the consequences of embodiment. The cerebellum is known for producing smooth movements. It's involved in motor control. It's involved in smoothing out motor control and and having these predictive models is very unique architecture allows you to do that. And O'Keefe and Nadal of course also had something to say about embodiment which is that the movement of an observer's body changes position but does not alter the position of objects. This is distinct from relative position. The body changes position of course as you move. It doesn't alter the position of objects though because you have sort of a framework for thinking about different transformational views on objects. So if you're looking at an object from the front or from the back or from the side, you can put that together and know that that's the same object. This is a graphical abstract from this Yang paper and cell. This is a zebra fish. It shows that the cerebellum and the inferior olive are involved in this location error signal and then it's also involved in determining velocity and also then you know the zebra fish is moving with a change in position. So it's locating the error signal. It's correcting the error signal and

generating a certain velocity that changes the position in an adaptive manner. And we can think of this as a control system where we have this input that leads to swimming behavior. So we have the neural coding, the behavior and these brainwide circuits that control this centered on the cerebellum, the inferior olive. And so we can think about this in terms of a map. So we have our cell cerebellar map. It's taking in information from different parts of this organism. So we have a lot of things in the head that are mapped to things in the hippocampus and then those produce outputs for those things. So we have output behaviors uh being uh mapped from the hippocampus to output behaviors. Um and this is very much like sort of a homunculus map that you see in the neocortex. Um and so you know we can actually replicate this in a vehicle. We can have this little area here where it maps sensors from different places to these maps and then an output to these sectors. So you have inputs and they get mapped to different cell populations which have uh sort of a mapping to a specific output. So in this case in vehicles or wheels and you have these uh sensors they could be light sensors or vibration sensors bringing in information. And so you know we can kind of classify our different uh inputs and our different outputs. And this is kind of what you see with the topographic distribution of output neurons and the cerebellar nuclei and cortex to somatotopic maps and primary motor cortex. So this just shows for different uh sectors like hind limb or proximal forelimb or distal forelimb there's a different distribution of cells for each of those types of output in the cerebellum. And so looking at this more closely you can build a start to build a network from these different uh cell types. It adds self, it adds a functional layer to these cell types in vehicles and it gives us this ability to map things outward as well as from the sensory system inward. There's this core sensor factor mapping where you get spatial biases in these inter neuronal networks. You get information exchange among spatially sensitive inter neurons and then peripheral processing networks which are shown in the different colors here are specialized sub networks for timing differentiating and integration of activity. So what we're trying to do is build a very small nervous system that replicates some of these findings. But next I'd like to talk about these two papers. The first is omnidirectional sensory and motor volumes and electric fish and then spatial planning with long visual range benefits. escape from visual predators in complex naturalistic environments. So, this first example shows the weakly electric fish. The weakly electric fish are interesting because they send out this electrical signal around their body and they sense prey in an environment where they can't really see anything. It's a very murky aquatic environment and they have this is what they call active sensation. So the electrical signal that's sent around their body from the electric organ

near the head to the tail where there's a receptor is this electrical field this volume. And so uh in the first paper that I talked about this they've uh reconstructed these sensory volumes around the fish at different orientations. And then these dots involve sort of prey that are in the electrical field so that the fish can sense the location of the prey so it can move towards those prey and capture them. So you can see that the prey are spatially located spatially uh segregated within this volume. um and then the fish have to make a movement towards that. And of course there are different perspectives on this volume. So if you look at the side view or the back view, you can see that there's a different distribution of things. And so one of the thing other things that they explored was this habit-based versus plan-based action selection. So once the electric fish senses that they're prey within the volume, they can go and capture the prey. But they have these two strategies called habit-based and plan-based action action selection. Basically, they've did these uh simulations uh where you could look at, you know, what was involved in this. And and I'll just tell you right now that habit-based is not predictive and plan-based is more predictive. And so habit is based on a model free approach. So habit is where behaviors generated by state action values are not related to immediate outcomes. They're not adaptive. They're just state action values. and there's no real prediction there. Um, and then planning is a model-based approach where behaviors are best based on action outcome knowledge structure. So, a knowledge structure that helps prediction. Future states and expected outcomes are modeled and this is very adaptive. So, you know, you know kind of what you're you what future states to expect. You know what outcomes to expect and it's much more predictive than the habit model which is kind of just output of behavior. And so these are interesting because we talked about in last year's talk by our group about the role of habit in planning in sort of agents and and thinking about uh cybernetics what they call goal-directed behavior. And so this is an important distinction that is also important in the spatial intelligence paradigm. Finally we want to talk about the view from embodiment in GSLAM. GSLAM stands for generalized simultaneous localization and mapping. So again we've talked about embodiment we haven't talked about GSL slam and so we'll talk about two paper GS slam papers one is by Adam Saffron and colleagues generalized simultaneous localization and mapping GS slam as unification framework for natural and artificial intelligences and then this other review on a visual slam for robotics evolution properties and future applications. So in these papers basically they think about um spatial processing as the sort of control system. You have a control mechanism U sub P discrete time steps T sub i then sensor observations O sub t . So you have these three components. Then it leads to an agent state X of T environmental map

M subt and that leads to spatial representation and behavior. So it kind of looks almost like a reinforcement learning model with input from previous states the environment and then it gives you the spatiotemporal representation. But of course there is this aspect of the classic Sydney Harris cartoon where he has he's very vague in step two and he says then a miracle occurs. And so we don't necessarily know what the filler is in this between agent state the environmental map and then defining those and then what what leads to spatial representation and spatiotemporal representation and behavior. It's a it's a hazy box. It maybe can see somewhat inside of it and I think things I've talked about in this talk at least from the biological standpoint can help. But we can also think about saffron's idea of latent slam. And this is a basis for map updating. So we're not totally lost here in slam. We can use these kind of latent spaces to help us simulate spatial behavior and then things beyond that. But uh latent slam is an evaluation between currently represented and novel experiences. There's what they call view pose matching based on place cell and grid cells. So they use the place cell grid cell model. So when current observations match information then loop closure in the representation happens and that's confirmation that that observation is in the representation in the weight in space. When no node can match existing information in the representation new nodes are inserted. That's this graph relaxation. So basically what happens is you have this latent space that grows with experience. And so you know you either confirm that you've been you you have this this experience of the wait and spaces or this structure of the weight in space is either confirmed or disisconfirmed with environmental information and then when it's disisconfirmed you add nodes to represent that. So next time when you encounter something similar you have that representation in place. And so a couple words about the loop closure algorithm. Um this is from introduction to loop closure detection in slam. Um this is where you have a form of error accumulation that occurs similar to path integration in animals. And so uh in path integration as well as for this form of error integration you count the steps along a path and when a landmark recurs the map is updated to reflect that information. And so if we state this kind of in a pathy way, we can say keep wandering until you find a loop closure. Wait a minute. I've been here before. So you wander down the beach and you keep going and you find a loop closure until you kind of go all the way down the beach and back to another point that you've been to before. So there are connections between this kind of weight in space approach and a cybernetic system. So latent slam attempts to recapitulate a neuroanatomically interesting weight in space model. Latent space allows us to compress a full state space where we don't need knowledge full knowledge of every state. In control theory, state space models

serve as a path of dynamical systems like we talked about with the Iranjanzadeh and all this about trajectories. In a cybernetic formulation, however, we can build upon latent slam by using a state space, a growing network and causal relations. So we can actually formulate a solution to this in the form of a cybernetic model and this is something I think we also talked about last year. We'll get to this in a minute but this is an example of graphical relaxation algorithms. So this improves your Eulerian path estimates for your network structure by checking if a shorter path can be found through alternative edges. In this case adding edges. So you explore this network. You see what the path lengths are for each traversal of the network and if um you know if it can be found great. If not it adds a node and you have this shorter path. But this is kind of the way you can do this in a cybernetic context. These are these complex referential or quasi causal relations for different control outcomes. This is from last year's um symposium where we proposed this uh cybernetic model of feedback loops that produce not just the past, present, future but also alternative futures and complex outputs associated with all of them. So you can model uh a feedback loop as sort of um going back to the future basically um as you have in a first order closed loop feedback context and then it produces a complex output with a lot of different causal uh consequences and and u with a lot of uh sort of uh causal implications but then you also have these alternative futures that can be generated from the same system. Another way to think about this in space is you have these Teletubbies that are, you know, there's one in the present, maybe they were here in the past, here in the present, and then you have these alternative futures, maybe a future state over here, and an alternative future state over here. There's a spatial location associated with each of them. This cybernetic model recalls Douglas Hoffstead's I am a strange loop where in chapter 4 he talks about loops, goals, and loopholes. And he talks about how self-applied operations can produce these circularities that are very interesting not only from a causal standpoint but from a spatial standpoint as well. And so we can also think about ergodicity and how that can also contribute to the sort of spatial learning. So ergodicity is this idea that if you visit enough locations in a space, you can produce a mean path that describes every location in that space. So the mean value is something that gives you full spatial knowledge of that that domain. And so this is a paper where they're using a robot to learn all the possible states of a movement path. The mean path is described by robotic systems. This mean path is optimal and this can actually help uh assist people with disabilities, people who can't move optimally in space. If you have a robot that can sort of explore all the spatial locations in maybe like a movement space like a workspace or in a room or something like that then you can assist people

when they make mistakes and help them cover up their mistakes um through knowing sort of the mean value should be at any location. So again, our our graphic here in our piffy statement, if you walk down the beach enough times, you can predict a path that matches the wap of the waves from a mean value. So you can actually track the path of the incoming tide and walk right along the water's edge if you've walked along this beach enough. This brings us to this idea of an ergotic ensemble, which is where if you sample a system enough times, any given ensemble is equivalent to the time average. That's sort of the formal definition. Learning increases information but an internal representation may provide erodicity. So an example of this is for native language speakers the language tends to be erotic versus English as a second language speakers where the language tends to be non-ergotic. And this is from a study where they looked at native speakers and English as a second language speakers. you have to build this strong internal representation characterized by sort of native language ability when described in basian versus statistical mechanics terms. So erodicity is usually described in terms of statistical mechanics. Individual cases are not equivalent to generalized maps. And so this is something interesting about this that we can use this kind of uh erotic example. We can use erodicity to describe spatial navigation but we also have to take individual experience into account. Sometimes um spatial navigation for one person is not erotic but for a population it is. So what have we learned? Uh the first thing is that a representational system that is anatomically explicit like such as the hippocampus and evolutionarily conserved so the hippocampus and vertebrates provides the hardware for quick efficient spatial learning. So the longer you have that structure around evolutionarily the more refined it can be the more erotic your you know if you acquire something in development then it's you know it's erotic and you have the hardware to carry that out. If you have to use sort of a an expedient neuro architecture for this to learn it or it's not something that is typically learned from development then you're not going to have as efficient a spatial learning. Number two, optic flow mediates spatial perception in terms of both the body egocentric and environment eocentric. And so we have these different frameworks for the body and for the environment. Optic flow mediates spatial perception in that way. Viewpoint representations or frames of reference also depend on sensory specializations established in development. So technically the sensory specializations are established in evolution, but you can have these things play themselves out in development and they influence the viewpoint representations across species and how the you know for different sensory systems for different uh neuroarchchitectures. They actually mediate spatial learning. Number three, path integration produces dynamical trajectory with complex parts. So

this is again path integration another form of spatial intelligence where you get this dynamical trajectory where you can decompose the parts of the trajectory and it tells you something about in you know uh periodic spatial intelligence or at least the behavior of spatial cognition and so path integration results from sensory and internal information as well as processes of regulation and prediction. Number four, a body can be characterized by a sensory manifold while sensory motor integration acts as an immediate source of spatial bias. So again, we have this the body plays a role and then sensory integration also plays a role. And so sometimes our body is getting visual information from one location and auditory information from another. Sometimes our bodies can act to bring those together, sometimes not. And so this is an immediate source of spatial bias. Action selection operates on the integration of immediate or longer term information. So we can have a simple model of immediate uh sensory information like in the habit case in the electric fish simulations or a longer term model of planning or sometimes we can use a world model like we talked about in the uh intelligence augmentation example where you have this model of the world that you draw from. So it's a longer term context. Number five, unified models such as GS slam serve as a framework for embodied information processing. This unifies the role of hippocampal path integration, circuit, optic flow, and the spatial bias to sensory reception. So, GSLM really unifies a lot of these mechanisms into one framework. Um, and then six, which is an additional thing, and this is kind of more of an insight. enough free or random navigation can produce a good enough cognitive map by relying on an ensemble of spatial information. So if we just let our simulation run forever, our agent explore forever, it can produce a good enough cognitive map, it doesn't necessarily need a lot of architecture, but the architecture deals with a lot of these other aspects of structure that random or free navigation may not solve. And so again with the neuro we're dealing with potential brain centers, connectivity mechanisms for spatial representations and cognition across animals. The simulate we're using boxes and arrows or cybernetic type systems, spatial distributions of sensory information, mathematical functions that describe behavioral output, simple biocybernetic agents, and models of sensory processing from the literature. So in future directions we need more complex models, more exploration of causality and systems context, expansive and reconfigurable networks and of course more synthesis of all these ideas. Thank you for your attention. All right, we're back with our next session. So, I will let Stephen introduce the speakers and thank you all for joining. Go for it. >> Okay. Thank you, Daniel. Thank you very much for having us here. Uh this is a wonderful opportunity because we all come from

different backgrounds but we're very much united in that active inference has a tremendous potential to help more people actionably in the relatively near term than we think is realized and we've been coming at it from different ways and learning from one another. So we appreciate this opportunity to share this u uh Jakamo I have to tell you before we started talking with this with this project your work on light touch and and contextual work um is something that I have quoted and that I believe has a tremendous relevance for my profession. I'm a chiropractor. You're a PT. Uh Jim, you're a body worker from a sematic point of view, massage, but you've taken Marco blankets in a whole new direction. And I'm looking forward to seeing your presentation on Friday. And Dave is our academic that's going to be leading us. Uh Dave Newuel is at UCCC. he's which is one of the premier chiropractic institutions globally and he's working with a new group of pain inference uh of active inference focused on pain and that's basically the name of the group and we look forward to Dave's work pulling us together. So with that Dave um take it away. >> Thanks Stephen. So um I'm going to attempt as far as I can to um chair this. We're going to try and make this relatively informal uh conversation. We have a few sort of kickoff questions that we're going to try to use to unpack each of our um guests here that Stevens has outlined in terms of their background, their interest, how they got to this particular point. Um um I I think Steven and I but I think the two of us did watch Carl's um plenary introduction. I must admit um you know listening to that I firmly put myself in the developing category when it came to um active inference and uh I've had some interest in this for for a couple of years and from particular background before that which we'll probably get into and all of us will um so but having said that um you know we've each come to this in our own particular way um and um and um And some of the questions that I ask and I suspect perhaps some of our um uh participants here will will ask as well will be somewhat rhetorical. Um and uh and so we'll we'll do our best to make it entertaining, perhaps informative and um and and see how it goes. So I'm going to really sort of kick off um with each of you really and and I guess I'm going to go I'm going to go with Jim first and go back that way through Jakamo and then Steven. Um and um you know that what was your journey? Tell us something about your journey in terms of being interested in active inference. Why it struck you as a framework that excited you? You know what's unique about this? Because it's pretty esoteric in I mean like it's not all over Facebook and the national newspapers. So, it's still, you know, relatively outside of those people that are really into this, you know, um a rather niche idea. Um so, so how did you get into it, Jim? And uh >> let me let me try to answer that. Um uh the So, first of all, uh Stephen, I I'm not sure. I just

spoke to Daniel online by email. I think that my presentation which is a recorded video um that I finally updated uploaded last night um in a fit in a rush to make the deadline. I think it's going to be available um at 22:30 UTC today, but um so I don't I don't know if I'm presenting on Friday, but it would be a video that's it's already recorded. Um and I go into everything in depth there, of course. It's about a 50-minute video. And um so my background is um I'm a I'm a uh my academic training is is historic history and social theory and and then my my profession or my love really is is touch. I just love touch and I'm one of those people and and and I'm a deep tissue structural body worker and I've worked with um the fascia with the my focus was in the deep core anatomy as I worked on people I came to understand that everyone is really dealing with what I call postural collapse syndrome uh which is a failure of the postural the core anatomy the postural system and so I've spent the last whatever 10 years really analyzing that and when I read Sharington so and I'm a geek so I'm reading all over the place and I'm trying to understand interoception at first which I got from somatic psychology and I thought oh everything is interoception then when I read Sharington um I understood that that that paradigm that that whole notion is completely wrong and it's proprioception and after that I connected with with um uh Carl Fris 's work on the topic and the work of the active inference community and the whole idea of the Marov blanket and a model of life. Now, Sharington was the one that pioneered that neural architecture, that model of life in 1906 was never developed until Carl Friston really took the bull by the horns. And um and I think that having a model of life is such a powerful thing, a perceptual model and it's a spatial model. So for for those of us into somatics, the spatial reasoning is really kind of the way we work. We're rather than a topheavy kind of intellect reasoning, it's more of a ecological kind of um openness to the ecological affordances that are provided and and those are deep within us. So So I got really excited. I just love the um I mean you know complexity theory the cybernetics and and um and as a historian my focus has always been on what I call the somatic tradition. Um two books were published in 1543. Copernicus was a mathematical one and that defined the mathematical tradition which has become ascended today of course quickly became ascendant um starting with Newton but the somatic tradition throughout history has been a vital one and that is asking a different set of questions than the mathematical tradition. about life and complexity and Vicilius um wrote his fabric the fabric of the human body also published in 1543 the same year as copernicus and that for me began that sematic tradition the only way to understand complex living structures is through complexity theory and I think that active inference is it's just definitely the most exciting and comprehensive and really profound

expression of complexity theory that I've come across. I'm a big fan. Um, that's kind of my background >> Brilliant. Now, that's really interesting. I've got a couple of things. We'll come back to that in a moment. Um, but we'll we'll move on to Jakamo, I think. >> Yes. Thank you. Yes, I'm Jakamo Rosetti and my background is from physiotherapy. I'm both a clinician and also a researchers. And during my journey of in research, I have had the possibility to investigate and learn some points from the contextual factors placebo and noibo effects. And through this journey, I have had the possibility to read some points regarding the active inference. And I find this proposing theory really interesting because it fits well with a real life clinical practice because it can help us to explain the unexplainable things that we encounter in the clinical scenario every day. So starting from placebo, nobo and contextual factors, we can conceive these elements as strategies that can have an impact in terms of active inference and can modulate the perception of patients and also have an impact in terms of therapeutic outcomes. But I think this really this this is a really interesting theory that we should develop and investigate well using a networking and different perspective from different fields. >> Brilliant. Thank you Jacko. >> Thanks Jacko. Um, Stephen, >> uh, I'm a doctor of chiropractic who practiced for 26 years. And in my somewhat schizophrenic profession, I played a number of roles, but uh, as an undergraduate in college, I was between psychology and biology practicing. I spent time in Prague with with Vladimir Yand if anyone's in that world uppercross lower cross syndromes was words that he created concepts that he created and also I was into yoga and in my practice we I realized that simply manipulation by itself was not fixing the problems once there was a motor pattern that occurred and so we came at it from a biomechanic and yoga point of view with a little bit of mindfulness I began lecturing and in 2003 and then in 2011 the course that I've been teaching was posture and motor control exercise integrated with chiropractic mobilization and manipulation. The book that caught me was a book by Bud Craig, how do you feel? A neurobiological moment with your intrceptive self. Bud Craig opened the door to introspection as a sense that was separate and distinct from proprioception and within a lot of the psychological research now the mind body research intraception has become a big topic and when I started looking at intraception then I ran into active inference and it's the same grounding because introspection is one of the inputs for active inference and it's something that is not being realized in in my mind in just about any clinical uh way. Jacum, I believe some of your work is one of the few that are touching on that that that linkage from a sematic point of view. And I believe that there's a tremendous amount of good that can come from creating agency in the part of people, giving them the ability to

have self-efficacy and to harness active inference as a framework to be able to systematically import that to people in different clinical situations. back pain to psych the psychological issues. >> Brilliant. Thank you, Stephen. Um, so me I guess um I sort of I've been teaching chiropractors for many years. I'm I'm now part of a wonderful university called the health sciences university in Bournemouth in the UK that teaches many health pray healthcare professions now um including the man all of the manual um professions um physiotherapy um osteopathy chiropractic we don't do massage as such but I have a PhD student who's uh who runs a school actually in the UK who who does massage um and I think over the years um I've become interested in the mechanisms behind how therapeutic encounters with manual therapists, you know, are are just doing the thing that they're doing. I mean, I think there is there's there's now, you know, plenty of evidence suggests that visiting a manual therapist has really positive outcomes, but I think one thing that's been missing over many years, very frustratingly, for scientists like me, is exactly what is going on. You know, people do all sorts of different things to patients. Uh whether it be manipulation, whether it be some form of touch, whether it be exercises, joint mobilization, all sorts of different things. And we know that um that things like, you know, exercise from exercise to talking therapies to manual therapies to a whole bunch of other things seem to work relatively similarly. They they work as well as each other in many ways. And uh so so I got really interested into why that was that was happening. You know, what are the mechanisms going on and and through that I I obviously came into the world of placebo and nobo like jakamo and um found that somewhat frustrating because there's so much historical ladeness that goes with the language of placebo and nobo. It's a sort of uphill battle to get clinicians to listen to you when you're talking about placebo and nobo in many ways. And um and out of that then um segued into contextual factors which are this these these things in therapeutic encounters that can essentially change the meaning of of what patients are are expecting to happen and um and really then um stumbled across I think I I I don't this is right remember there was a paper by Angara on capt who talked about predictive processing as being a part, you know, potential mechanism behind placebo. And I think from that sort of just fell into the active inference framework um community and uh you know was so enamored by the breadth of that framework, its parimoniousness, the potential u width of things that it could start to explain. you know, notwithstanding that it actually had a you know, it it told a story of a mechanism of life, let alone clinical interactions and um and uh I was so enamored. We we just set up a center of pain and active inference research at the university. So that's how I am now. That's how

committed I am now to this framework. So um so yeah very similar stuff and um so one thing I would say would be and I think this again it's one of these rhetorical questions you know why do clinicians in your your viewpoint need this rather esoteric new um uh 45,000 ft view of of life framework you know because I I'm sure in all of your professions you know they they will say Well, you know, this is all very well and good, very interesting, but you know, you know, my patients are getting better. What What would you say to that? Anybody? There's a book by Simmonson, begin with why. I believe that active inference is an underlying why for a lot of different things that has not been effectively articulated nor communicated. I would say very much that the effort to to talk about it from Carl Friston's point of view from the presentation today that are things that I don't know a lot about personally but there's congruences and I think the more that people in different professions can look at those congruences and pull those truths together addresses some of the issues that I believe are coming for the active inference community which is trying to spread wider And so having a common why for different schools of thought like we have here within one trench manual therapies and making those uh congruent with other schools of thought to become more of an accepted truth is this is the way life is I believe that's because that that's the underlying u to me fascination and promise of of active inference ultimately. >> Yeah. So a unifying mechanism really that's I would agree with you that's tend to be what I've found actually segueing out from that or going on to you know using that as a platform then can can each of you and I'll start with Jacma can can can each of you sort of then tell us a little bit about how it's informing your own work particularly because I think you all practice still um how it's informing that on the ground practice practice focus people in front of you real world stuff um and uh and you know uh because I think that's really important it's so easy to talk about the sort of high level stuff that we're we're talking about and listen to people talk about that you know even not not that I can do this but um Stephen and I were talking about this even the sort of all the maths behind it which completely discombobulates me but the you know what what is going on on the ground so so Jack give us give us an idea how it's how it's in you know what's happening in your practice. >> Yeah, I think that active inference helped me but also all the other clinicians for many reason. First of all, it can symbolize an over an overarching framework that help us to explain why different treatment strategies can help and have an impact in terms of therapeutic outcome in our patients. independently from which kind of treatment we decide to apply, but it works as an general framework that help us to understand what happens in the clinical encounter. And as a second reason is that help us to explain why sometimes the

different treatment solution that we apply with our patient doesn't work because there is a great complexity in the real life that is somebody unpredictable. For this reason we need a framework that help us to explain what happens in the clinical practice because you know when we treat the real patients that sometimes we are challenged because the patient has a very strong prior strong belief that are really difficult to be changed. So in this using this model we can for example use the different element presented in the therapeutic arena name as a contextual factors in order to modulate the perception of patient and help them to challenge their prior using our worth their rituality our techniques our verbal and nonverbal communication and all the other cues presented in therapotic encounter. s that can be combined with a specificity of our techniques they can act as a boost that have an impact for our patients perception for for so in this reason and for this reason I think that the active inference can really help the clinician to understand what they are doing and also understand the patient response in the clinical practice and for that reason there is a strong need of more research and more awareness among clinicians regarding this topic. >> Yeah, that's great. I'm gonna go to Jim next. >> Hi. Um yeah, well, you know, um the title of of my my my video, my presentation is exerceptive bias and um somatic belief updating in the global model. And so I'm going to um >> unpack that for us, Jim. >> Say that again. >> Unpack a little bit of that for us. >> Oh, I'm going to exerceptive bias then is an innate natural setting in the system. If we think about the receptors on the outside of the of the of the organism, right? We have distance receptors, touch receptors, taste, smell. These are distance receptors. And we um the part of the brain that's associated with exerception is also the surface, the cortex of the brain. And interoception then would be the viscera and then that would be the insula. Bud Craig's famous um insight there is the insula. and but but for propriioception it's going to be at the at the root here in the cerebellum and the spinal cord. Now so exerceptive bias is just a situation in which the organism is running its outer sensors and you as soon as something is born into the world it has to immediately look for for warmth for shelter for kin for food for safety and that never stops. So there's a certain innate bias in the system. There are an awful lot more inputs coming from the outside than from deep within for sure. But exerceptive bias then can become a control bias or a sort of a a vicious cycle, a positive feedback loop if we're repressing the inner sensibilities which um it's the postural system. So it's all balance, it's all integration, it's all stabilization, centering and grounding. These are the chinesiological functions of the system. But they're also the cognitive functions. And without those then we can jump into you know basically we're just thinking too much and we're stuck in our head. So

when I sit down, when you sit about my work, right? Like for me, a classic example of my work is like I'm over at the cafe and someone sits down and and you know they're talking about their pain or their issues and everything and it's like let's get out of our heads here and I say come on over here and sit down and let me work your arm or your hand or let me rub your shoulders or and I do this in their coffee shop and um of course you know with consent but um you know getting out of our heads is sort of a huge huge important theme and as manual therapists that's our specialty is coming into the body and as a manual therapist I see my work both theoretically and also in a tactile sense is as a somatic educator helping people to understand that interior perceptual system and come into their bodies and have that grounding and centering that's available to them when they need it or when they want it. Um you may not want to be doing your grounding and centering if you're in heavy traffic, but you definitely don't want to be um dysregulated internally either. So there's it's that that underlying system is always running. But what I think we're seeing in the world, the surprise that we're seeing in the world is overreach. We can call it technological overreach or the anthropocene. and that the society that we've built is really very too topheavy, too intellectual as organisms in the world, we need more balance. And so that's kind of my focus um as a somatic educator as a Yeah. >> D can I say something? >> Yeah. Go on statement. >> Yeah. Yeah. Um, a point that I believe to making active inference clinically actionable in the field on the ground with clinicians is making it something that is what you said before Jakamo a boost to whatever else we're doing. And I would even amplify that by making it a synergy because it's not just I think 20% better. I think it's more like potentially 200 that's percent better. I it can make a dramatic dramatic difference because it cannot just be in the moment of the uh therapeutic encounter whether you're doing a manipulation or I am however we both do it. It's connecting with them as a person that and Jim I think this comes back to what you were talking about. Um yeah >> but a point that I think needs to be made that ironically enough Carl first said said earlier what he said was how we see the world and how we react to the world is what he was talking about but uh um active inference being and when he was talking about seeing I don't think he was talking about visual visual I think he was talking about the gestalt of the self of the person and my add to that my my riff on that would be how he seatically and percept conceptually feel the world and how we react to the world and to that in terms of making it actionable something that is a very much an intrceptive insight is that it needs to be moment by moment it can't be a cognitive thing where now you know about it so so you do it the interaction needs to be now and now and now very much correlating the the contextual communication which David I I

like I I teach things using something that that would describe But I never I've not used that word in in how I lecture. But that's exactly what makes sense to do because contextual means in this context. This is what I'm doing with you. And the synergy between feeling that on a sematic basis both proreceptively and intrareceptively and they're connected but they're not the same. Um has a much greater um salience from a sensory point of view and thus relevance in terms of shifting behaviors. Yeah, that's fascinating because in a way um and I think we've all come to this in different in different angles. You know, I think um one of the things that has frustrated me with with with the profession that I've been most associated with is chiropractic, but I know it's also something in osteopathy. uh I don't know much about bodywork and massage but I I know that there are similar conversations going on and and I think there is something similar perhaps in terms of physiootherapy as well and and what I guess what I've seen through the lens or the telescope even looking at the wrong end of the telescope that that I've seen is um too much um uh body centric explanatory frameworks. Now what's really interesting about this discussion we've just had here is and and I would say that that Jakamo and I are probably you know if you're going to look at a spectrum sort of a cartisian spectrum between mind and body with mind at one end and body at the other um I guess that Jakamo and I have become very interested in the mind stuff and I know that you know we've got some colleagues George Estz and Mike Mikeer Mthaka that will um that will very much be Steven and Jim's end of the spectrum which is which is very bodybased. Um but the there's something about the bodybasedness of certainly the manipulative professions perhaps where um you know there's almost a complete um not a complete but you know a tendency to sort of not worry so much about what people are thinking and I think that's that's why we've headed in well certainly why I've headed in that direction. On the other hand, there's something about a body focus um in a different way to finding a lesion. I think the old traditional explanatory frameworks are around, but you find a lesion. It's very biomedical. Find the lesion, you fix it, you get rid of it. You know, if you're if you're a surgeon, you take it out. If you're a manual therapist, you fix the lesion in some sort of way. But that's not the body nest that you and Steven are talking about. You're you're actually talking about a very different bodiness. But would you say that all of us are actually trying to find some position in the middle where we feel that we can get these professions that get these clinicians to just be more balanced between these two things to get real holism in a way. I used to hate back back in the 70s and 80s the word holistic came into vogue and I despised it because it quickly became a mantra for selling all sorts of snake oils and gadgets and gizmos. I disagree with the question. It's not

mind or body. It's not it's not even mind body or body mind. It's one gestalt. It's one thing. I don't care how much you're focused on and your your mind. If I drop a brick on your toe, your focus is going to change quickly. And that's part of being human in a moment-by-moment basis. So from clinicians using that as leverage and making what we're doing with them something that is giving them agency that they can understand in their in their language that correlates moment by moment and leaving them with a psychological feeling of success of a positive a positive response. So yeah I would agree with agree with that. >> Yeah Jim you had something to say and then Jack thanks um thank you. Um I I wanted to say that um I think that probably for me at least the singular contribution of active inference which is also um I point out in my talk and I go into it very extensively uh Sharington's original um his probably his greatest achievement is his the model of life the model a neurological he calls it the neural architecture of the animal as a whole the perceptual model of life that is scalable and it's scalable from from abstract systems like the Marov blanket to the very intuitive example of the cell to the vertebrae body and and um and other kinds of systems but but it certainly makes Hi Daniel. >> Yeah, it's >> Daniel mute your mic. >> Um, so um the model of life I think is is a way to look at things. Uh so it's three perceptual systems with three different fields in the body. The outer surface, the inner circus, internal reception and then the core proprioception. all of which are minded and embodied at the same time. And this is just a different way of seeing things. I think it's probably the single um most exciting and powerful insight. Um in addition to the notion of active inference itself, a kind of a active inference then takes the mind and the body and brings them together. It's active and inferential at the same time. You can't break apart the sensory motor loop, the perception action cycle. And this is a real shift in thinking so that that when we touch, we're touching a person. We're we're touching a person in different ways at different levels with with boundaries set up within those levels so that I can be touching you more superficially or touching you more deeply. I can be touched more superficially and be touched more deeply. I think it's an exciting massive I think I think historically speaking it's a civilization level paradigm shift. >> Yeah, I agree. I do agree. Yeah. Yeah. No, thanks Jim Jack. >> Yeah, Dave, I totally agree with you regarding the challenge between body and mind theory because we should also and always consider that the things are different if you you are living in the lab or if you are living in the clinic. So if you are living in the lab, you can ask about something related to the body or to the mind. But if you are living in the real life clinical practice and you interact every day with a patient, you know well that your life is composed by moment by

moment interaction where you apply some tools that are related to your clinical background and expertise such as manual therapy exercises, physical therapy agents and concomitantly you modify the context using different types of touch for example modulating different types of verbal communication and nonverbal communication in order to synchronize your body and your person with your patients in front of you. So in this if you consider this real life situation we should stop to oscillate like a pendulum from one extremity to the another but we should try to be balanced in order to strength both uh the positive elements of the different theories and in this sense I think that active inference help us to have a general framework that can help us to explain what really happens in the clinical practice. So I think that if you are living in the clinical practice you encounter different challenge every day and the models the framework of active inference can help you to um give an interpretation of meaning of what the patient uh present in terms of pain dysfunctions beliefs prior and try to challenge them. >> No that's that's great. I um one of the things that I I thought about is you know that there is a sort of rhetoric that goes with um you know an integrative approach to whether we call it holism whether we call it integrated medicine or integrated approaches and it feels to me that you know um like you all of you have said in different ways that active and I agree with Jim I think it's a paradigm shifting idea um and of course it's got a long history it's not you know as as much as Carl has been the sort of contemporary uh horseman that's take this forward, it's obviously come come all the way you could say it goes all the way back to Plato's cave and and and you know for sure. >> Yeah, exactly. But but the um so it's interesting what you were saying Jim and I think the cuz you said everything is minded. I like that phrase. And um it does that speak to this sort of hierarchical um models at all hierarchies? I know that um Mick our colleague um and uh I don't know what Carl and the guys would think about this but I know that there are other authors that have suggested that you know their everything or from cells upwards have a Markoff blanket and they also have a model of themselves and they act you know within their own environment to confirm their own existence by by by generating that model and acting in such a way as to keep that generative model going. So, you know, it sort of gives the idea that there is a sort of mind or a model at at every level. And I guess what we're doing we I guess we've got somewhat interested in different levels of the hierarchy of those models perhaps, you know, we could say, you know, um we're interested in prior because we're we're interested in sort of high hierarchical models of meaning and culture and what this really means and but yet there are all these models all the way down. I think you actually talked about the core being a being a sort of model of its own that had its own markoff

blanket and so on and maybe if yeah maybe talk to that and I know Stephen has got some similar ideas as well so you know pitch in. >> So I um I love that uh thank you for that introduction. Um the thing that I played around with um in my talk is the idea of of a double blanket theory. Now we know that there are two blankets. A model has to have an internal model that you know there are two blankets in the model. But the inner blanket tends to be um subsumed by by by just the calculations of of of its parents and and and um and siblings and stuff. And and I don't know much about statistics. I think it's fascinating, but I'm a historian. Um so what I the way I play around with it is I I call it blanket politics and the inner self has now interoception is a little different. It's a little more amorphous but the proprioceptive it's a very concrete sense selfidentity. It has very concrete intentions and goals and a concrete view of life that's different from the exteroceptive the outer blanket. The inner blanket and the outer blanket of course are integrated profoundly integrated in terms of of proprioceptive prior and exteroceptive you know sort of um maybe posteriors or actions there there's no distinction but at a certain level um it's possible for there to be blanket politics and I think we can talk about that in terms of Friston's dis uh connection hypothesis but also I like to think about it as a cultural theorist in terms of um the double bind, Gregory Bateson's double bind and from there starting to really think about things like like psychopathology and schizophrenia. But the point being that we really have different selves and there are different agendas involved. And if we're really too maybe too internal, but that's rarely the case, far too external, then that can set up a kind of a um a situation uh like the double bind that I call blanket policies. I want to say one thing about the relationship of the mind and the body and just throw this out there. For me, the relationship of the mind and the body is like a rider and the horse, except that you're both the rider and the horse. And the horse knows the rider. The horse knows where to go. 2:00 in the afternoon, we're going up the mountain trail to the spring. 11 in the morning, he we're going downtown to go shopping. And if that rider has a heart attack or falls unconscious, the horse is going to turn around and head right back to the paddocks um to get fed. and and and the horse. So, the mind knows the body, the body knows the mind. As manual therapists, we're really not fixing people. Of course, we are fixing people. I fix people. But if we want to really get um get into the semantics, the body heals itself and the collective body, the collective society also heals itself. The point is how to get out of top- down control and start to really post. you know, when you post on a horse, you get into the rhythm of that canter and the whole thing works smoothly. How do we really find that um that connection with our patients >> and enable and help and assist the body to heal it?

Sometimes there has to be an excision, a surgical excision, but in general, what we're trying to do is really facilitate the self-healing processes of the organism. Yeah, that I what comes to mind is before I go on to Stephen and then Jakob as well. Yeah. >> Um that really what we're saying in is in dysfunction of whatever that may be. Whether it results in pain, whether it results in um anger, whether it results in um um anxiety, you know, whatever that that there are lots of situations where the body gets in the way of the brain and the brain gets in the way of the body and and in a sense, you know, that what again it goes back to is there some sort of way where, you know, using your analogy where the horse and the rider um notwithstanding that they are one, but they actually start to function as one. And then there's a and the dysfunction is when they're not functioning as one and they're sort of getting in the way of each other. Anyway, that that's a great idea. So, Stephen, >> yeah, humans are unique in terms of the hierarchies of complexity that we've been talking about, which one of the things that attracted me to complexity theory because in the lab you learn about lower animals, ants, there's earlier today and the truths of those things up to the mammals are truths for life. But humans have a number of uniquenesses both on a biomechanical as well as a cognitive uh uh framework that both interact to connect mind and body. We're the first creature on earth to ask what is it mind or body. No other creature has posed a question. with the first creature on on earth to have to be an oblate biped that is a mammal. Birds are oblate bipeds but they fly. Our superpower is cognition and thinking about things and being able to have choice. So in the rider or the horse, the rider is the only one that can say you know what um I just remembered something. Someone asked me to do something. I forgot about that. Let me go back and and and change directions where the horse will keep on following that pattern. And for most of our lives, we keep on following the path that we followed before. And that's one of the challenges with any kind of rehab jakamo with with patient compliance that we all know about. And that's why from an active inference point of view, getting to people in the moment is where there's so much leverage. What everything that that you said is make is being present with them and relevant to them which means addressing what they're perceiving with what you're doing and having that affect what they're perceiving and building a virtuous spiral to take this to a whole different level. And when you look at it from even another point from a overall wellness besides a pain point of view, all the quadrupeds their their locomotion and their diaphragmatic function are linked. And when they're breathing and and living, they are breathing in the same sink. Humans are different. We've disconnected our breathing. Moreover, we're the only creature that can intentionally I'm going to stop breathing and hold my breath. Other animals

don't have that as a volitional and we've used that ability to control and then modulate to develop language and language has become everything else that we've done from a cognitive point of view. So I think that beginning with the fact that we're getting vertical and vertical but the first leg that we talk about is postural balance. It's not just posture. If I'm standing here like this, what looks crooked may appear crooked, but an overall biomechanical level, it is physically balancing from a biomechanics point of view and from a headspace point of view. Your posture is how you balance both your body and your mind. >> Yeah. No, that's great. Jakamo, did you have anything to add to that segue outwards, which is entirely my fault. I do apologize. >> No, no problem. I was thinking about a previous conversation regarding the rule of touch and as a clinician I'm thinking that for example when you decide to apply a therapeutic amotherapy techniques for example and you are informed by evidence that you can modulate short and medium terms the therapiotic outcomes you have a tools but active inference help you to explain how to apply it if you want to have an impact in term of patient interceptions for example When you decide to apply a manual therapy techniques and you use for example a very hard pressures, if you are not in a moment synchronized with your patient, you and you are not aware, you are not in a listening phase, you can adapt, you cannot adapt the touch. So it's a a very it's a fundamental step because you have a powerful tool but if you if you apply it in a painful way to your patient there is a risk to have a worse outcomes. >> So you need to modulate the touch in order to get the positive outcomes and you can do it only if you are synchronized with your patients. >> So this is a dance. >> Yeah it's a dance. It's fundamental. If you want to change and try to challenge the prior of the patients, you be at first be synchronized with your patient in connections. So would you say >> yeah go on go on go. >> Yeah that's great. So would you say I mean the question a rhetorical one um you know how do we stop some hardedged um scientific u manual therapists have been doing this for years. How do we stop what we're talking about just sounding like new age woo that that is one of the problems here obviously we don't think like that but I mean in a way our talk about dancing about holism about you know horses and riders and whatever do you think that it is that is the role in a way of this framework that the active inference framework is is something that is entirely >> scientific it's we'll talk about empirical evidence in a moment when we finish but But that that stops it, you know, um feeling as if we're sort of we we're being very woolly because actually isn't woolly whatsoever, Jim. >> Yeah. I I want to say that it is woo woo. >> It is woo woo. >> Yes. And that's very important. And so there's two fields in the model. There is the surface field and then there's the deep field. Now the surface field has

language, volition, intention, communication and it has a kind of consciousness that we associate this is who we are. um we identify that as reality and the it's the evidence that it produces is other evidencing and but there's another field and that field is haptic and that awareness is a haptic awareness and haptic awareness in the deep core doesn't have access to all of these different um expressive mechanisms that exteroception does. It doesn't have distance receptors. that doesn't have language. It's a pure haptic or um uh a touch sense that's generated. So that's very important because the body can go over here a little bit too far, but that system is going to recalibrate and reorient and establish and reestablish stability and balance. It's a very important function, but it also has a form of consciousness and awareness. We look down on it because it's not verbal, visual, and symbolic, but it is a form of consciousness. And so when you walk into a yoga studio, you know, the first thing you establish is epistemic trust, uh, uh, a meditation, >> we would call that therapeutic alliance, wouldn't we, if you were in some other clinical setting, >> perhaps? but going just a little bit deeper into that system and working in that quiet field, that deep field. And um there really is a kind of a um the semantics are different in that language. And I think one of the things that active inference is doing is trying to create a semantics or a sort of a grammar and a vocabulary which you can hear in Carl Friston and others talking that we ourselves are starting to practice using that really expresses that that innate active inferential >> um potential that comes from the deep field which is woo woo. And we need to be able to affirm that for what it is. It's the deep field. It isn't the exteroceptive field. But but there needs to be kind of um balance and integration uh between them. I think that is a lot of evidencing work that needs to be done for that. >> Yeah. No, that's great. any before we move on to something because I think we we will wrap up and I just wanted to wrap up with >> um I think we got an hour so and I know that Jack Jakamo's got to go so I just wanted to to wrap up around that and what do you think are the barriers that that get in the way of active potentially might get in the way of active inference in our fields be you know gaining traction or or perhaps not just our fields but in in the scientific community generally particularly around anxieties around limited empirical support for example and um you know accusations of non-falsifiability I know Jim has got something to say about this because we talked earlier on and I brought this up but I mean just let's just go can we start with Steven what what do you think do you think there are there are a few hills to climb here in terms of that that scientific support >> I think that we have more than the seven hills of of of of Rome here. Uh there's many different ways of climbing it. U one of one of one of my efforts that to correct you, I I I no longer practice. I've not practiced

since 2008 pretty much. Uh but I've lectured and taught and that's given me a viewpoint of the world and the professions. Um I um I spoke this morning about with um Tampa General Hospital where we spoken to PTS and OT's over there and I think Jakamo a lot of the things you talk you've mentioned before P OT's are doing as much in the moment by moment more than PTs more than chiropractors now in a lot of the daily practice of what they're doing. uh to to answer how it would look differently. I'm speaking in two and a half hours um about the paper that we did in uh that was published in frontiers of neuroscience intrceptive posture pictures and taking a picture of someone's body because posture is important not pathologizing it not making it a disease that I'm going to fix cure but observing their perception internally of what it feels like to stand tall and capturing that in a systematic way that you can do it with an objective to be able to observe what the body's because I think that there's going to be different techniques. I pref on a personal point of view I may prefer more active oasis technique and or light massage or jakamo you might prefer light m the other way around and there's ultimate variations within all of these sematics. Uh I I don't like wearing wear wearing turtlenecks. It doesn't feel good to me. You don't like wear it's not a matter of what's right or wrong. It's a matter of what works for that individual. And your point I totally agree with. We need to connect with that person for their body, for their mind, for their priors. >> Yeah. >> And moment by moment help to reshape their perceptions towards their actions and behaviors because otherwise it's academic which is fun but it's not real and that's the challenge. >> Yeah. So, so there are many there are many doors into the system at many levels and and and you have to figure out the particular door that you're being faced with. So um yeah uh Jack and then we'll end with end with Jim if that's okay. Anything to say about your worries about the uh empirical support around this framework you know will it will it just fade away and become like you know the the ether hypothesis you know back when and then discover that it doesn't exist. I'm sure it won't. What do you think? I think there are different issues and challenges up to now. For example, that we don't have too much paper produced convol involving patients with muscular skial pain or chronic dysfunctions and this is an issues that we hope that evidence will cover in the futures. Another issues is related to the lack of knowledge of clinicians and researchers. If you don't know it, how can you use it? How can you investigate it? And the third possible issues is the bias of the try to defend themsel using the previous frameworks and asking themselves why should I update my way of thinking, way of way of acting. But we know that uh science is growing and networking share different perspective and try to see the the things from a different uh point of view. So I think that the

model is really interesting but now what we should do is to move a step forward inform evidence producing literature and also inform clinicians with some tools and also some suggestions in order to help them to apply this and integrate the model in their clinical practice. >> We have a great opportunity and responsibility but we should move it forward. Yeah. >> Yeah. Yeah. No, agree. And I think I think all of us in our own way are trying to do that. Jim, I'm gonna give the sort of semi last word for you and and then we'll we'll finish up. I mean I think um I think in terms of active the active influence community so long as it remains heavily exteceptively biased it's going to continue to to be kind of insular and and and um there there are going to be major gaps in in in the theorization. I think that um the impact that active inference ha can have on the clinical environment, the clinical experience is not different than the impact that complexity theory is having the biocschosocial model and also the structural model of the body. This is a major revolution. This together with the active inference the a civilization level paradigm shift is very multifaceted and there is a cultural expression of it as well in movement movement but the um understanding of the body as an integrated whole I think the most important single discovery in anatomy is the rediscovery of the fascial matrix and understanding how isolated muscularkeeletal disorders are part of a a of a global model, a a structural uh system. And this is something basic to what structural body workers do. And the book behind me is um a really great source of this kind of new synergistic or I would even say cybernetic anatomy. That's just a new revolution that hasn't been developed. Just like active inference itself, much of the terrain has not been mapped experientially. And so I think that we're really um we really are um at the very beginning of something that's quite exciting. >> Yeah. No, I I would agree. Um Daniel has been very kind and just ask us one question. I I I know Jakamo has to go. Um well, maybe I could just ask this question to you very quickly, Jakamo. Um, and I think we've sort of answered it to some extent, but if you can think of anything else, but he says, uh, you know, how would the encounter setting look different? Um, in what ways would this be technical, algorithmic, educational? Certainly, education interests both of us, Jack, I think that would be something we would say would be would be important. But, um, any any insights just very quickly into that question? I think that active inference help us to apply what we just know. For example, in terms of contextual factors, we know that we should use verbal and nonverbal communication and the use for example of therapeutic touch. But using this new framework, we can decide how to apply it. But this strategy arise from a step-to-step evaluation, one moment to one moment evaluation. So I'm interacting with the patients. If I notice a particular anxiety, I try to

modulate it using a verbal communication that is not a rigid way of interact but it can change and can be modified and modulated on the base of patients response. This is an example. >> Okay. So, so perhaps one way to to to sum that up is that although clinicians are already dancing with patients to some extent that they they could dance better um and uh and you know perhaps be as responsive to the person standing in front of them as as as how they feel they should be moving their uh their feet as well in this thing. May maybe you know if you can think of the most complicated dance to do maybe that's what clinicians could do better. Jim, one last thing, Jim, and then we'll have to go. A >> terribly quick little point. I'm just looking at the little chat. Um, in what ways would this be this change of the clinical environment be technical, algorithmic, educational, etc. I want to add the word somatic, >> right? >> And I think the somatics is really the um the the the area for pioneering right now. Um, and in all that that means on a very profound level. >> Brilliant. Fantastic. Daniel, are you there? >> Yep. >> So, I think I think we're done. Um, that that was real fun. Thank you so much for that question. I think we've been uh we've enjoyed it. I think I think uh this sort of stays up on YouTube anyway, doesn't it? So, maybe people will still have a chance to watch this um beyond this this live session. Is that correct? >> Absolutely. Same URL. >> Thank you so much. >> Thank you so much for the opportunity. Right. >> Yeah, really fantastic opportunity and and thank you for our um uh our somewhat holistic way of approaching this and uh um uh and uh yeah, I think we all really enjoyed that. So hopefully people will enjoy that in the future. Thanks so much, Daniel. >> Thank you. Bye. >> Thank you, Daniel. >> All right. Bye. All right, next up is going to be a pre-recorded video by Sonia De Jagger and the title going to be described in the video. So, 20-minute video by Sonia. Here we go. Hello. The title of this presentation is active self-referencing. The concept of self-reference might be that which joins complex self-modeling with reference problems in formal languages across a spectrum. And before beginning, I want to thank Chana Dobson for the conversations that we had leading up to this. Unfortunately, she cannot join this presentation. That was the original plan. So what I will be showing are a series of speculative proposals uh slightly all over the place but uh hopefully interesting um and in keeping in line with the theme of the presentation. There will be no referencing to other authors and sources. Um these are tacit in the background and hopefully if you are interested enough and you would like to know more about this you can reach out to me um and perhaps later on to myself and Shana as well. uh we will try to write this paper for which she was going to cover uh some of the mathematical aspects that we were interesting interested sorry in exploring. Having said that let's move on to

the presentation. So in the context of thinking ourselves as predictive actively inferential organic creatures, we can understand reference as shared models encoding possibilities for action. Anything that gives us any kind of information is capable of orienting ourselves uh in a specific direction. So all linguistic, logical etc. reference are inherently ambiguous. We need to actively infer. reference require context for concretization or else they remain unactualized plans, imaginations, memories, whatever we want to call them depending on the context. And this becomes very important when we're going to well if we're going to think about self-reference because in a lot of the situations during which we can contemplate self-reference there is a sort of suspension of um action as we will as we will see. So simple reference with relatively easily concretizable possibility spaces such as let's say the reference cat are I would dare to say obviously less computationally challenging than complex ones such as negation for example or the concept of reference itself. Here we can think about situations that we have um decided to nominate as paradoxes, contradictions. Um and in some contexts I have really stretched uh the boundaries of this sort of conceptual space and and um included all sorts of um problematic toological statements and and situations during which things have to somehow return to a let's call it a set point for now. Um all of these phenomena are cognitively challenging reference moments during which bifurcations during which information can be created during which something can be contemplated and possibilities for action become possible. Um these are indeed moments during which we are challenged by the problem of reference because as things uh need to actualize themselves when there is no immediately apparent possibility for action things get rather tricky. Of course, we can think about this rather well-known situation where we have a statement which refers to itself and in many different ways but also in one simple way contradicts itself in what it is stating. Um, its actualization is rather awkward or its actualization isn't. We could say so here we have this pointing to the sentence this sentence what we are pointing at what we are referring to is what follows after this. So this when we're looking at this this right now this is what is true now as we move forward because we are inclined to do so reading is something that kind of happens before our very eyes once we are able to read we cannot but not read almost uh this points to the sentence this is true now this sentence is true now or sentence is true now and Now this is true the fact that it is a lie. So what just happened actually in many ways did not actually happen. It did not get actualized but it is actualized before our very eyes. So what is happening here? We can also just simplify this and we could go down many routes. Um but we could just simply remove the sentence bit of course and just understand that this is something that says

that it is not this is a lie and that is a problem. One of the problems of ambiguity and uh sort of the actualization of possibilities and statements is that this goes all over the place. Of course, different types of contexts will lead us in different directions depending on what it is that we are depending on the context in which we find ourselves. So obviously all reference as mentioned earlier are ambiguous. Everything requires context for disambiguation. And in the space of sort of in the metacognitive space of thinking rather abstract future possibilities, um these things of actualization versus nonactualization become confusing for the actively inferential agent again. So this situation is contradictory by being what it is because the future as we know affords possibilities which we can sometimes calculate as probabilities sometimes. So what is remarkable in an active inference context is this predictive challenge right that we encounter when we enter these self-referential loops when reference kind of models reference itself when we have to say something about the very sort of fulcrum the very the very substrate which is enabling um the capacity to witness possibility. So reference and this kind of sort of metacognitive capacity to self-reference or of self-reference is an incredibly powerful device for witnessing and or creating new information. This is the discovery versus invention challenge which if we are given to entropy as we know it is or can be conceptually collapsed as one phenomenon which is the inevitable recombination that we are given to of systems stacking upon systems um which we can only understand in terms of uh how they orient themselves towards something. That is how they you know uh minimize uncertainty for themselves and remain being what they are through these aforementioned reference processes as a minimal condition. So what predictive function does self-reference serve? What are we what are we arriving at when we are able to witness this? What is it that that that we are doing when we engage in the witnessing of of reference as as that which enables the very witnessing, right? Is it a foundational feature of selfmodeling or is it a side effect that we kind of inevitably developed because of the capacity to to actively infer to sort of be in a situation and need to experience moving on to a new situation. So again well as we were thinking about it this this would amount to the proposal that um kind of basic self-reference is a prerequisite for selfhood where selfhood emerges gradually evolutionarily uh developmentally through the self-s surveilling and synthetizing of one's own information processing. And this is what becomes confusing or contradictory in formal contexts again in sort of purely witnessing contexts that do not yet actualize. For example, when arithmetic says something about itself because being the creatures that we are, we did not always have reference. Uh it could be could be speculated because well in general we are used to self-

reference being an information creating endeavor. We inevitably actively infer ourselves towards something always there is always orientation. Um whether we are highly aware of it and have very um concrete mapping of what it is that we wish to follow or whether uh we are simply let's say uh inevitably given to whatever actualizes itself before our very eyes for us because we continue to exist. So when something is incapable of stabilizing the informatic result that we usually expect from referential exchanges, no inferring is possible. No action is possible. But what can also be said under speculative investigation here is therefore the concept of self-reference as that which enables entity uniqueness because it creates information and in general given to language as we are most linguistic um or communicative phenomena in general are representable as as reference within shared generative models. There is always the expectation whether um I mean there are many ways in which this could be said but because language exists between creatures and we expect that meanings are you know relatively shared u reference often point to more or less um the same thing. The actualization of things even if we cannot parse it ourselves is happening elsewhere. That is always the expectation. That is what language is sort of doing for us. But reference itself. So the substrate that enables this linguistic modeling cannot be modeled as a reference within its own system of course and this is much like the self which remains computationally intransparent to itself. A thing cannot fully account for its entire structure. That's uh quite a simple and at the same time incredibly complex thing to realize the loop that is then created. What is what is this this loop of self-reference and sort of something being able to maintain itself as a thing that has to check itself um for existence permanently pretty much. This is sustained by agents first modeling themselves as agents of course to effectively act but this is always incomplete and and um well complex in many ways as as has been mentioned. So framing it as such self-reference can therefore be interpreted as the process of representation of thisness or individuation hexity a thing becoming aware of its capacity to act. So act uniquely creating information each time. It's sort of the the what what we can hang on to, what we can um understand as remaining the same in the face of um eternal unpredictability unfolding before our very eyes. We need to hang on to something and that something is the self of course and that is sustained by the self being able to check itself for existence. Um that's sort of self-referential thing that that anything that is that we can call conscious uh and in many cases and capable also of metacognition that is that is what we are doing and when that function is not performed because we enter uh other hierarchies of stacking reference upon reference upon reference. uh we enter loops where it doesn't seem as if we can create or we can witness novel

information. So the sentence is a lie for example that is either we can contemplate that as something that is creating a lot of new information because it's creating this sort of back and forth paradox where we need to go um back to the beginning to the end back to the beginning to the end uh in order to understand what it is or try to understand what it is that that thing is doing. Um but there is a kind of symmetry that doesn't work for us for predictive purposes there. So um we we kind of witness these formal tensions and there is no active information giving resolution um in this sort of self-referential self proposal that we usually witness as being capable of giving noble information because we are given to uncertainty. So maintaining this recursive process um sustains the experience of continuity through memory and planning which cannot be symmetrical um but can be intertwined through through reference points. These are the things that we hang on to again for uh sanity and for all cognitive systems lived experience necessarily involves the continuous rehearsal of self-predictions as already mentioned and the permanent updating hereof and again as already mentioned this becomes extremely complex uncertain um if not unmappable and literally unmappable in dynamic communicative in the in a dynamic communicative landscape where we rely on others self-referential processes not only their selfhood let's say but also processes which depend on specific set points um and and sort of formalizations in the hierarchical systems that we are a part of. So a self model which is capable of self-reference as linguistic communicative capacities updates all the time its predictive self-referential tension with itself which creates new information and this effect is mirrored in the larger communicative landscape through other processes of self-reference. So because we witness this we also witness it as happening in examples such as this sentence is a lie or as I like to formulate it often this lie is a sentence which um I would be happy to talk about but there is not uh time for doing that here today and of course this has implications for understanding metacognitive prediction limits and the emergence of perspective and selfhood in in all actively inferential agents because what I think I would like to say here is that the reliance on the idea that things are actualized elsewhere. We might not be um let's say capable of witnessing what um our neighbor says about an event that happened when we were not there. But we are able to understand that as having happened or something which might happen. So the mere idea that actualization is possible and that this is actually what language gives us the sort of representation of possibility through a shared uh linguistic uh niche. Um that enters a very strange new domain when we uh create well new new language systems, new formal systems which actually take reference elsewhere and do something entirely different with it. I hope that

was not too convoluted. Um these are all work in progress thoughts and uh things very much at the edge of what a brain can afford. Uh and again as mentioned thank you very much to Shana for the conversations that we had leading up to this and uh it would be really cool if in the future we can um finish this paper with uh with my half which is what I just presented here and then her half which would be uh some mathematical aspects of uh of considering this. Um, thank you very much and especially to Daniel who holds all of this together. Um, and to the active inference community. That's it. Thank you. All right, next up is going to be a quick 5 minute lightning talk by John Clippinger. Here we go. Hello, my name is John Clippinger. Uh, in this lightning talk, I would like to talk about the end of extractive economics. and arbitrage towards a holistic economics through adaptive inference, value creation, and verification. So, there's some big ideas I'm going to be throwing out here. Um, and it'd be difficult to get them all in in a really coherent way in 5 minutes, but I am building on an earlier presentation called aic finance uh active inference and dynamic portfolio management that you can look at that provides the fundamentals for this. But what I'd like to do is really sort of get these ideas quickly out there um towards a new kind of economic system, the whole new kind of economic framework. Um it's one that that really goes from looking at different opinions of of arbitrage and establishing markets and bidden ass um towards uh a measurable uncertainty reduction uh rather than speculative opinion and combining basian inference um and thermodynamic accounting and multi-agent coordination. It proposes a verifiable and adaptive energy efficient architecture financial systems capable of aligning real capital with real world outcomes. So what's the problem that we're trying to solve here? And it's sort of fundamental to everyday economics and finance and that is that you establish value through bid and ass. So it's what people are willing to pay versus what they're willing to buy for. What's the spread? What's the difference in opinion? So prices are really tied to differences opinion. is not some kind of underlying objective measure and these can be manipulated because there's real incentive to to create spreads in order to make the difference in the margins of value. Um and so there's an intentional information asymmetry that's always been exploited. So liquidity is not necessarily equal to value. I mean you manage liquidity. You've got to create liquidity in order to create markets and you can change and manipulate liquidity by through interest rates. But these are just ways of adjusting it within the context of different kinds of opinions. Not saying there's any underlying principle of assigning value there. And what it does is it creates an incentive to manipulate value and hence you get bubbles and you get rug poles and you get a lot of the problems that we have right now with current markets. So what we're arguing for is

core replacement. Price is the capacity to reduce uncertainty for a unit per unit of energy. So you use active inference to treat every agent as a system that spends energy to reduce predictive error around targeted outcomes. We then price the asset by the expected reduction and uncertainty normalized by energy and capital used. Now these ideas are pretty radical in the context of current economics but they're not as radical as you might think because when the interest rate swaps when you have fixed rates versus variable rates or floating rates there's something called the *basically so far* which is secured overnight financing rates which is a broad basic measure of the cost of borrowing cash overnight backed by collateralized by US treasury. So there is this mechanism that allows you to exchange things of different kinds of value proposition in order by virtue of the standard and now this is set by regulation but we're arguing that this can be set by basically a scientific method of assigning a value around energy prediction reduction and information reduction. So it creates a way of incentivizing markets. If you look at markets as Markov blankets, basically Markov blankets of nested agents, they're trying to synchronize around a common outcome that minimizes uncertainty and aligns them all around a common outcome and stabilizes it. So you can create have a welfare index. Who can have some kind of outcome or set of KPIs around a project that allow different players in a marketplace to get the signals they want in order to achieve a shared outcome that is beneficial to all. So this is the holistic notion of a market. We're not it's not adversarial. It's cooperative. It's alignment of interest. A multiscale alignment. And you really have posteriors as prices. Each agent publishes a verifiable posterior. Each dollar jewel reduces x units of uncertainty around a desired outcome. Settlement by evidence. Basically self-synchronization. Now what this does is obviates the need for the fundamental market mechanisms like bid and ask liquidity pools. So you have decentralized exchanges. You have monetary policy which is adverse to fiscal policy. This combines them all in a new kind of metric that is very powerful. So if you're interested in this, please give me a note. I'm at [uh johnfirstprinciplesfirst.ai](https://johnfirstprinciplesfirst.ai). That's fp1.ai. Thank you. We're back with entropic motivation, a panel discussion. Alex Kefir will facilitate and share some slides. Thank you all for joining. Go for it, Alex. >> Yeah, thanks. Thanks a lot for joining me on very um looking forward to this discussion a lot. Um, so yeah, since we have an hour and these are like this is like the deepest theme I can think of to discuss. Um, I didn't want to spend a lot of time on introductions. Um, but just really briefly for people who are who are following this. Um, so I'm Alex Kefir. Um, we've I'm really thrilled that everyone has decided to join uh today's panel. Uh we've got Daniel Palani who again I'm not going to give lengthy introductions here

but um Daniel of course has developed the empowerment framework for understanding u motivation and agents. Um Roirez Ruiz has worked on the maximum occupancy principle uh which we'll discuss today as well. Um Adam Saffron is um I I think of Adam as like a active inference and like physics inspired machine learning kind of generalist but I also just love discussing philosophical topics with Adam. Uh similarly for Dalton um Dalton has also done some work on well you all know Dalton uh here but uh on the free energy principle developing some of the mathematical foundations and also with a particular emphasis on its relation to um things like maximum entropy and maximum caliber so obviously relevant here um and Sander uh Venda Cro uh some some of you have only met over text so I apologize if I didn't pronounce things correctly but um Sander is a psychologist who works in the um predictive processing kind of actor inference space as well and um I recently came across some of his writings. There's a great blog called there is no reward. Um so I thought that each person here would would make a great contribution to this discussion. Um so I wanted to just say a couple of introductory things and then we'll move away from slides and just get into a discussion. Um so the overall theme here in my mind anyway is um what role the question is what role does entropy in particular like the maximization of entropy under constraints as we'll talk about but like what role does this play an intrinsic motivation so you know entropy is one of the probably most uttered words in cognitive science in this uh you know uh 20 certainly 21st century um so you can define it many ways but basically I just mean the you can Think of it as the average surprise or um the spread of a probability distribution or the amount of uncertainty associated with the distribution. Um and of course it has a thermodynamic meaning as well and we may get to that. Um and by intrinsic motivation I mean really this is just about motivation but I'm I'm just to try to focus ideas um there are many ways of thinking about how creatures you know artificial and biological or natural naturally um arising might be u motivated and intrinsic motivation again many ways to define that but I think of it as just kind of separating trying to isolate or distill or understand aspects of motivation that don't boil down to like the kind of thing that's modeled in very vanilla reinforcement learning. So that might be a straw man, right? But I mean there just like learning sort of to pursue some uh reward or like learning associations between certain actions and reward. So um the question here to for me at bottom the question here is like is there a an aspect of motivation in intelligent agents and living agents um that that doesn't just boil down to to reward seeking behavior? And that also might might encompass things like um fitness in the in the context of natural selection. So we'll talk more about that. But um I wanted to get right into some

questions here. Um so and please feel free to interrupt by the way. Um yeah uh yeah so so as I mentioned um I think one useful lens on this which has featured in my recent work is to think of um to look at the connections between particular accounts of what intrinsic motivation amounts to um and how it might be related to the idea of constrained entropy maximization which is important in a number of ways in cognitive science and in physics. Um but so I I recently have been looking at these three frameworks, empowerment, active inference, and the maximum occupancy principle. So, um, Daniel, since you've joined us here, um, I won't ask for a general introduction to empowerment, but I wonder if you'd want to say just a few words about what what empowerment is, like what it means formally and and how it um how it relates to how you think of it relating to um motivation in agents. >> Okay. So the basic definition of empowerment is essentially it's a measure of the potential information that an agent can inject into the environment by its actions and pick up later by its sensors. And it has multiple interpretations. There are many converging ways of looking at that. One way to look at it is to maximize the um possible choices that the agent could have in choosing its future that it can distinguish itself. So it's not just able to choose different futures but it must be also able to distinguish itself that these different futures have been chosen. So the the idea is basically that if you're in a situation where you have more choices, you are more likely to be let's call it vitality. It's a way of maximizing your vitality, maximizing your options to run away from death, which is zero empowerment. Um there's an interpretation in terms of evolutionary niche that your sensory motor niche mean is if you have high empowerment, it's well well matched where the actuators and sensors are well matched. If they're not well matched, you will try to match them better. Um there are of course other criteria like energy conservation so on. But uh assuming you have a given sensory motor equipment, you will try to go somewhere where they are best matched where what you do and what you can distinguish um are maximally compatible. Uh there's another control theoretic interpretation. Basically it's combined controllability observability measure and there are many others. You can think of it as mobility in games. If you look at deterministic games you could look at it money is a form of empowerment. It's a potentiality. It's not what you actually do. That's in a nutshell. Um, of course there's much more one can say, but just to give you a quick upshot what empowerment tries to capture. Oh, thank thank you so much. So, right, so you can already see how a lot of those things that you mentioned don't just boil down to simply seeking reward in a more basic sense. Um but so obviously I want to open up more for discussion but um I thought it would be good to maybe since we have people who are sort of representative of these of these

theories to sort of lay them out first as as background. So um so so for active inference um anyone who'd like to jump in but but Dalton I think you you partic in particular have done work on again on sort of anticipating a bit um pointing in the direction of of the sort of duality between active inference and um entropy maximization under constraints. So would you want to we're also at this implied active inference symposium so it's not as if you really have to introduce active inference but would you like to say anything to frame how you see these um these related being related at this at this point? >> Uh sure well I can say a couple of things about um the particular view of active inference that I think would be most productive to take in these discussions. Um one thing I want to to point out immediately is uh when you have uh an active inference agent uh you are looking at something that uh looks as if it has the policy of minimizing expected free energy. So the actions that it takes in order to maintain its set points are uh appearing to minimize expected free energy and and indeed if you build an agent and code it up such that it is an active inference agent then you can do this explicitly by hand. Um the interesting thing about the expected free energy uh is that agents that minimize expected free energy they both pursue information gain and also goal satisfaction. So there is a sort of uh balance between some kind of intrinsic drive or epistemic value uh and also a reward-seeking extrinsic behavior. Uh and these are are bridged in in one quantity. Uh this kind of um on the face of it has a lot to do with empowerment in the sense that it's it's not simply exploitation. Uh it has an exploration element. It's it's saying also I want to learn as much as possible about my environment uh so that I can improve my future uh suite of possible actions or possible modifications of the environment. So I want to explore in such a way that I can eventually bring about an environment that might be even better for the goals that I want. Um as opposed to simply preparing a state of the environment in the goal that I currently want. Um more broadly if you look at the free energy principle itself uh this same kind of dichotomy is there when you have a complex system you have some kind of typical behaviors some kind of uh states for which the uh system is characterized uh by which the system is characterized. say these these characteristic states are things that we think of as in humans lielike states in control systems they are the set points that make the control system uh be the sort of thing that it is. Uh but within that if you have a system that remains organized for long periods of time it needs to have some kind of flexibility to respond to the perturbations that it's receiving from the environment. So a system that very rigidly stays within a very small set of characteristic states is actually not very good at organizing uh because it is very easily dispersed by changes in the environment. So again there's a kind of

trade-off between maintaining a very rigid uh set of of characteristic states uh those are your sort of constraints uh but also exploring within those constraints what is possible and reconfiguring as the system goes along. Um, so I I think empowerment uh and constraint maximum entropy have a lot to do with each other in the way they treat this balance between exploitation or reward seeking kinds of behaviors and exploration or uh dispersion kinds of behaviors. They both allow a certain uh flexibility within some kind of confidence bar. Uh, and I I've thought a lot about this. Um, and I know Alex uh and Daniel and others have thought a lot about it. So, I'm interested to see what insights come out of this panel. >> Thank Thank you. Thank you, Delton. I've been throughout both of these these um sort of discussion introductions, I've been nodding vigorously. I don't know if people can see because we're looking at the slides, but we'll go to the cameras soon. Um, thanks. Uh so uh um Jorge would you like to um introduce anything about maximum occupancy in terms of the um sort of yeah well just what what what the framework says but also like um uh what what its motivation is in this context. >> Sure. Um um yeah so I'd like to I think it the maximum occupancy like uh fits very well with what people have said in and the theme of this um discussion uh looks like the main motivation for this uh work was to think of how people usually think about like curiosity you know um and exploration as you know means to an end you know like uh exploration is very good because then you can uh reach higher rewards in the long term by just exploring are in the state space and action space. Um, so we wanted to flip that around, you know, because like uh reward functions are in reinforcement learning for example are like uh when you're engineering, you know, is trying to uh have a robot do something. Um, you have to really try to engineer that reward function. Um, and there's no real principle way to think about where reward functions come from other than what other people have said before in the past like how reward functions can be evolved and and so on and so forth. But uh so we wanted to turn that around and say well rewards what uh organisms think about as rewards are actually means to an end, right? And and or should be more like means to an end which you can think of many ways like survival or whatever. But so we wanted to develop some kind of objective um that has the like minimal assumptions you know um so we we came up with uh this objective of maximum occupancy like just from inspiration of the second law of thermodynamics and Carl Fen has said it before of how there's a duality of uh like max path uh maximize path entropy and and minimize entropy uh for the organism. Um so yeah so it it comes with very well with this theme because you know like just maximizing entropy for example in this case maximizing the future entropy of your actions and states doesn't really

make sense without constraints right so it it just maximizing entropy of your actions and states is not an interesting thing to do if you just random walk um all the time um but when you actually think about constraints of organisms that have evolved then you have very interesting things where they go away from those states that are uh as Daniel has said like low empowered or in this case you cannot access future action entropy and state entropy in the future if you die for example. So it has this survival kind of instinct um trying to work with its constraints to still maximize um path entropy. Um that's basically kind of the the inspiration in what the maximum occupancy is about. But um yeah, if you have questions, uh yeah, >> great. No, thank you. That's that's wonderful again. Um so yeah, and I think that that that kind of perspective that maybe things could be flipped on their head so that um these sort of curiosity or novelty seeking um behaviors are in a sense perhaps more fundamental than um reward seeking uh is is a really interesting idea that's that's I think been a major inspiration for me in wanting to do this. Um so before we we move to sort of phase two of this um which would be excuse me to consider some specific questions although I don't want to over constrain this because there's there's um a lots to say um um I guess Adam or Sander do you want to say anything on the on sort of the over overarching theme of intrinsic motivation or any of this? Uh any comments you'd like to make at this point? Adam first maybe just alphabetically. [Music] Uh, no comments at the moment. I'm just enjoying the discussion. >> Awesome. Um, yeah, Sandra, >> you have a few comments. Um I I I I see some of the discussions that have been like historically been taking place in psychology and and whether like like this need for cognition or stimulation or exploration what what place it has within the other so-called basic more basic needs. This is this is pretty interesting um that I mean that um there's now with this maximum entropy or with the empowerment of maximum occupancy this possibility to invert indeed that picture and to to to not see the kind of um knowledge drive or stimulation drive or need for cognition as like the old one now. something that is alien to the the rest. But that is really front and center of of of this picture. And that's I think there's evidence in psychology, but it's never I mean the the kind of evolutionary basic drives have have always been a dominant thing. So, and to see that there's a framework that framework that that doesn't deny that picture but but that can make it more rich is is for me very stimulating. >> Awesome. Thank you. Uh I'm going to I think stop sharing slides now. Um so um is this is this good? Can we all see the cameras now or are we Yep. >> Okay, cool. Um, so yeah, so I did, you know, there were some other just slides with with little prompts and notes, but I I think I'd rather have more of a personal interaction as we do this if possible. Um, so I I

just wanted to go through some some So there's a lot I wanted to get to today. Um, but thanks for for laying out the sort of um basic um ideas and frameworks here. Um, so in in compelling terms. Um so I had I had three themes in mind to discuss and again we can go as far off the rails as we want. Um that's the entropy part of the of the of the equation. Um but but the first theme uh that I had written down here was on sort of the degree to which um these two aspects of in of motivation. Okay. So intrinsic motivation usually seeks something that's not just reward in a conventional sense but so we have rewarding behavior or risk minimization and then everything else. uh and the everything else um I tend to want to um place under the heading of maximizing entropy but I also don't want to frontload that assumption too much but um I guess the question is to the extent that we have at least two de dimensions of motiv motivation in general that that fall out of many different frameworks for understanding motivation like uh is is one of them more fundamental than the other and we've already heard from a few people who want to flip the conventional wisdom on its head in a and say that curiosity or entropy seeking behavior is more fundamental. So maybe we all agree on that. Who knows? But I'm curious whether anyone has a a view on that. Um I could call in particular people. I feel like a Yeah. Yeah. Daniel, go ahead please. >> Um I would say from my perspective I think the arbiter is evolution. So in evolution there is no real thing like reward. Reward is a fiction that we employ to compute um behavior to compute desirable behavior and it requires the concept of utility which has a fulfill number of axioms which we know is not that easy to get if you actually operate with real beings, real organic beings, human beings. we know that naive uh utility theory doesn't perfectly work there. Um so I would say ultimately what you want to understand is how does the concept of seeming reward or seemingly reward driven behavior emergent evolution. Um concept of goals for example is also an emergent concept in my opinion. It is coming through the fact that certain behaviors have to go through bottlenecks. Like I'm not talking about information bottlenecks here, just behavioral bottlenecks. For example, when you catch food, the food is localized. So it is a bottleneck to catch the food. Um it looks like a goal, but the reality is if the food would be all around you, you wouldn't care and you would not it would not be a goal to catch food. I I always give the example this your tapeworm sits nice and warm and is constantly flooded by food around it. It doesn't the food acquisition behavior is not a specific goal because the environment is nutritious as it is. And I do think that's what misleads us as humans. We like to categorize things but this is an emergent phenomenon. I don't say it's not a real phenomenon but it's emergent. It's not primary in my opinion. Thanks, Daniel. Um, so, so that's it's I'm glad

you brought up evolution. Um, I think so, so I take your point that, um, yeah, that reward is perhaps an emergent concept or emergent notion. Um, but I think I think one question that I have here and I will I will not take too much time. I I promise. One question I have here which which informed this this this idea of of having this panel is that I I think in a broader sense like ev if we take evolution to be sort of producing variability and then selecting according to some fitness function that the fitness function has received a lot of attention but the the producing of variability is is is maybe the other dimension of or aspect of this process. Um, and that might have a lot of structure or it might not, you know, I mean there's Anyway, uh, so do do does anyone else want to comment? I was I curious. Yeah, I don't want to over prompt people, but yeah. Yeah, I it it reminds me of like the studies on single celled organisms that when stressed that seem to upregulate their mutation rates. So that that there's just more bets made and and that seems like a more random walk that that might end in a lot of um yeah attrition but but will also give like the necessary structure for the future for the new environment apparently. So is that what you mean with the exploration and evolution? Um I sorry no please I mean I mean this the increased stress right what it means in my opinion evolutionarily it means that you have a higher chance of dying out and so um is basically a change environment that um actually reduces your vitality and to counteract it you create just more options so that at least one will survive right so you don't do that when you're in a good shape right you don't take risks when when you're doing Well, you keep the risks under control. You only take major risks and vital risks um when when there are many chances that you might not succeed. >> So I say >> you can understand it from that. >> Yeah. Sorry. Yes. >> Yeah. No, sorry. Um just that in humans it seems to be the reverse, right? When we play or when children play, it's when they're wellfed, when they're they don't have stresses, when so it's reversible. >> Sorry, >> it's reversible. This is a reversible risk. That's not really risk. It's just free learning. Uh taking risks, um when I when you say you increase mutation rate, you take irreversible risks. Basically, it's like having a a company and the company is in a bad shape and then you start taking bigger risks to make it survive than you would do if it's doing well. >> Mhm. >> But play play is different. Play is a is a safe um it's actually an excellent point. So reversible is important. Yeah. >> Yeah. Yeah. I see. >> Cool. I'm I'm curious whether so um I don't know in what perhaps I'm reading into it but uh in your work on maximum occupancy I tend to think of of this as somehow modeling like the desire is to stay alive in a way sort of for its own sake right it's it's you want what you want is to be alive and that is to occupy state action space path space so I don't know if you further comments that I might make. >> Well, yeah. I mean, I

think that in this case, surviving is kind of a consequence of trying to occupy your future action and state paths. Um because when when you die, you can't, right? So, but it's not the primary drive, let's say, of just surviving. Like surviving is a very degenerate kind of reward function, right? Like if you survive, there's just so many ways to survive. Um um but uh but when you are actually doing some kind of maximum entropy on actions and states in the future then you're you're encountering some some you're you're you have very minimal assumptions of what you can do but you're constrained about those things you can do. Um so um yeah so that's kind of what I would say about like survival but coming back a bit to towards like what is kind of more fundamental um I think it would be more fundamental with regards to what what you want to achieve right um so like if you're if you're an engineer um that wants to just make robots that do stuff um you know like uh if you can engineer could reward function Um then you could say that you know intrinsic those intrinsic motivations that you give uh you could give uh to robots um they serve the function to um maximize work or in in that case kind of intrinsic motivations is not as is not the the driving thing. U and and people you know have argued that reward is enough. Uh but kind of the idea of of maximum occupancy was like well reward is not necessary. Um and and I think that that the the the what what we wanted to argue with this work was that as Daniel has said like rewards is kind of like an emerging thing. Um it's something that makes you want to achieve uh other something else. uh and and in that sense um that kind of extrinsic reward I mean um it's funny cuz I like I we organized a workshop about like intrinsic motivations in um in this RLDM conference and and kind of like what came up with it was like well all rewards are intrinsic rewards and and in that way um I agree um but like these rewards are kind of something that has evolved over time to make you achieve something else uh or to help you achieve something else. Uh and so in that way I would in that sense I would argue how that um there's something there should be something like a principle um that makes you uh over time or perhaps evolutionary time uh develop this uh reward functions uh these extrinsic rewards that people call um >> Great. Yeah. Um so so maybe a more a slightly more focused way of asking this question is so I mean in in a sense it's it's all down to evolution but like um curiosity for example like curiosity and um like things like epistemic drives and information seeking behavior um with respect to that question like one question I had is well Do we must we view that as in the service of obtaining something else? Now, I guess we've problematized the notion of rewards somewhat here, but like is is that sort of information seeking always in the service of obtaining some goal? Is it always a means to an end or might it be like a more fundamental aspect of what

drives life and agents? And I'm um that that's that's I I think under some guys I'm I'm I'm tempted to suppose that there's something fundamental about that this other component of motivation, right? So so there's the constraint, right? Then there's the entropy maximization. Um but so so um I I think Dalton maybe I'll call on you here. um from from an active infant perspective, it seems like in a sense these two things are on a equal footing at least rather than everything being in the service of risk minimization, >> right? Um I I suspected I might be called upon soon, so I I intended on volunteering myself uh just in case. Um before I answer that, I just want to echo something uh that Daniel said uh which was that I'm I'm very partial to the view that uh reward is a useful explanatory fiction. uh and in particular I'm I'm quite partial to the view from the free energy principle where we say if a thing is existing for long enough to be observable to be modelable at all as that thing then it must look as if it's seeking environmental states conducive to its continued existence and that that appearance uh gives the the impression of desire or it gives the impression of some reward some some goal seeking behavior. But if you take something simple like chemotaxis, uh, something like a bacterium or an amoeba, it one could argue it would be a mistake to anthropomorphize that as seeking reward as opposed to merely following a gradient because if it weren't following that gradient, it wouldn't exist to be described at all. um what is fundamental what is physical is the maximization of entropy. So in in the end uh even um the imposition of constraints is something that we do as a modeler. We say here is a constraint that phenomenologically describes our observations. It gives us the right answer. Um, but does that constraint exist in reality? That really strongly depends on what your sort of metaphysical or ontological position is, right? Um, so it it becomes a little bit more subtle than than saying actually that uh constraints exist as opposed to there's something that that matter to us as the modeler, right? Um, so that that's sort of one component of the answer. Uh the other component is um it is often useful to describe agents uh broadly construed as balancing both of these things uh because often times there is that impression of reward-seeking behavior and when engineering agents when building things we may want to give them reward-seeking behavior. Um so active inference is telling a story where both of those things are given uh equal um waiting that they're they're regarded as uh equal contenders in a competition. And this proves to give very succinct very explanatory descriptions of a lot of the dynamics that we observe. Uh so if if you're in the business of flipping the story on its head and saying if there is something to be observed for a long enough period of time then it must be somehow balancing preparing a preferable environmental state with exploring its options to decide uh what could possibly be

preferable. uh if you're in the business of telling that story, then active inference is a a very useful way to tell that story because of the way the two things are balanced. Uh but I think it's a somewhat deeper philosophical question whether all systems or some systems or no systems actually do this. Do they actually have the policy of minimizing expected free energy or is it a useful model of what we observe as the dynamics of such systems? Yeah, thanks Dalton. Um, I guess I have no problem sort of unconsciously taking a realist stance on that question, which is often a blind spot I have. But, um, I think, okay, I mean, but granted, it's a it's a good point though. But so so regardless of of how sort of much epistemic humility we want to adopt about our models of situation um it does seem it does seem that well you've you've you've raised the second major kind of question I wanted to address which we'll see how long we want to spend on it. Um I think we're each of us in this panel is is is um sort of immersed in the you know thermodynamic descriptions of things to different degrees but like um the the second question I wanted to ask is well what it's probably there's no single simple relationship although I would like there to be but what's the relationship between these formalisms which describe after all they're describing sort of um objectives for agents um and just entropy constrained entry maximization, things like that in the physical sense. So, um I don't know that just asking the question in such a flat way is going to be a good conversation starter. But I mean, Dalton, you brought up um active inference agents look as though they are um seeking to minimize free energy, expected free energy in particular. So um and actually so so let me just just say something here. Um so I think like in a sense if you think of agents or um sort of inferential processes like minimizing instantaneously minimizing variational free energy that's an easy that's it's a very easy very clear illustration of this sort of you know um minimize energy while maximizing entropy because like in Helmholtz's free energy you have the the energy which is the average internal energy of the system minus the entropy and so to minimize that free energy you want to maximize the entropy right um or minimize the energy um or both at once. But so but in a lot of um these frameworks, we're not really talking about just that. We're talking about policy selection. And like I know I mean also I think this applies as well to empowerment like Daniel. So you're right, you think of empowerment as it's not just you want to choose it's it's an objective for choosing states, right? So that you maximize the sort of the empowerment that you'll be in once you'll have once you're in that state. Um and so so neither of these things um policy selection under expected free energy as an objective or empowerment uh bears a direct relationship I guess to you know maximizing entropy in this Helmholtz's free energy sense like just maximizing the entropy of states under under constraint

directly in states. So I'm wondering if there's anything to say. Okay, the the overarching question >> I mean empowerment is is can be used as a surrogate for a utility function. Okay. So um while it depends actually on this this channel capacity it's a state function and it's defined by the pathways available from the state and and how much you as an agent can influence that. um free energy and other models do depend also on distribution. So there is a bit of more additional um structure that they um have and it probably models different things too. So one of the things that we need to um uh realize is that um empowerment people always say oh we use it for exploration. Exploration is incidental. It's not the main thing that empowerment does. So yes, sometimes it can lead to exploration, but that's not um it's a spin-off if you like. It's not its main agenda that there are tools which are explicitly explorative. >> Right. Right. So, so one question I had here is sort of like of the various things that you might pursue as an agent, uh, exploration is one of them. Um, empowerment I when when I when I try to reduce it to a bullet point, I think of it as based on things you've said, I think of it as like sort of maximizing the variety of futures you can realize or something. Um so that's obviously different from conceptually from exploration in terms of trying to get information about things. >> Exploration comes into play if you need to develop your model. If you have your model, there's no point in exploring anything, right? So when you don't have a model, exploration can increase your epistemic uh empowerment. So empowerment can change by what forward model you have. So if you have a very detail forward model, there's no nothing to explore and you just maximize empowerment straight. If you have a bad epistemic model where you don't know exactly what your actions will do, there's an advantage for you to explore what actually actions are going to do. So empowerment will uh invite you into exploring the states, but not just any states, just the ones who are most promising for maximizing your options in the future. So, as I say, exploration here is more side quest rather than the main issue. Um, which I don't think it's necessarily bad. We we sometimes feel like um if things are not primary, that's a bad thing. But I I feel that evolution doesn't care, right? In evolution, there are lots of these side quests that turn to the main quest. So, for example, curiosity. I would for example speculate and this is pure speculation that curiosity might be a side quest um to make higher organisms ready for changes in the niche. Um so because curiosity is dangerous. Curiosity killed the cat. The reason why it's they say that but maybe for higher organisms which are which cannot compete by larger numbers um where they're sensitive to to to environment. It is a distinct survival advantage to have curiosity in the population on population level. On the individual level, probably not. I don't think that on

individual level exploring Antarctica is good for your offspring, but on population level it might be. >> if I could if I could add something just before you move on. Um, I I do want to uh note that I think it's interesting Alex mentioned uh variability a little earlier on. Uh, and it's interesting to note that uh variability is itself a way of increasing the capacity for side quests. You know, you introduce a whole bunch of different mutations and see which ones persist, see which ones increase fitness. So it gives you a kind of test bed for other beneficial uh things that are not simply about uh increasing uh reproductive ability uh but are maybe unexpected ways of of of increasing the duration for which an organism persists and the amount which it can reproduce and pass on its genes. Uh so there there's kind of a uh a link between this discussion and the earlier discussion which I found uh quite uh compelling. I I also wanted to um open if if that's okay. Um just to to note perhaps I don't know if it's kind of technical more but um something I've I've noticed from both empowerment and active inference um was that this this idea uh that um if there's you know if you have perfect knowledge of the world let's say or your model is very certain or whatever u and there might be ways to keep you exploring so that you're never like converging but let's say that you do and that you're learned both of these two things kind of stay put forever, right? And like you could say, well, the environment is always changing so you need to go back and like you know your internal states change so you go back and fulfill your model or you minimize your surprise about your model. Still, I feel like there's nothing fundamental about exploration if if correct me if I'm wrong in that sense of uh as Daniel said, like it's a side quest to fulfill those things, but it's it's it's like if you converged, you stop and then that's it. You leave your organism there. Um and and so that's that's why I wanted to like kind of contrast that to what we've proposed which is like well no you keep doing stuff as if you're allowed to kind of thing. Um and and and in that sense that kind of keep can give you more robust robustness in your um um in your organism or in your artificial agent more in a fundamental way. But uh but I yeah I I that's kind of my question. >> Yeah, Daniel, please go ahead. >> Um I think this is a very good point. Um the question is when is it okay to um you know dry out in a in a given niche or when not? And my claim would be the following and this is this is purely speculative here. uh that the reason you want to continue exploring when you found a good maximum is because you have no idea what happens to your environment. See, you sit on somewhere and everything is fine, everything is happy. Uh you and you live in this this this perfect paradise and you don't don't look around. What happens is you lose information about the neighboring states and in so in other words in the language of mathematics you become only point oriented rather than neighborhood

oriented and I believe that this is um makes you very vulnerable if there is a major change. So you will not be ready for any major I have some examples but they're bit too long to explain here but essentially you're not ready for um changes where the local environment would actually tell you look something's going to happen. uh but because you said only on your point you never notice and that's why I think uh we when we stand as humans we never stand still we constantly evaluate the ground constantly u and and anything we do has a certain variability even the most perfect sports people they should just reproduce exactly what they always do and they never do and I it sounds like a like a bug but it seems to be a feature of nature to never do the same again. >> Okay. Yeah, I like that. Um it sounds to me though that it's like um it's an emerging thing, right? So it it of like Okay, it's just because you live in a world where things change and things have evolved to look for for other things around. Um perhaps it's an even empirical question. I don't know to say like well if actually you have a very very stable environment do you have do you evolve beings that do not export >> I I actually claim you stop evolving and we have examples for that and um there are frog species which are so specialized they require very specific setup to continue um you disrupt the environment a bit and they're finished right um and you see this in nature but Mostly you don't because in large scale echo webs um you don't have the stability. It's always something that intrudes your in your world. But you can see that in economy too, right? An economy that is relied oh we always relied on the same things working fine and suddenly something major there's a major change um that happens. You now have the problem um you're not ready. So um bubbling um as I think Adam said bubbling is the right thing to do constantly to some level. You don't want to waste too much on it but you have to keep doing it. There's an advantage to that. This this is this is very interesting because my one of my intuitions here is that it just sounds like it sounds as though there needs to be a reason for this this babbling. And of course it's very useful but but I I guess I I wonder whether just being alive like that's it is babbling like that's that's the fundamental thing. So there's always you know if you planned and you can always you can have these goals that pull you from the imagined future but then there's also something that just pushes from behind right which is sort of the essence of like being alive and I I I guess I want to link those two things but they are quite different. planning to stay alive, right? Planning to maximize entropy on infinite time horizon is still quite a different thing from just doing stuff. >> I don't think you randomly bubble babble, right? I mean, one of the problems with with many exploration strategies is they are um undirected. You you don't care about knowing what the lint under your your bed is doing. you most of the

time you're not interested in that there specific things you're curious about. >> This this is true. Um so well okay that sort of conveniently brings me to one question I had was this this sounds it's a funny thought but like is there any reason to think that so evolution is quite a general concept of course right it happens at different scales but like evolution on the phoggenetic time scale how you usually think of it. Um is there any reason to think of it as having some kind of curiosity component? Um exploring state space um you know in the similar sense to how an individual might I mean maybe that last thing makes it is is not helpful. You were nodding earlier Daniel. Um so I mean you could think of that you know from the point of view of like multiscale active inference like oh nested agents all the way up and down but um that's a very thin sense of agent um but um I mean it so I I think I was reading a article by um Terrence Deacon and he mentioned sort of um something about Anyway, something about the fact that evolution is driven by I mean this is just obvious evolution is driven by sort of variation. Uh so you know if you're implementing a genetic algorithm the obvious thing is you need the fitness function and you need the modification functions. Um so it's it's it's just sort of a question about whether and I mean Delta maybe you'd say this is a matter of how you model things but um if we're looking at processes of evolution on the on the time scale of you know speciation and things like that. Is there anything that we could view as like a structured process of exploration? Anyway, it's a question and um I don't have the answer. So, I'll let that sit for a second while I dig for um well, if sorry, is any comments on that? Adam, thanks, by the way, for doing some chat jockey. you've been quiet, but I really appreciate the what you're adding to the chat. if I mean if >> I mean I I can make a comment. I just didn't want to hog the hog the panel. evolution thinking of evolution as an agent is a very interesting thing. It's not an agent in the usual sense because the interface is very vague and and and distinct. But I do agree uh that yes evolution has a way of learning has a way of I think just recently I read a paper that evolution does have a um ability to improve its learning mechanism. I would be surprised if it doesn't do that. Um we know that Lensk's experiments are a nice example of pre-structuring um evolvability pathways. um this long-term experiment that he did I don't know 20 years ago I think it was this paper uh really really nice paper fantastic experiment I do think that we we can think of evolution in this as long as we don't say oh it's an agent with clear membrane creation or something we can think of it as a kind of as a kind of cur curious learning mechanism uh But we also have to understand that evolution unlike normal agents doesn't have that constraint. It has to work under constraints of physics and chemistry but um energetically is quite unconstrained. Number wise it's quite unconstrained. So it can be

generous with that. >> That that makes sense to me. I mean well thanks for thanks for partly so I'm not I'm not crazy or both we're both crazy or to to think this that there's some there's >> that's all right. Um so so there was one more kind of big picture thing that I wanted to get to today. We have about less than 10 minutes left. Um and that was just about rationality. So again, we'll see if this sparks interest or not, but um I think of I mean, so rationality again has many senses, but like um you know, often when you're thinking about agents and behavior and motivation, thinking about practical rationality, like I can reason my way to obtaining some goal uh which is a large part of what we've been talking about. Um so the simple question here is just well in so far as motivation intrinsic motivation does not involve that in so far as we have like entropy seeking behavior and it's not just in the service of obtaining a specific goal. Um is it what's its relationship to rationality? Is it irrational? Is it irrational? Does it have to do with bounded rationality? Um that's the question. I'd kind of like to think there's something super rational about this that it's not it's not not everything we do has to be in the in the pursuit of some of some >> Yeah, I I agree. And yeah, I think from I mean I've been struggling with long-term illness and it's made me realize that the ends that you described in rationality, the goals are like for healthy people like for a sick person, it's just about expanding your action radius, gaining ground and just being pushed back and and like expanding your world in in a very literal sense and gaining some freedom really. And that's the basic thing. And of course, you have to be eating eating or or be well fed and and and but but it's somehow natural for me to see that the freedom and the empowerment really is is primary. Um and only once you have like discovered the goals, you can start to optimize. So, so you have to form form the goals and and and that's true, I guess, some form of maximum entropy or or empowerment. Um, but but yeah, it it resonates with me to to think that it's very much primary. Uh, yeah. Um, thanks. That's that's that's that's interest resonates with how I been seeing it as well. Um, so, um, yeah, I'm not sure I'm not sure I have much more to add to that actually. Thanks for Sandra. I was going to ask you to chime in and I think that was a good moment to do so. Um >> yeah just to get a bit back to sorry to I don't know whether I can articulate it but like somehow like this this primacy of like the maximum entropy it seems like I mean I've been reading rolly who's taking like this kind of more perspectival more interactionist view of of um thermodynamics and and maximum entropy and and this seems to map onto somehow onto empowerment for me that it's a a relative measure perspectival measure. Um I cannot articulate it because yeah I'm not on top of the the physics of it but it seems to be that there's a link between the kind of like the blind entropy of course you have where where heat is

spreading where where gas is diffusing. You have like the the structured entropy where where constraints are used to to better dissolve gradients. And then you have like some kind of empowered entropy where where it somehow reflects on itself. It it it models its own entropy predict production. It it starts to um model that and optimize like the expansion of the of the paths the possibilities um with this agency then of course. Um but that seemed to be like like very simplistically in my mind like the the steps that that are are taken and that's why I I agree that I think this maximum entropy is with the link to empowerment is is more fundamental than um yeah than reward or goals or um but I don't know whether that that's uh I is very philosophical and um sketchy. >> I I am a philosopher by training. That's that's that's what that's what that's that's why I'm here. So, I appreciate that. I mean, this this is the kind of I wanted to just just dig deep and see what people It seems that um we have a couple minutes. um probably wrap this up, but it seems to me that like well so the reason I maybe we have time for one quick thing like I kind of want I admit I want there to be a connection between physical entropy maximization and entropy maximization under constraints and in agent aential motivation because they're so it just seems like a bedrock sort of thing like why do we keep seeing this um this sort of um structure in which you have um a spread of of trying to maximize sort of possibilities under under under constraints um from from many different points of view. And so although there's no simple relationship I guess when it comes to things like policy selection under you know um like pathic integrals of expected free energy or or looking at empowerment per state um it does this this is the reason that that I thought it was an interesting theme to explore under that under that heading is that it seems that there's something fundamental that might unify these different approaches although that said they are still different approaches in many ways. So, um, I think I'm going to awkwardly conclude this. Does anyone have any closing remarks? I really appreciate you all being here. This is this is I I have never done a panel before and I I I'm glad I did this as a panel rather than just, you know, trying to um um badger people with my own views. So, this has been really great and I thank you so much for for joining everyone. >> Um, yeah, >> thanks for organizing. Really, >> thank you for organizing. >> Yeah, thank you, Alex. Perhaps I'll close out by just saying um you know I have a lot of thoughts about that last question you asked uh and I think uh obviously I don't quite have time to say uh them all or even say some non-trivial portion of them uh but I think a fruitful thing to do uh would be to look into the world of stoastic thermodynamics where a lot of work is currently trying to bridge information and physics uh in order to describe uh complex non-equilibrium systems. Uh and and probably uh the relation

between maximizing entropy in an information sense and maximizing entropy in a physical sense uh will fall out of something adjacent to that. Uh hopefully we as a as a community as a field uh get closer and closer to that. Uh hopefully you and I get to talk more about it as well. I know I've I've promised to do so and we've both just been a bit busy. >> Oh no, I'd love to. Thanks. Yeah, I'm looking forward to all these continuations. >> Yeah, but thank you again uh for the panel. It was really wonderful discussion. Um I'm sad it was so short. I feel like it could have been much much longer, but uh >> what can you >> We'll do a retreat. We'll do a retreat. >> Thank you. >> Great. >> Yeah. Yeah. Yeah. >> Thank you. Thank you. You all have a lot of I know excellent writings on the topic. So it is fun to see that kind of come out just a little bit tip of the iceberg. So all right, farewell. Oh, All right, welcome back. We're with William Garter who will be presenting on NGCLARN V3. You may remember a previous symposia where William presented as well. So this will be great to catch up and hear about this toolkit and some of the cool things that people can do to apply and do modeling in it. So thank you William to you. >> Yeah. Hi. So my name is William Garter. Um I am representing the neuro adaptive computing laboratory from the Rochester Institute of Technology in New York. Um I'm here today to talk to you about NGCLearn specifically the V3 which is our upcoming release of NGCLearn. Um it is a fast and modular computational neuroscience library. Uh and with the acquisition of V3 it just gives you a bunch of new tools to work with. So first off I just want to give a very brief overview on what computational neuroscience actually is. So computational neuroscience at a glance is basically just the study and investigation of functions of the brain uh brain functions using mathematical computational tools and theories. Uh and what this means is that uh instead of going uh to math first and then trying to claim that we are you know using things like the brain uh we go to the brain first and then try and model the processes that happen in the brain with math. Um your brain is not actively doing a bunch of matrix multiplication but if you look at it from a high enough level and squint properly it does look like it's doing matrix multiplication and so we can approximate things that way. Um, and we are very focused on the bioplausibility of all of this. Um, and it's all grounded very much in the physical studies of living organisms. Um, so there are studies that look at how neuro pathways work. There are studies that work at how synaptic plasticity works. There are studies that look at memory. Uh, and these are all just the various things that we look at in computational neuroscience. Uh now what this kind of means is that we need tools to work on this. Um because the brain is complex. Uh for example, if when our brain is interpreting the world, uh it is receiving lots of data um like gigabytes and gigabytes

and gigabytes of data coming in constantly. And what this means is that if we have a model, we need to uh obviously we're just going to be looking at a small paired down version of the amount of data that comes into our brain. So we're always working with a slightly suboptimal picture. But on top of that, your brain isn't ever idle. So in traditional deep neural like deep learning, uh you take an image, you feed it into your model, and then you get an output. Um, sometimes we do things what's known as recurrent neural networks where you give it an input, it goes and it processes and then you give it another input and it uses the output of the previous one and the new input and it processes. That's closer to what your brain does. But the problem there is all of the little internal parts of that network don't have values. Um, and so like we want to make sure that we're representing things with biology. Sorry, was there somebody speaking? Okay, sorry. Um, and so then the this biological interpretation also lends itself to how we get inputs into our models. Uh, so for instance, the focal point of your human eye can only really process things at roughly 60 frames a second. Uh, your periphery processes things a lot faster than that, but the focal point doesn't process much faster than that. Um, but even in between those frames, it doesn't sit idle. Um and then obviously different sensors in our bodies uh impact uh sense things at different time. For instance, the periphery, the uh rods in our eyes process things a lot faster than the cones in our eyes. Um, and so then we work inside of this biological framing that I was alluding to earlier where when we model our neurons in our when our when we model our neurons uh we want to make sure that they are stateful and that we are working in a more uh time aware system. Um, and we do this because your brain is a time system. And so we still use a lot of the similar words that you're going to hear in traditional deep learning. But for us when we say we are talking about a synapse, we are talking about the actual modeled connection of the synapse. Uh when we're talking about a neuron, we are talking about voltages, currents, membrane potentials. We are talking about the physical biological aspects of a neuron. And what this also means in our biological framing is that the way that information is transmitted from uh point A in our network to point B in our network. Um it can't just send a floating point number because your brain doesn't send floating point numbers. The way neurons communicate in our brains is with uh pulses. Uh for we call them spikes. And so this in and of itself makes things a lot more complicated when it comes to modeling because if we're sending discrete pulses over time, we have to do a full forward pass over our model to figure out where pulses happen, send pulses, and then we have to do that again and again and again and again. And basically what it looks like at the end of all of this is we have about a 200 layer recurrent network where it's just going down for like

200 time steps. But really it's not a recurrent network. It's just constantly processing new information and constantly giving you some output, but that output might just be null. So all of this comes together to basically say that everything works better as just one giant state machine. So we have our model, we're calling it S. And then when we take a step in time, we move to another state. And depending then on what we do and in time, either we can move down, left, right, whatever. And these states are constantly evolving. And most active inference models, a lot of just computational neuroscience models in general, they all have this stateful uh framework that they work in of where if you just pause the entire model's processing of data, there's always something going like there's always values that exist. Um and the main thing to keep in mind with all of this is that these different initial conditions and state initial starting state uh vastly change the model's output even if it's shown the same information. So that's kind of the overview of computational neuroscience state for modeling and kind of setting up our problem space. Now this is not a like this isn't a field that came about 2 years ago. This is a field that people have been working and tinkering around in for a little bit of time now. Um, so that might be the question of well, somebody probably has built a built something before and they have. Um, I'm going but the thing about building uh previously built models is that they kind of generally operate as black boxes. Um, so because building these large stateful systems incur incurs a lot of overhead, most libraries that exist today hide a lot of what's going on. and you have some very specific tools that you can use to uh modify the internals of your system, but you don't necessarily have all the tools that you want. If you want to come up with a new learning rule, maybe you don't have all of the values that you need for that learning rule, and there's no good way of modifying the system internally. And so testing new components and everything is not easy. It's not trivial. It's something that takes a fair amount of time. And if we're testing models that take a long time to run, it means that if we're it can you can just lose days, weeks to just trying to get a model to see if it'll do anything and you don't know whether or not it'll work. Um, and so what the, and just as a reference point here, many some of these alternative models take up to like 8 hours to do an epoch over emnest for a moderately sized spiking neural network, which is a lot. Um, and so this is kind of why we went to like our lab uh went to building our own computational neuroscience library. And so that is called ngcarn. That is what I'm here to talk about today. And basically this is the little highlights reel and then we'll get into exactly how it's working. But it is a public pip package. You today right now could go and type in pip install ngcarn um and you will get a version of ngcarn. I will tell you right now you're going to get 2.0 which is a

perfectly functional version but it will not look exactly like what I'm talking about today. Um, ngcarn at its core though is a state machine that sits next to uh Jax which is a it's effectively the linear algebra library for numpy uh that was the backbone of TensorFlow before it split. Um, and Jax allows us to do things that are known as just in time compilation. uh which allows us to take a bunch of Python code that we know is has fixed sizes and fixed types and just produce effectively a C code snippet that we can run on GPUs, TPUs, CPUs, whatever. Um and in doing this our models in some preliminary results you can get from something that takes about eight hours written in TensorFlow to something that takes about 15 minutes on the same hardware written in ngcarn. Um and then the advantage of this is that you can also make it also allows for the creation of new methods, new models, new components, whatever incredibly easily because as long as you follow a very simple coding paradigm, everything will interface with it perfectly. So let's get into what some of that looks like at a high level. So if we're going to build a model in ngcarn, we do have to slightly change how we are thinking about modeling a network specifically. It's for us it is just one giant state machine. But if we just treat it as one giant state machine, it gets very hard to keep everything organized and not end up putting values where they're not supposed to be or just, you know, moving all the data around. So we use components inside of our state machine which here we're calling a context. We use components to kind of hold parts of the state uh together and modify only a sub part of our state and then be able to communicate with other components which just also encompass a different part of our state. And this whole compartment uh sorry component model uh just allows for us to keep easy track and it's basically bookkeeping that allows the end user to say I'm going to take these same values and do something new with them. Similarly inside of a component we have a compartment model and this is where data is stored and is moved around. Um and this is what allows us to very easily move data from one component into another component uh without a lot of overhead. So more specifically uh com looking at components here here we have one for what is known as the leaky integrate and fire neuron. Now specifically the math that happens inside of this isn't particularly relevant. Uh but know that this is one of the most common neurons used in computational neuroscience for modeling. Um what it is is it is effectively a uh it's effectively a capacitor where it slowly builds up current uh passes a threshold discharges emits a spike. Now what this means is that there are some fixed parameters. If we look here the number of unit the number of neurons in our like layer of uh leaky integrate and fire neurons is fixed. We're not growing new neurons magically as this is going. uh the that voltage threshold that it's slowly building up to that's also fixed. Uh

we have some time constants, we have a resistance value and we have some voltage rest and reset values. These are all fixed parameters. These don't change in time. You set them up once at the beginning of your model and then you just call run. Now they can vary um from component to component but as far as time is concerned they are fixed. Then we get to the compartments which are the variable parts in time of our component. So here we have current as an input into our component. At a various time we will have a current. Similarly if we pause at a different time we will have a voltage and at that same voltage maybe we have an output spike and we also have the last time that we spiked. So we keep a track of all of these values that change in time. And this allows us to think about our components in two steps. We have here's every we have here's all of the math and the logic and the fixed parameters. And then here we have a couple of values that we need to make sure are updated at the end of every time we are called. So that way we keep up to date with our state. Finally, how do we connect these together? So if we have component A and we have its parameters which are our fixed values and we have our compartments which are our variable values and then we have component B which has different parameters and different compartment values but we want to take the output of component A and we want to make sure that it gets put as the input to component B. Oh, sorry. Uh that just means that you're just going to take the value here and it just gets passed along and it's done and it go and it's traverses along what is known as a cable. And so inside your model, you can take any compartment value and you can wire it with a cable into any other compartment value. And what that means is that when component B looks for the value inside this compartment, it will just know that it actually should be looking at the value inside this compartment. And so this is how that we can keep things nice and fast is there's no duplicates of values but effectively it traverses along it. Now with component X Y and Z down here we're doing something a little special. We have a summation. So now it's entirely reasonable to assume that the inputs into our component uh need like we could have a variable number of inputs. Uh, one of the best examples of this is if you have a recurrent connection inside of your model um that are both feeding into say the current value of your LIF neuron. What that means is that you would have going down your model, you would have the current coming in and then you have current from somewhere up in your model also looping around. And basically there we might want to add these together. We might want to subtract them together. We might want to have them multiplied and have some basic arithmetic operation done to them. And so we allow you to do that along these cables is you can basically bundle these cables together and then just say hey add all of these together and

you can do that and this compartment here knows how to do all of that. So when you go and try and see the value of this compartment it will automatically collect these values add them together and use it. And you as the end user don't have to make sure that that happens. And if for instance we unplug component Y from the model, it just runs. If we add in a component alpha to our model that we want to add to this summation, we just add it into we just wire it into the uh we just wire it into the compo compartment like anything else. It all just works together. Now, this shows us how to set up our model, but we're a state machine, so we actually have to be able to transform through states. So, transitioning our model through time. So, here we have our state transitions across a component. We're going to look at these at a specifically a component level and then we'll look at them as as a whole system as our model. Uh here we have component A. uh it has our fixed parameters like before and here we have two different state transitions that we want to account for. We have our reset transition and we have our advanced transition. Now there's this little helpful key up here where anything that's a solid line means that that depends on those values. And if it's a dashed line that means it updates those values. So we'll notice here inside of our reset function it depends on this fixed parameter and this fixed parameter. And then what it does is it just updates every compartment value inside the component. Now in component in the advanced state of our state transition, we'll notice that it takes this compartment value, this compartment value, this compartment value in as well as these two fixed parameters and then it updates this compartment and this compartment over here. And you can set these up in as many as complex of methods as you want here. any amount of math can happen. The only thing that we have to keep in mind is that these values are fixed and these values change in time. That is the only thing that we need to keep in mind with this. So if we want to do this across multiple components or what we call a process on our model, basically what would happen is we have some overarching process here. We just have it called advance. And what it's going to do is it's going to call advance here on compartment on component A. It's going to compute all of its outputs. Then since these are wired together, the when this when when we call advance over here on component B, it'll actually look at the output of component A and use it for component B. And then it will compute its output and put it inside its output compartment. And if we were to keep chaining and have a component C over here that was wired out, it would just the D it would just keep going. And so this allows you to chain as many of these components together inside like a forward pass as you want. It's also important to note that these do not have to be the same type of component. They do not have to be the same type of transition. So here there it's all named

advanced, but as we'll see later, it doesn't always have to be named. It doesn't all have to be named advanced. Yeah. And so then um yeah, and it's always uh deterministic the order that all of this uh data gets transmitted in. Um it will always go in the order you define the components in. So you don't have to worry about race conditions and anything like that. So finally in order to actually get speed up from this because right now we've just written basic Python code uh we actually have to do go through what is known as a compiling step of our process. Uh and this makes use of Python's abstract syntax trees. uh which I am not going to get into uh the complications behind getting syntax trees to work but basically we look at your code that you have written in the generic Python classes uh and we tweak it just a little bit and we'll see how it tweaks it just a little bit uh to produce code that is fast and efficient for Jax's just in time compilation. uh this is the true source of ngcarn's uh speed up is that it is uh it can take code that is written in a very easy to read uh effectively format and then convert it into what is needed to be efficient for uh Jax's just in time compilations format. Um and so basically we just take that whole process and we send it through our compiler. Now the compiler as an end user is not something that you need to ever actually worry about. But there are advant but with V3 and this is what we're getting into here with V3. There are some things that you gain from that you gain that allow you to look and really inspect at what is our model doing? What is our process actually doing? Uh which is very helpful for things like debugging. So what does this actually look like in code? So starting at the top, if we want to define a compartment on our class, right? So these these all have self. So these are all inside of a class. Uh we can have inputs and it's just compartment with some initial value. Outputs compartment with different value. Weights compartment with a third value. This is how you make a compartment. Now there are a couple extra parameters that you can pass to compartments that are useful for things like plotting and whatnot, including like compartment names, units, all of that. Uh then we have fixed parameters which are just exactly like you're used to defining values in Python classes just `self.shape` equals `shape` `self.trc` scale equals `resist` scale whatever you want to call it. You could just define them like regular values. Then when we want to define our advanced state for our transition here uh we have our outputs which is our compartment up here. We're setting it to a matrix multiplication of our inputs and weights multiplying it by some resistance scale value and adding biases to it. uh which is not displayed here, but there's a bias compartment as well. And what you'll notice is this is just the weight. This is just a weight matrix. This is just inputs, weight matrix, outputs. And we'll notice that this looks pretty similar. This outputs set might look a little weird, but just it if you

think about this as just an equal sign, this looks incredibly similar to regular Python code that you're used to writing. The only thing that's a little bit new is that it has this little compilable flag above it. Now, you might think that there's a lot of work that goes into that at compilable decorator wrapping around this, and I'm going to give you a secret. There's not. All that is doing is just adding a singular flag to the method. It is doing nothing beyond that. So, then we get to Okay, so now we have our component. We have our state transition as an advanced state. How do we use these? Right? Obviously, that's where the complications got to come in. So, not really. Uh, so we have contexts, which if you remember are just our global state. Basically, everything has to be inside of a context. So, we have with the context your context name as some variable and then you just define them like you would regularly define Python objects. So, here we have our static synapse, which is a weight matrix from our inhibitory neurons to our excitatory neurons. It has a shape. It has uh a weight initializer which uh for we provide the ability to just get distributions uh generated for you. Uh so here we can use our distribution generator. We can get a constant distribution and we can say that we want it to be hollow. Uh which is the main diagonal is all zeros. Uh and then when we go back we can say nope this one we actually want it to be an identity matrix. Um, and there's uniform g like there's all the different we have most of the various distributions that you're probably looking for uh supported in this. Um, and all this is actually doing is just returning a method. So these all get added to your model automatically, right? And if you're looking at this, you might go, well, we still don't see where all of the compiling happens. Where is this big step that gives you all of the speed up? Um and I will tell you it is just whenever you leave this width statement you have to add no extra lines of code to your project it will automatically compile all of the all of the methods. So we will compile this advanced state method as soon as you leave this width block. Uh so you it becomes incredibly easy to read. You don't have to remember to call compile. You don't have to do any of that. It just happens automatically when you leave the width block. Um, another thing that can be useful if you're building really large models and uh have wanting to have nested functionality or repeated functionality, uh, you can make sub compon uh, subcontexts or subm models uh, inside of your main model. So here we have our components. Previously we talked about cables and how we would connect the two things together. Uh, this is the incredibly complicated syntax to do that. So basically here we have our Z1 excitatory neurons and we have their spike output or just S. Uh and then we wire it into the inputs of our weight matrix. That is it. There is nothing more that needs to be done with this. It just is basically shifting it just directly into there. Similarly, we can take the

outputs of Z uh Z our weight matrix here and we can shift it into $Z1$ inhibitory current matrix our current uh compartment and it just goes and if we want to do that fancy summation operation that we were talking about it is just summation with your compartment values and you can just keep tacking them on here. It's not two, it is as many of them as you want. Uh we also support products, we support negation, we support a lot of other um values here and you can also define your own mer you can define your own bundling mechanisms too. So now we've got wires, we've got components. Uh now the last thing that we talked about was a process. So here's how we make a process. This has a little bit more looking Python code, but it's incredibly uh repetitive. So here, so there's two different types of processes that we have. we have what are known as method processes which are basically a process whereas we are going to go over components and we are going to be calling transitions on them. So here we are going through all of the components of our model and we are calling their advanced state method. Now theoretically if we wanted to call something like let's just say we were using somebody else's component and they didn't name it advanced state they just named it ADV because they you know wanted to shorthand it you could just type ADV it will work these do not have to be the same name similarly the name of the component of the me of the process does not actually mean anything relative to these values here then uh what we might want to do is we have this advanced process which go does a forward pass over our model. And then let's just say we want to add that we have an evolve process which basically would be the equivalent of our update step over our model for all of our weights and so on. And we call that evolve down here. And then we want to join these together. So we only have to call one method to do both of them. Uh and that is what a joint process is. Is these are joining other processes together. And you can nest these as deep as you want. You can make multiple joint processes all in one. You can make you can call an advanced process multiple times in one of these if you wanted to do advance advance evolve advance advance evolve it will all work and it'll just keep going. So this is the processes setup and these also compile right we talked about that these also compile and similarly to the width block of our model as soon as we leave that width block. These are defined inside the width block. As soon as we leave that width block these also automatically compile. You don't have to worry about getting things to compile in the correct order. All of that is taken for care of it automatically for you. So what does this compiling actually look like under the hood? So here again we have our weight matrix for our forward for our forward step our advanced state and we're going to compile W uh $W1$ EI here. Now, it's important that we remember that the spike output of $Z1E$ is being wired with a cable

into W1EI's inputs. So, if we look up here, we have advanced state. The name changed a little bit. It's now circuit_W1i advanced state. That is because this is the name of our model. This is the name of our component. And this is the name of the method that is being compiled. And notice that the input has changed. The input is now ctx or it stands for context. Um so this is the global state being passed into this method. Then what happens is this self.outputs outputs gets changed to circuit W1EI's outputs is equal to and we just it's just this directly mapped over with we'll notice that the fixed value here becomes a global fixed value here and that self.inputs inputs because circuit Z1E.S because it's wired here. And so this just all goes through and so you give it a context. It mutates the context where it needs to and then it gives you back the context. So this is how we compile a method. Now for convenience sake of the presentation, this is uh I've got syntax highlighting on for this. Practically speaking, you will never actually see what is down here. This is all done dynamically. and store it entirely in memory unless you get it to print it out yourself and then if you want syntax highlighting you got to copy paste it into a Python file and get your ID to syntax highlight it. But practically speaking you never need to actually do anything with this. Um but it is however useful when you're debugging to look at this and go oh if this doesn't if this isn't updated if this is still just W1 EI's inputs we know that it's not wired to anything. And so then we could actually debug what's going on in our model by just looking at what it's compiling to which is incredibly useful um because if you just like forget to add a step uh you can see it because the output just doesn't have the step in it. Similarly if we have an evolve process so I did this one because it's got slightly less components in it where we have just W1 right so this is just a heavy in syninnapse that's doing an update. What we'll notice here is that here is our con our compiled process where we have a context and loop arguments. Now we're not using loop arguments but this is where you would actually give it uh parameters that you are passing into the forward step of your model. So this is things like dt. This is things like the current time. This is things like the image you're looking at and whatnot. And so then it goes through and then I don't have the original code here, but you'll have to take my word for it that this is compiles and looks correct. But importantly, we'll notice here we have this W1_MP compute update, right? What this here is this is another method that exists on the heavy and synapse for doing its update. Well, we also we we successfully capture all of those too. Uh you can make as many sub methods as you want as long as they all are just take you know they take in self they take in values it doesn't matter and you can call them and it all just builds together perfectly. So how does this actually look when you want to like use the process loop through the process what

are you doing? So uh for convenience you can call your model uh dot you can call your forward process get keywords. So for us, for this that'll give us t and that gives us Δt , right? Uh you don't have to do this u but this this will tell you all of the keywords that it is looking for to actually run your component uh run your process. Um and if you're missing any of them, it will complain at you. Uh and then we can pack all the inputs together, right? Because you know we're going to be looping through all of these. We need to actually make all of the inputs that we're going to call. Well, luckily um that's really easy cuz you just call pack the rows of all of it and you tell it the number of iterations and then you can do one of two things. You can give it a lambda function or just any function really uh that takes in x which is your iteration number and produces some value. The next thing you can do is you can also just give it constants. So Δt equals 1. And so what this will do if we say iterations is 10, this will give us an this will give us basically a matrix that has t is equal to 0 1 2 3 4 5 6 7 8 9. And it'll have Δt is equal to 1 at every single time step because it's a constant for the pass. And then what we can do is okay so we have all of our inputs now. Uh now we want to actually loop through all of those inputs. Well you can just call scan which is very similar to Jax's scan. if you've used it um if you've used any scan method I believe TensorFlow still has a scan method on in it um where basically it'll just loop through all of the inputs that you give it in effectively a for loop but it is doing this all compiled in C uh so we don't actually want to write our own for loop this is dramatically faster um and the output of that will give you hey this is our spike output which is something that we set when we compiled all our method together and our final state of our model that we can do something with because as it runs through it doesn't necessarily actually change your model. It just tells you this is how your model would change and then you can choose what you want to do with that. So there are some current papers being written with this some current applications that have been written with this. Um so we have a neural coding frameworks for learning generative models uh done by Auroria and Kefir. Um this one uh makes use of NGCLearn. uh we have an uh we have uh reproduced uh Dan Cook's model of unsupervised learning of digit reconstruction using spike timing dependent plasticity uh we have my own personal paper with my adviser called time integrated spike timing dependent plasticity uh that is also made in this model uh and then we have a contrastive signal dependent plasticity self-supervised learning and spiking neural circuits also done by my adviser Dr. in Arabia. Um, and you can find all of these models and more actually in a sister library uh known as the NGC Museum. Uh, so this is a library that our lab works very hard to keep up to date for at least our projects uh for the latest version of

uh NGCLearn and it houses all of the biomimetic and brand inspired computing. And if you write a paper in NGCLARN and you want to add it to the museum, we are happy to add it to the museum. Um, the whole point of it is to be a public place to share code because there's nothing more annoying than reading a paper, looking at the coefficients that they claim to have used, reproducing their work, and it not doing what it's supposed to. So, this is a place where we can store all of that. It's also a place where we have some historical works. Uh so that way if you want to get a baseline from some historical paper uh but don't want to implement it all yourself we can do that as well. And so with that V3 should be releasing later this month. Uh I will say with Thanksgiving it might get pushed back to early December but we are making great strides and it should be up soon. And with that I will open the floor to questions. >> Very cool. So people in live chat can ask questions. Um I wrote some down as well. So just to stay close with the theme of the talk, what would you summarize are the major differences between NGCLER 2 and 3. Uh so the main difference between NGC learn sorry it's going to flash for a moment is this right here. So in ngcarn 2, there was an very specific setup that you had to do when defining this advanced state method, and you couldn't use self, which meant that you had to do a bunch of pattern matching yourself to make sure that all of the names were the same and that all of the inputs were correct. Now, because we're making use of abstract syntax trees, you don't have to do any of that. So you can just throw compilable on a class method that has self all over it. Um, and we just extract that all away and turn it into the global context that you actually want to be referencing. >> Yeah. Awesome. That's a that's a big >> It's also faster. I don't know if I mentioned that. V uh V2 is roughly 10 to 20% faster. I'm sorry. V3 is roughly 10 to 20% faster than V2. um which on the grand scale of things isn't that much faster but it is faster. >> Yes. Very cool. So maybe just to to kind of ground that um in terms of of expert time or the size of models or networks that you could reliably render and execute. Where does this move the frontier? like what was is that was that a limiting step or was that preventing large systems from compiling or what? >> So I think one of the things that it really stops is I think the time element stops people from getting into the field. Um, pretty much everybody who has been working in the field for a while, um, has their own little custom whatever that took them 5 10 years to get, you know, to work and iterate on that gets them to something that's fast, right? Because they've taken the time to figure out how to speed it all up and optimize it. But if we have new students, if we have new like new PhD students, if we have students in a class, they don't have five years to optimize their codebase in order to do preliminary tests. Uh what we allow for that is we just allow you to kind of be a little hands-

off when it comes to how this is optimizing and becoming quick. Um I will also note that this does not require fancy GPUs to actually run quickly. Uh that 8 hour and 15 minute benchmark is done on my laptop with no GPU. That is the CPU speed up. Um the GPU speedups are equally as fast. Um but you don't need it uh until you run out of CPU memory which is always a constraint. >> Thanks. Okay, continuing with the questions. So does NGCL learn allow a different scheme or ontology other than the neuronal like you had the block diagram with the voltage current spike variables. So can you specify arbitrary stateful models in in that block? >> Yeah. So here where we're defining this this is just a class. So this right here is a weight matrix right. This is our dense weight matrix where we have we it's just class weight matrix and then you do have to inherit from Jack's component but like that's that just gives you all of the backbone that you need um but doesn't actually require you to do anything. Um you can define as many compartments as you want as few compartments as you want. You can define any number of fixed parameters that you want and you can define as many state transitions as you want with as many different names as you want. It will just all find them and work. Um the only thing I will say is that because this is a giant state machine. Um obviously if you have tens of thousands of changing values at every forward pass I it's going to get slow but that's not a fault of the library. Any library is going to get slow with that because just changing that many values in Python takes time. So, do those internal calls or nested calls have to remain within the NGC semantics or where would an external API call or another routine or program? How would that be brought into the design of what >> Jax's matrix multiplication call? This right here, it's just Jax's multiplication matrix multiplication call. You can just call them. >> It doesn't it doesn't care if it's not a class method. It doesn't try and do anything to it. So if it's not a class method, it'll just call it. Um, now I will say that if it calls something that mutates things that it probably like because this is all built on Jax, Jax generally speaking expects things not to be mutable. Um, now this is a copy of what is passed out above um, just from the way Jax works. So like we can mutate it here, but like you don't you want you want your methods to return values. Um but that is a Jax specific thing. Um but practically speaking here we have this is Jax's numpy ver uh form uh matrix multiplication call of these two inputs. Um inside of ngcarn we have an entire um ordinary differential equation backend uh that you can just reference like it's just ngcarn. OD whatever OD you want and it'll just work. And like that's not something that's like written specifically to work with the compiler. It just allows it to function like you it just doesn't care if it's not a class method. It's just it just works. >> Interesting. But it might hang for a while if there was call local llm method inside. Would it

spawn a whole Python thread? >> Uh so it's not necessarily going to spawn threads in that sense. But yeah, if so, for example, if I have uh if I'm doing a matrix multiplication of two gigantic matrices that is going to take like 30 seconds to do without any of NGC learn around it, just using Jax's optimized matrix multiplication with JIT compiling on a 30 by 30 m on like a a matrix that takes 30 seconds, it's still going to take 30 seconds. We are not speeding up the linear algebra. uh we are purely speeding up all of the Python that sits on top of the linear algebra because it's really slow. Writing values is really slow and so we are trying to make it as fast as possible without rewriting uh the accelerated linear algebra library or LAX uh which is what Jax is built on top of. It's the NumPy bindings for all of LAX which is what everything uses. If you come up with a linear algebra method that is faster than LAX, I highly recommend you tell them because they're entire mission is basically making it as fast as possible to do linear algebra on a computer. >> Cool. Interesting. Okay, a note in the chat and then you could add a comment or I'll ask another question. Note to audience members, NGCLearn does also support convolution, deconvolution, sparse synaptic transforms. etc. In addition to a bunch of historical spiking and ratecoded neuronal cell models, NGCLearn also supports generic iterative differential equation integration, i.e. Oiler midpoint RK4. >> Yeah. So I I assume that's coming from my adviser in the chat. Yes. >> Um >> yeah. So yeah, I don't I did not have space in this presentation to include all of the historical like all of the components that we support. But if you are used to using generic founded uh computational neuroscience components, there is a very very good chance that we already have them in the library and you don't have to write them yourself. So pretty all of the all of the and that includes both rate coding and spiking like you can do rate coded with this. Um I will say that it might be overkill for some rate coded models but it would work. Um, also technically speaking, you can do back propagation through this. Uh, Jax does do back propagation like like gradients and whatnot. Um, spiking neural nets themselves though are not differentiable. So, I'm not going to get into covering it here and we don't truly have a tutorial on it because it's not something that we're trying to do. Um, but it is possible to do it and I have gotten it to work fairly easily. >> Okay, I just have a few more questions. So often in in modeling whether sort of a monolithic >> mono composite type model or more of a synthetic intelligence approach like nodes that have different kinds of programs running there's this continuum between taking a bit of a more problem agnostic architectural approach like off-the-shelf neural network architectures or domain specific architectures and then just kind of throwing in the training data going from there. So in that situation the the semantics of your model and

what it does are being shaped kind of from a flat prior with the training data set. And then on the other hand, sometimes there's there's very situation driven modeling even into the toy modeling space where people are looking to encode the semantics of like anxiety level as a parameter itself that might be undergoing some kind of bayesian updates but on the interpretable parameter itself as opposed to that more machine learning driven approach without >> looking to constrain the how the topology of the initial network relates to the data set and the function. So when you're designing these kinds of systems, where does it exist between like a super nuance way to specify a neural network that's agnostic and trained sort of like a convolutional neural net or do you encode in more direct fewer parameters of interest?

>> So uh I'm going to say that you can do both. So we are specifically a library. We're not we're not gunning for a specific method here, right? I'm not here to tell you that method A is better than method B. Um so with this sort of thing with like my personal research and my use with this um a lot of the time is driven by I am looking at the so I've done a fair amount of work with reinforcement learning and similar spaces like that um in the context of computational neuroscience. Um and what that kind of leans to is very much thinking about well this is what my sensor data looks like. So, let's make sure that like the encoding scheme that we're using actually matches up with that cuz like sure I could go take some generically large LLM and just throw data at it and hope that it works. Um, but that's not necessarily going to be efficient. Uh, computational neuroscience as a field of research also has a sister field of hardware research called neuromorphic computing. uh which generally speaking uh imposes a lot more constraints on things. Uh so when I'm building models, I tend to work with a lot more constraints of physical constraints uh for like value types and you know magnitudes and the way information gets moved around. Um so you can definitely do it all the different ways here. I would say one of the nice things about ngcarn uh though from the standpoint of trying to come up with a new component for instance is the fact that you are just coming up with a new component. So, if let's just say I wanted to go out there and I wanted to make a new neuron, right? Because what I was looking at advocates for me making a new neuron. Um, you don't have to do a bunch of extra steps to get it to interface with everything else. Um, you just have to define the logic basically and wire it up correctly and it will just integrate into your model. I'm not going to tell you that it's going to work as far as like practical side goes, but as far as coding side goes, that's all you would have to do. I hope that answers your question. >> Yes. And then here's my last question. Um or semipen ultimate and then I'll see if there's any in the chat. But how do you see or go from the forward design phase and

generating synthetic data and outputs from a priori specification? How do you also work with the empirical parameter estimation? Whether you were looking to design a sort of model level property like learning or or are you fitting this to spike train data and looking to fit it at a more micro component level? >> Yeah. Yeah. So some of that so it's I'm going to use the the hallmark frame of it depends. Um so if we are in computational neuroscience, if we are looking at a toy problem that was not originally designed to be working with spiking nets, the biggest example of this is emnest, right? Emnest is a bunch of static images uh that have floating point values. Uh we don't have a bunch of we don't have like like we're not static models. Our outputs aren't static. Our outputs are just a bunch of spike trains. So we do have to represent that somehow. Um and so we uh it is something that in ngcarn we do try and offer some uh support for doing things like that uh for helping converting collecting whatnot um in the models that we've built in the museum uh that can show like yeah here is one way to do like clustering with this. This is one way to assign labels to output neurons. So we do that sort of thing is where it's like okay like we have to make the synthetic data effectively work to something. Um but the or I guess it's not quite synthetic but like we have to basically make our own labels um that mean something to us. Um and then on the follow-up side of that when it comes from like do we work forwards or backwards from either the real world and synthetic data or synthetic data towards the real world. Um I would say that good science does both. Um sometimes finding a real world problem that an idea fits to is challenging. Um and other times you have a real world problem that you're trying to solve. Um and therefore you definitely should be working backwards from it. Very cool. So now the actual last question any first you can add anything you'd like and just where are you at in your overall research journey? What are you looking to have done by the end of this phase or just what's next for NGCLearn and and how you want to develop and do that and other things. >> Yeah. So I am in my fifth year of my PhD. Uh so I am slated to defend my dissertation fall of next year. Um I really like working in the space of research tools. Uh it is something that I have a huge passion for doing. Um and so research tools and more recently specifically research tools for hardware applications. So I mentioned a little bit earlier uh there's the whole sister field of putting spiking neural networks onto physical hardware. Um I have done some research in that recently and it is a lot of fun and I enjoy doing it. Um as far as more immediate plans of stuff um there's a survey paper that we are currently working on that should be coming out soon uh in relation to spiking neural networks and computational neuroscience. Um, in addition to that, uh, coming out either December this year or January next year, uh, there should be a white

paper actually out for NGCLearn on its development, the thought process that went behind it and everything as long as well as that. Um, beyond that, um, I have writing my dissertation on the to-do list and probably one more one or two more research projects in there somewhere that we have not figured out yet. Great. Yeah. lot of exciting reasons to explore NGCLERN. I have just cloned into it on another computer. So I look forward to uh learning more about it.

>> If you clone into it right now, if you are using it through pip, it will be v2. If you clone it from the GitHub repo, if you just check out the v3 branch, uh you will get all of what we're currently working on. That is the dev branch for version 3. Um but the release branch should be updated within the month. >> All right. Okay. Thank you, William. Good luck with finishing your PhD. >> Thanks for having me. >> Bye. Okay, that was all of the live streamed presenters for the day. Now, there will be several pre-recorded videos. First pre-recorded video is Steven Wineer. Interceptive posture pictures for active inference informed neuromuscular skeletal NMS therapy. 43 minutes long. Here it is. Intrceptive posture pictures are a tool and a methodology for practical active inference informed neuromuscular skeletal therapy as well as for people who want to move well as they age. Intrceptive posture pictures IPs are a tool and a methodology for active inference informed neuromuscular skeletal therapy. Muscle, bones, joints being able to move well. which is why it's also an amazing tool for anyone who wants to move well as they age. I'm a neuromuscular clinician and educator and to me active inference is a framework to connect and shape people's mindsets towards healthy behaviors in both body and mind. Now I understand that AIF is part of a broader understanding for PhD scientists, researchers and others looking at how to model from a computer AI point of view as well as understand from a personal point of view how humans work in a whole new light. I believe that for clinicians, patients and the public, active inference can play a role in preventing the progression of I hurt my back back pain. and I overdid it. Neck pain, common stress, going from acute to chronic and especially when chronic low back pain is the leading cause of disability globally and neck pain and tech neck and normal daily stress is part of a sematic pain that a lot of people live with in their lives. Active inference can help clinicians and people help themselves. In addition, it's also a way to mitigate full risk for aging boomers, as well as to address mild concussions, things called MTBI or PPCS, persistent postconussive syndromes after accidents or sports injuries, as well as helping everyone including you to connect with your intrceptive sense and your sense of balance, your sense of your overall body alignment to have the zones of postural mass balance. balance and have those zones of postural mass, those posture zones move in everything you do. Balance, align, and move. We

call that BAM. My name is Dr. Steven Winger. I'm a doctor of chiropractic. I practice for 26 years. For the past 20 years, I've been lecturing around the world about teaching people to move well. I'm a diplomat of the International Academy of Neuromuscular Medicine. And I'm very proud. Last year 2024, I published my first peer-reviewed paper in Frontiers of Neuroscience. It really was a very cool thing because I had wanted to but never went into research and I had gotten a whole new understanding of what it takes to be a PhD level researcher. I've written books. I've lectured literally around the world. And when I saw the word cloud for this conference, the first thing that popped out to me was energy principle and clinical and the clinical application of this is something that I think it's tremendous and it's not yet been realized especially in NMS care. So the title of our paper, introspective posture awareness and accuracy, a novel photographic strategy towards making posture actionable came from my work teaching which works with well this is some of the disclosures that I do need to talk about. Bodyzone.com which is public health awareness and ideas largely focused on posture and posturemonth.org or which is a public health awareness campaign and strongposture.com which is a self-care campaign and tools for professionals mostly to give to patients to work along with those patients. My main interest now is ceup.com which is online continuing education for professionals. We teach doctors, therapists to incorporate the ideas of using a grid, the posture's own grid, which is a tool to objectify a simple posture picture. And we certify people with this in with posture practice. Disclosure, these are things that I do have an economic interest in. What you have an interest in is your posture. And everyone does. Posture isn't good or bad. That framing is wrong. It's old. Posture is functional which means it's strong or weak. And the overlay with active inference is that maximizing the symmetry means reducing the prediction error which saves energy. Energy is the thing that life is all about. Efficient movement that generates a fluid flow is not wasting energy. And every unique body selects their movement strategies that match their internal models expectation of what's going to occur next moment by moment. And that's why good static posture and dynamic motion are not the same for everybody because not only is your body different from mine, but our experiences are different in life and what we've done and who we are. I'd like to start to get you thinking about your body and how your body is basically working in what you do. So my question for you is as I stand here right now on an overall biomechanical basis and biomechanic means muscles and joints how they're working together right now is my body balanced. I've asked this question around the world to many audiences and people invariably say no he's not balanced. No. And you're probably saying that right now, but remember the

question on an overall biomechanical basis, is this body balanced? And the answer is of course it is because I'm not falling down. Which brings me to the point, your posture is how you balance your body. It's what you're doing to maintain verticality. Humans are the only oblate biped on the planet that is a mammal. birds also obligate bipeds some other creatures in different places but humans are the only mammals and mammals are unique because we've learned to do a number of things one of which is maintain our internal energy at the same temperature uniqueness of human bipedal moment-to-moment balance requires inference to stay vertical maintaining that takes a huge amount of the free energy that we have and conserving energy is a priority for anything that's alive. This is a law of nature. If there's an inferred prediction that has a higher energetic cost or a greater inefficiency, the organism is going to have a negative perception of that action and tend to shy away from it. When that persists for a while, it can be felt as stress, fatigue, pain, or just feeling bad. Human activities require accurate balance control. From simple standing to complex sports movement, falling down is something that we don't want to do. When you watch children learning how to walk, they fall down and they bounce back up because they're made differently. But as we develop, we become less resilient to falling down because we've learned how to stand tall. And our bodies develop in that direction. which is also why losing your balance in daily life is a early warning signal for problems breaking down your health. And you may say, "Well, I'm not old enough to have that problem yet." But if your balance is weakening, you're getting a big flashing yellow light to do something differently. And balance assessment of how your body is standing both statically and dynamically is a sensitive indicator of neurologic health. In 2012, maybe 2013, I saw a TED talk by Bud Craig and it got me interested in introspection. I picked up his book, read it, How Do You Feel? And it sounds like an emotional touchy feely book, but the subtitle really got me. It's an intrceptive moment with your neurobiological self. The premise of this based on a neuroanatomist that spent his time literally tracing where nerves in the body go is that the insular deep part of the brain is where your somatic feelings of your body emerge from. And there's a sense that drives this called intrception. And the next question becomes well what is intrception? I mentioned before that mammals are warm-blooded. Mammals have to control their internal temperature. And that began a process of controlling what's happening inside our body to different degrees for different mammals in different ways. Humans do it differently. And we've developed cognition in ways that have allowed us to socialize and to project our feelings from ourself onto other people and to have abstract thoughts about those in cognition. In our hope progression over this past six or seven

million years, we've developed a very unique cognition and very unique body and very unique abilities to connect with our body. and focusing on intraception is what the NIH the actually the NCC division of NIH in the 5-year plan from 2021 to 2025 said we need more research about intraception because it is a huge and unexplored factor in promoting effective health and in addressing many syndromes and diseases of modern life. the stress and the many problems developing from the incursion of technology into the activities of daily living. You probably know what I'm talking about. The definition of intraception from NIH is the representation of the internal world of an organism which includes the processes by which the organism senses, interprets, integrates, and regulates signals from within itself. Now, if you're on the biologic side of this, this is basic. If not, I'm going to do some really fast anatomy to give you a sense of why this is so important from a processing point of view. The insula develops in humans from the silven fissure, which is an infolding that's on the side of the brain. That line that you've seen going left to right is the sylvian fissure. When you were developing from an embryo, you went from one cell to a ball of cells to a tube of cells. And at the top, there was an explosion of cells that started to fold in. That stuff on the outside, the cells on the outside are the cortex. And that cortical infolding went down and got buried and became an anatomic structure called the insula. And because it's been so deeply buried, it's been hard to study. So anatomists have not really looked at it as we began mapping things. Well, some of the newer technology has let amazing insights occur about this over the past 20 25 years. And it's these insights are exploding. There's a ton of research going on about intraception because when you look at again you'll notice that above that horizontal line there's a groove going up and down called the central sulcus. That is the line between the blue and the red. And that line between the blue and the red is the line between the sensory cortex and the motor cortex. And the sensory cortex is where information coming in from. And the motor cortex is where information going out. On the coming in and the going out, there's what called a homunculus. Basically, it's a sematopic representation of neural connections. You may have seen pictures of a big nose and a big big lips, big eyes and small arms which represent the pointto-point correlation of the nerves in that part of the body. They're different for the information coming in and the motor control going out. That is why there's a different sensory homoculus and a different motor homoculus. roughly the same eyes, two eyes, one nose, but the proportions of the neural density is not the same. When you look underneath, when you open up that sylvian fissure and look down between the parietal lobe of the brain on top and the temporal lobe of the brain on the bottom, you look at what's underneath it. That's where you see the insula. And

the insula has a front and a back, an anterior and a posterior portion that controls things in the body. It is the most uniquely complex part of the human brain according to some people. And it's a really good argument because it ties things together. There's three different emunculi. It's not just only the sensory cortex. It's not just the motor cortex that has a representation or a pointtooint. It's also the insula. And that insulin has homeostatic as well as autonomic as well as allostatic input into the brain. If you know what allostatic is, it's another big thread in what's occurring in life sciences. And one way to look at it, which is conversation for different day, but if you know what homeostasis is, the ability to keep things level within the body, allostasis is to homeostasis as prediction is to reaction. Alistasis is predicting forward, which is one reason why active inference in my mind is destined to reshape a lot of life sciences. The lowhanging fruit almost is the biomechanical side of because when you start getting into some of the the hormonal issues, it becomes far more complicated. Oh, one more rather important thing from Bud Craig's book, emotional feelings emerge in the insula. Craig writes that the insula optimizes homeostatic sensory motor integration and homeostatic behaviors. Masio and Carvalo wrote a paper about intraception and the origin of feelings and they talk about how the intraceptive nervous system maps the homeostatic state of the body and that intraception is structurally and functionally different from extraception. Different pathways, different processing, everything's different. It's a different aspect of your sensing your body. It's a different way of reacting because of that different sensation coming in. Which is why a 2024 paper, what is the role of intraception in the symptom experience of people with chronic pain? A lot of words really well put together, but the bottom line is in common sense. People who are more accurately connected to their body's inner feelings have less pain less often. People who are inaccurately connected to their body's inner feelings tend to have more pain. This is why accuracy of sematic bodily prediction is so relevant from a pain point of view as well as from a addressing the causes of the pain and empowering the person to work with and connect with their body more effectively and to feel better about it. And that emotional side of it is one reason why intraception has seen far more attention in the psychological sciences working with emotions than in the physiologic sciences. One of the directions within muscle and joint back pain care has been what's called biocschosocial to work with the biological and the psychology and the social because these things affect how people deal with their pain on a simplistic level of I'm under stress because of bad home life to a far more complex level that ties directly into active inference of incorrect predictions because of priors, things that have happened before that then make me look at how the world is going to act towards me, which

then shapes how I act in the world. And these things are not conscious. They might be, but they don't have to be. In fact, it is a really strong argument to be made that active inference is how we feel. Emotions are inferences about causes of your bodily state, which is your body's best guess about, well, why you feel the way you do. It's okay for me to feel this way. And feelings in general are intraceptive predictions. Your brain predicts the state of your body. Now, pain can be predictive error signal. If something hurts, there's a torn tissue or a broken broken bone. That's a prediction that if I move, it's going to hurt. And that's true. When an expected and an actual body state doesn't match, you've got a prediction error. Feelings aren't passive reactions. They're active inferences on a moment-by-moment basis about what's happening inside your body. I'm afraid to do this. It's going to hurt. We'll try it like this. I thought it's going to hurt. It's going to hurt. Oh, it didn't hurt. When you feel your posture more accurately, when you feel the muscle tension and the mechanics of what's occurring or the pain more accurately, when you're experiencing your brain's predictions about your body more accurately, then those direct sensations coming into the nervous system are more accurate to reality and you tend to have less pain and you're empowered to do things that can then help you have even less pain. One of the biggest issues with musculoskeletal problems is people don't move because it hurts. So they become less able to move. So they move less. And this is a problem that plays out over a long period of time. But hopefully you've been living for a long period of time. And I know many of us hope to live for a much longer period of time. Interoception informs active inference. Which is why looking at how someone's balancing their body on two feet, their posture. I'm balancing like this. That says something from a biomechanical as well as a social point of view. I mean, our language says it. He's kind of shifty. He looks crooked. He's a standup guy. That two lit balance symmetry and accuracy of prediction has a dramatic effect. In fact, it's a proxy in my mind for how your one leg balance is expressed. If I'm balancing like this on one side or like this or like this, whatever it happens to be, those motor patterns will be directly reflective of how I move in everything I do. And this is one of the profound implications of active inference. Biomechanics is more than the physics. It's also the predictions and processing that's occurring within my body about how I'm going to move. The old view is that muscles respond to muscle loads and the nerves tell the muscles what to do and that's all there is to it. The active inference point of view is that muscles activate based upon the predicted load and they also preload in that prediction which means that well I'm just tight like this because your body is preloading based upon actions that have occurred between your predictions and what's happened and the predictions may be

stress maybe threat. The fact is that people that live with anticipatory muscle bracing, have a protective guarding based upon reactions are moving in patterns that initially will compensate would move differently which is one of the concepts we talk about that are functional. You can change it if you focus on it. But ultimately it adapts. This structure changes. When I use this muscle more and that muscle less, this muscle gets stronger, that muscle gets weaker. Your brain distributes the load based upon the capacity of your body. Is it the predicted capacity or the actual capacity? It's the predicted capacity, not what's actually possible. It's what you think you can do. Even after the injury, the heel tissues may keep on being protected because you've learned that pattern that outdated prediction is carrying forward in the express pattern of how you're moving your body. And that's why you end up with chronic load syndromes going on where you're putting more load on something. It I'm just like that way. That's the way I move. I don't know. I just that's just me. that asymmetry of function accelerates breakdown and asymmetrical development creating a negative spiral. That's the model that we look at and I believe that this is very much the active inference model that is creating a more accurate way of looking at the world. George box is a noted British statistician has a great quote all models are wrong but some are useful. We came out with the idea of standardizing how to take a picture of posture that many clinicians have used but adding something to it that one leg balance concept as a clinical framework that applies active inference principle to motor control assessment and exercise to enhance outcomes in neuromuscular skeletal rehab as well as chronic pain management. All you need is a camera. Take a picture in four standardized views. Looking at somebody from the front, from the side, and then from the front, again, balancing on one foot, standing tall, and that's important, and standing on the other foot. These ideas align very much with the word cloud that active inference is something that ties into the energy principle and it is practical from a clinical point of view. In addition, it's something that the public can do and that can be researched. And this is why I'm talking here right now because if you're doing any kind of research in the biologic sciences and you're working with subjects, adding a posture picture to what you're doing is potential window into their body's prediction from an intraceptive sense because that's where a lot of this information coming from. The senses for your body for introspection are not the same sensors as the senses for proprioception. How your body is moving. The goi tendon and the muscle spindles are dependent upon load and motion. But just feeling something is a different kind of a feeling inside. Intraception is thirst, hunger. It's one of the internal body feeling. Your body's sense of state is an internal body feeling and the nerves for that go on different pathways up the spinal

cord and it's processed in the insulin which integrates this information with other parts of the body before going out to the motor cortex to guide the motions on a more gross level. And the whole one leg balance idea is so important because your walking begins with balancing on one foot. Every step you take you're balancing on one foot and alternating. Which is why if you're doing that asymmetrically when you're standing statically you've burned in that pattern every step you take. That's why in my book stand taller live longer our second posture principle was balance. The first posture principle motion your body's designed to move. Active inference drives those movement patterns. Movement is about the whole body. Motion is about one arc of motion within the body. prior beliefs about how your body should move within a global movement unfolds from all the little motions of the joints in that kinetic chain of your body. And the predictions about the sensory consequences of the action drive what you do and the effort to avoid pain trumps minimizing prediction errors, minimizes energy use. Well, it's harder to walk like this, but it doesn't hurt when I walk like this. You'd be amazed to see how some people walk into the office when they're in an acute pain and what they manage to do to work out to have an affordance of some kind of a cane or a crutch in different directions. Oh, as a sidebar, I don't like canes. I don't like crutches used indiscriminately. I'd much prefer a staff to keep the body vertical than a cane to let the body fold forward because that tends to lead very much into now you have a scooter to go around and now you have a wheelchair to go around it. Effort keeps the body moving. Compensatory patterns about how your body tends to move is your body's best inference about how not to have pain and how to move given your body's current biomechanical constraints about the muscles what they've learned and the nerves how they've learned. In other words, functional compensation drives structural adaptation and then structural adaptation constrains further changes in patterns of motion that change functional compensation that change structural adaptation in how you move and how you balance in an ongoing spiral of those five concept. Taking a posture picture, seeing how somebody is standing with the instruction of stand tall, show me your best posture, take a step or two in place, just stand tall. And then taking a picture of that can be powerful to engage them in making a change in the body. At the end of this class, I'm going to show you two basic and simple exercises you can do to strengthen your balance and your body's alignment. The protocol is to first have someone stand tall with best posture. Take a step or two in place. Look at them from the front. Take a picture. Face to the left. Take a picture. Face back to the front. Have them stand tall. Lift the left foot up in the air and lift the right foot up in the air. It's really easy. One step at a time. And what you're looking at is where their head is, where their

torso, where their pelvis, and where their feet are. And we get into this in detail with clinicians. This is not the place for that. This is from the paper. And looking at those four zones of postural mass, how they're balancing and the symmetry, especially on this, the front view of left and right leg coming up look something like this. A simple set of pictures. This comes from a paper about changes in interception in mindbody therapies for chronic pain. This is a systematic review and meta analysis. Intraception is the perception of physical sensations from inside the body. When they're incorrect, there tends to be more pain. Correcting them reduces the pain. And if when you're looking at an intraceptive posture picture, if someone's always leaning to the left, if they're lifting their left foot and their right foot, and they're always leaning to the left, that's saying that their perception of where their body is in space isn't correct. Especially if they're not hurting. If they're hurting and it's compensatory, that's your turn. But when they shift the way that I am, that's saying that I have changed my body so that I am incorrect at where my body is in space. And that's why systematically focusing attention to strengthen that weak link, which is what we'll do at the end of this class for just a minute with two exercises can make such a big difference not just in how your body looks, but in how your body feel. Some examples. Young lady standing tall. Left and right leg coming up. She looks pretty good. Here's a young lady that's also standing tall. You'll notice the pattern of how she's bouncing her body on the left leg coming up and on the right leg coming up. She's going way off to her left side with her body with her right knee is going out when she's raising her right leg. Left side she's keeping much more vertical. The ability to maintain that verticality says something about how her body has learned to move at a very young age. She may have had an injury. This came from a clinician. Hopefully, the patients followed through and she's done well. But the fact that a child this young was brought to a doctor says something. By the way, look at that first picture. Look at the child's feet. How they're rotated off to one side when she's trying to stand tall. That gives you a hint that there's been some asymmetry trained in that body. And it's something that especially at a young age can be improved. Here's a young man told to stand tall. You'll notice he's internally rotating his hands so his palms are facing backwards. That's not standing tall. That's not attention. That's a child that's probably been using a whole lot of tech. A clinical posture picture is an experiential perception of that expressed verticality of I'm standing tall. And this is why it's an expression of information coming in from the outside and the inside. and your perceptions and predictions. The predictions part of this is so important. Correcting posture is not about telling someone to stand taller. It's about making the body feel the predictions differently and reprogramming that

step by step. This is why active inference is such a powerful concept. Those compensatory patterns, the things that can be changed are inferences about what's going to happen in the body. A lot of times this is called dysfunction. The simple way of looking at it, if someone's hurting, the waiting of the inputs coming into the body are shifted. You're not thinking about it. Your body just reflexively is changes. Your brain is such an amazing tool because it can adapt to different situations. Humans have a great adaptation potential. And there's an old saying, your greatest strength is your greatest weakness. Our flexibility means that we can be trained unintentionally into moving in ways that we don't want to be doing when we think about it rationally because that's what our body has been trained to do. Our attention focusing on one link at a time, one piece of the puzzle at a time can modulate biomechanic efficiency. It's not a one shot I fixed. It's one thing that can be changed. Which is why we talk about incorporating biobehavioral strategies, the NMS pros to help people live well in a tech world. Don't want to let technology ruin your brain or your body cuz they're connected. Before we go on to showing you exercises that I'd like you to do, I'd like to strongly encourage you have someone take your posture picture because the emotional impact of I thought I was standing tall from the front, from the side, and on one leg balance is powerful. Clinicians know it. I know it from working with thousands of people. And what I'd like to ask you to consider once you've done this for yourself, because if you are a scientist, you're looking at this from a scientific point of view, as we all should be. If there's any part of your research that you're using human subjects, I'd love to talk to you about helping you incorporate more systematic implementations of an interceptive posture picture as another way to get a window into how that body and individual is predicting actions for their body and how simple actions can change that by working with that person and engaging that person, by educating them about their body and how it moves. This is what we talk about in stand tall of longer. Your body is designed to move. That's the first posture principle, motion. The second is balance. Posture is how you're balancing your body and it's also how you're balancing your mind. A lot of the implications of the word posture is not physical. It's your view on the world. How you're looking at things. Where do you stand? And how you balance your body in space and over time is what we're retraining when we're focusing attention moment by moment on a portion of the body and how it relates and links to other parts of the body in order to balance. Because your body moves in patterns of motion that follow the path of least resistance of how you balance and align to move in patterns. If it hurts to move this way, if it's mechanically constrained, if that's the habit that's caused muscles to get tighter and weaker in one direction

and stronger in another direction, that's going to be the way that your body learns to move. Initially compensating you learn to move in the pattern that you teach it. But then over time where this becomes locked in things become stiffer. It doesn't move as well. Ultimately you can't move in that direction. I just move like this where changes in posture and changes in motion cause your body to physically change. It becomes a spiral. It keeps on building on itself over life. My firm experience and belief and hope and efforts drive towards the knowledge that active inference informed education linked with individualized self-care is an approach that builds both cognitive understanding that ties together things like pain science and cognitive behavior therapy on a base level. Not therapy from a point of view of your childhood and things like that, but being able to control your body in an iterative ongoing way, arc by arc to strengthen predictive accuracy to improve sensory perception to have more accurate processing and therefore prediction of the motor expression of how your body moves and then creating a way to track that just by taking an interceptive posture picture. We have a new course that I literally just finished. Integrating intraception and biomechanic prediction of where your body is is part of what we're creating as a biobehavioral approach to working with neuromuscular skeletal professional. It's also a way for psychological professionals that are working with chronic back pain to help and to be working with people who work with chronic back pain from a neuromuscular skeletal point of view. That's the big focus as well as neck pain and concussion issues. I'd like to show you two attentional focus exercises you can do yourself once you've taken a posture picture to see how your body is to try to connect your perception of how your whole body is balancing with the outcome of if you can balance better. You feel your body is balancing better and you can stand on one foot and stay there for more than 5 seconds, more than 10 seconds. And as quantified by an interceptive posture picture of observing the deviations of your body connecting that balance one with two parts of your body alignment where your head is and where your torso is. We call it forward head posture. You may call it tech neck. But if you spend a lot of time in front of your computer, which most scientists and researchers do, you'll probably find that your head is more forward of your body than your parents' head was when they were your age. in terms of some of the issues going on. If you take a picture of your kids, how they're standing and how their resting posture is as opposed to yours is now, and if your kid's posture is worse than your posture is, it'll give you some insight on where the world is going. This young lady is a model. She had no back pain, no neck pain, college student. We did a event taking posture pictures. I told her to stand tall, show me your best posture, and just relax. Stand tall. And this is what she did. Any way you

measure where the head is versus the torso, from the back margin, from the center of mass, her head is translating forward. And you'll notice it's also going up. It's translating up. And that translation up is what happens because the muscles in the back part of the head become shorter. they compensate shorter ultimately adapting shorter to restrain and constrain how the head is balanced on the torso. So looking at those four zones of postural mass, those four posture zones of head, torso, pelvis, lower extremity, the very first alignment exercise is wall lean. And it's really, really easy because all you're doing is having somebody lean against the wall with their feet a foot away from the wall so that their pelvis, their torso, and their head align with that vertical wall. You can do this at home with any vertical wall. But you want to take your feet away because you want to reduce the confounding variables of feeling where your head is over your torso. So, first aligning the pelvis and the torso against the wall and keeping the head level. Look straight ahead. What you're dialing into is your perception of is your head truly level in space. Have someone take a clear picture of your body from the side. It's a way to discern functional compensation from structural adaptation. So looking at her over here, you'll notice this. She was standing looking straight ahead, thinking that her head is level. If you looked at it from the back of her body, the margin from the back of her torso to the back of her head is quite a bit forward of it. And if you look at the face, it's looking up in the air. When I looked at her like this and I said, "Levelle your head and push it back towards the wall." That's what she did when I went up to her and said, "Pull your head more forward. Level your head and externally put her head to level and let her push it back to the wall." And I made her stop at the limit to explore the limit of that functionality, what the Marco blanket is, of that prediction and stopping there. This is where this body truly is. Then told her to take five breaths. She was able to do it with a lot of focus. And that focus on keeping it level at the limit is what it takes to change bodies. Having her do that for five slow breaths three times a day will make a difference if she's being good about it, if she's being correct about it. And that's why working with someone to help you, could be a family member, to keep your head level as you're pressing back gently, not hard. Don't force it. Because you want to remold tissue that's sensitive in the body, fascia. Don't push it. You want to be gentle. Gentle works far better than hard in this instance. You want to feel the tension. And when you breathe into it, it'll give a little bit. The imperative, what we call the must or the must's try, is to keep the head level. If you're looking at a side view, if your forehead is behind your chin, you're probably tilting up. If you're away from the wall, putting something behind your head, a pillow, an inflatable ball to get a feeling of where the wall is and pushing back against that very gently to

build the awareness to retrain that control. Awareness and control of your body in your environment is the center of the posturemonth.org or ACE campaign which is what we're behind to improve people's ability to connect with their body beginning with the awareness of the body and then strengthening the balance your posture just how you balance your body. This is what we're looking at with an intrceptive posture picture just looking at how someone's balancing. How do you strengthen that? Holding that for those same five breaths when you're balancing with your body aligned to your body's perception. So, finding a wall, standing tall, and dialing into what that feels like, and then locking in your knees, standing tall, engaging your core, open your chest up, look straight ahead, standing tall. Kind of feel how that feels. Touch the wall. And keeping your outside leg locked, standing tall. Simply lift the leg that's between you and the wall. And you don't want to lean into the wall. You don't want to lean away from the wall. You want to focus on standing tall, but you do want to barely touch the wall. You're going to want to say, "Okay, I can do it without it. I can do it without it." And the moment you do it without it, you've lost that feeling of inside your body. The moment you go from, I'm maintaining my balance, you're coming away from the feeling of standing tall, standing tall, touching the wall, lifting the leg, and barely touch the wall. How light can you touch the wall and standing tall? and focus on the standing tall because the information coming in to your body from that little bit of touch is the link to reality that lets you strengthen how you're standing that will then help you strengthen your balance. Tying those two links together. Simply tying those two links together is a huge first step in strengthening balance and alignment of your posture. And it's also an indicator that you can use when you take a posture picture, an intrceptive posture picture to validate the alignment of that objectively externally. Doing that along with people's bodies internally is a way to create changes in how people use their body in everything that they do. A lot of times people don't want to do this because I don't want to see it. It's too too scary. Herbert Spencer was a noted British philosopher and sociologist. The profoundest of all infidelities is the fear that the truth will be bad. I hope to hear from you. I hope to work with you. If you have any questions, drop me an email. My name is Dr. Steven Weiner. Thank you for having me here. And I look forward to working with the active inference community with a clinical perspective to help people stand taller and move well. Have a great day. All right, we're back. Next is Observer Theory by Sam Sanchol. This is a 31 minute pre-recorded video. Here it goes. Hi, big thank you to the Active Inference Institute for having me today. I'll be talking about observer theory in the context of books really. I'll start by outlining the history of these ideas then explain how this extension provides

a first stab at a general model for any observer sampling of computational universe. This framework proposes a resolution to discrete versus continuous dilemma. It handles cross domain causation i.e mind body problem and begins to quantify the novelty of surprise observers see. It also formalizes a terminal object that stops infinite regress in this model and enables sampling utilizing the category theoretic formulas developed by Gard and Riwal. This extension of observed theory is built on the foundational work of Steven War from Jonathan Goris and start with a big brief explanation of the ruler's physics project posits the underlying fabric of reality is a vast screen space of all possible computations. Observers like us sample reality from normal structure. Recent formalizations have frameless in category theoretic terms. A key insight is that observers are not godsight views i.e. there is no view from nowhere and there is no true objectivity. Wolfram has pointed out that observers leverage something called computational reducibility which basically means they pick out regular lumps in bru and equivalence relations between them to compress information. Essentially, we only see what our brain can simplify. Wolf from also emphasized that we act as if we're stable. We assume we persist through time and stay roughly in the same place, which let us treat space and time as coherent backdrops for anything we say about observation. These subconscious assumptions are exactly what we capture by positing this background object and by formalizing the observer in this From this model to get reality as we experience it, you need an observer to sample the brad. Here they formalized second order cybernetic systems within the structure. The rouad updates state to state and observers sample between state updates creating a single thread of experience. They coarse grain a wondering amount of computational complexity to something manageable which outputs everything you see around you. Furthermore, observers hold an internal model that they update through their cycling which helps them predict things within the real or in other words find pockets of computation reducibility. Notion of a background object is a bit vague so I'll clarify. We can use a simple example of an observer looking at the sky. That person will use fixed references like the horizon to interpret what they see. In physics, it's usually background independent. But here we introduce this object to capture what an observer takes as given. This could be their internal world model. It could be what they already produced from the ruby. Wer emphasizes that observers make strong assumptions. For instance, we assume we persist through time and observe roughly the same things every moment. This is akin to having some background reference that doesn't disappear. By explicitly modeling a background object, we codify the idea that observers always compare new inputs against unassumed stable context. Historically,

Wolf's writings on observer theory note that observers correspond to patches of regularity in reality. In other words, observers live on stable maps in the grand computational universe. Before introducing new ideas I'll summarize minimal observant. Think of it like a very basic robot or a thermostat. It has three jobs. One, set something from the outside. Two, store and update an internal record of what it says. And three, act on the world based on that. The authors led by RCA stress the simplest observer must have a B in distinguishing itself from the Safari. For example, the thermostat senses temperature, compares it to a set point in its memory, and turns a heater off. This act of observation is reduced to a feedback loop. In their words, a minimum observer perceives external states, updates internal configurations, and generates an action for a closed feedback loop. This simple model captures the functional essence of observation without any assumptions of consciousness or scale. It serves as a foundation for our larger model for generalized obser. Now embed this minimal observer into the full rub. In the extension we define an observer as a function s takes r to r . But what does that mean? Well, we know that Rudiad is the full category of every possible computational universe. But observers have limitations. They can only sample some parts of this. The fun XO captures this starting. It takes each state or transition in R and maps it to what the observer actually perceives. In effect, it forgets or forces details the observer lacks the capacity to distinguish between the result is row the rat for the observer and that's a function that and that takes the fun so and applies it to the full rad. It implies the limit of the accessible rulad for that observer. Different observers can have different functions and thus experience different worlds. In plain language, the fun encodes the observer's hardware and bikes reality. This makes explicit Wim's idea that our physics is a slice of the determined by us as observers. Next, we'll spell out the constraints to define the observer. Any observer has finite computational resources which we call boundedness. So it can only process phenomena of limited complexity. Formally, we sensate sample only if the bandness exceeds a threshold which could relate to a processing limit or an observer's memory. Likewise, instantaneous or two brief events vanish, we include a persistence threshold, meaning that the state must last long enough to register for the observer. Finally, an observer doesn't care about absolutely everything. It only notes relevant features. We encode this via a boolean. Only if the relevance is true is the information from that state integrated into the observer's internal model. For a human, we can take the example of a bright flash. It might be relevant to them in the dark, but random noise might not. Together Bandon's persistence and relevance filter the ruler and the observable rat to a much smaller set of possible states that the observer can act can actually

experience at a given time. Field of observation is simply what the observer actually perceives the time. It's a subset of the observable really that passes all three filters. In other words, this field contains exactly those states that are small enough, stable enough, and meaningful enough for the observer to see them. You can think of this field as the data in your current sensory bite. For a human, this field might consist of objects and colors you can see in your immediate field of view. It's not everything in the world. That would be the observer's total accessible reading. It's what you're effectively attending to. Now, this field is the basis for everything else in box. We define information content and information integration over this field. The diagram on the page shows that in the context of the RUDI ad. The pink lines show the field and the red dotted lines show the full Ruby ad that the observer could access. I as a subset and it's the bit they're focusing on right now. Now we quantify novelty or here surprise of each piece of observed data. We define the information content as the total entropy of a state X less whatever the observer can already predict or has compressed. You can also say made computationally reducible from prior observations. So the information content is essentially the net new information the sampling X offers. If it's zero, the observer gains no new information. If it's large, then it is novel or surprising. I this directly mirrors FEP but generalizes it to any observer. In other words, our observer is trying to minimize the information content on average. We tend to seek environments where surprise is limited. Why is that? It saves some of our bounded computational power. We'll use this information content to build a measure of the overall information the observer sites. Get a big picture of sense of the observer's experience. We define integrated information as the sum of the information content over everything the observer's view. The formula tells us how much new information the observer has coherently integrated at this moment and the moment being the state update time within the frame model. It's analogist to integrated information in consciousness studies but again applies to any class of observer. If integrated information is high the observer is experiencing a highly structured interrelated set of inputs. If it's low, the inputs appear to be more random or distributed. Integrated information aims for a concrete observer centered measure of how complex and unified the observed information content is. In practice, maximizing integrated information in this formalism is seen as an observer's general telos akin anthropomorphically to having a rich and meaningful experience of the world. One of the biggest puzzles is how discrete computation can give rise to the continuous space and time we experience dayto-day. Our answer is that the continuity is an observer con is observer constructed to save the computational boundedness of a given observer. We formalize this by

giving each observer a measure indicated by U_0 over its field. Intuitively, this weighs different samples according to the observer's sensitivity to them. For example, humans weigh central vision more heavily than peripheral. Consequently, the observer's perceived continuum is given by integrating the discrete information content against the measure. In simple terms, the brakes filling in the gaps. This aims to resolve the discrete continuous gaping of the model. The underlying road is all frames and the observer's mind interpolates to get a smooth movie. Wolf noted that when done properly there's no conflict. Space time could be a construct of observers called spring. So with this we can comfortably use continuous math in the observer's perspective knowing it emerges from underlying discrete expiry measure function. making the scoop of continuous idea more. Let's look at a few anchorages. Imagine watching a movie the film is a sequence of static frames, but because they're shown rapidly, you the watcher in the cinema perceive few fluid motion. Similarly, the road provides discrete frames of reality in its hypograph and the observer's mind generates motion between them. Another example is the blind screen vision. Each eye has a spot with no photo receptors where the optic nerve leaves directly. If you position an object there, it disappears. You instead just see the back if nothing missing. Your brain literally fills in That's the observer theory extension in action. We don't perceive raw data. We perceive an interpreted scene. You could also vote Newton's bucket. We see water's surface fall under rotation because our perception assumes a fixed background. These examples illustrate the continuity and cohes in the formalism. This is captured by the measure integrating the discrete samples. The theory posits that continuous reality is an observed construction to save Q . Let's move up from the mechanics of observing constructing continuity to what that continuity produces the layer domains of reality as experienced by our observers. The paper proposes the observer's operationalized continuity not in a single domain but across four nested layers each expanding in degrees of representational freedom and coherence. In short in simple language this means that higher domains are more informationally dense and back by less rules than lower domains. It should be noted at this point that this decomposition can be done in any way a researcher chooses even within the domains that I posit or for entirely different sets of domains suitable for other classes of service. The following decomposition utilizes humans cognitive layers i.e. it's an morphing but others could work. I'll walk through the layers one by one. The layer with the most rules is the physical dome. Here computation appears tightly coupled with causality and energy conservation. Think of classical mechanics field theory and graphs. Everything is closed and coherent and information of evolution aligns with notions like closure. This is

where standard physical computation lives. where off physics lives at least until you get back to the point. Next, we'll move to what we might call proto subjective space. The valuation or domain into these systems with observerdriven sailings such as attention, prioritization or inference. This is where concepts like active inference emerge. Here the observer doesn't just track physical states but values complexity creating meaning from a computational structure. The third layer is the symbolic and it's the world of form abstraction. The symbolic domain allows for rule-based transformations independent of their instantiation. Things like language, formal logic, and mathematics. Observers in this domain don't just respond to inputs but create representational continuity chaining symbols into expressions, proofs and meaning structures that can be detached from any one physical or emotional such valuational context. For observers like us, this domain enables coherence across the physical valuation documents. It can describe physical systems or encode structures. is the seed of meta theoretic insight. Lastly, we reach the edge, a space of maximal modeling freedom that we've that we've turned the minimally constrained domain. This is a domain where the usual coherence rules of symbolic logic break down. It's a place where we can talk about paradoxes, infinite regress, nonwellment sets, things that we can imagine but that we can't really instantiate or can't hopefully describe in words. It approaches the boundary of what we can access. From a categorical standpoint, this is the domain that that gets closest to topological closure. Observers are capable of traversing all paths and equivalences subject to identity even those incompatible in lower domains. This is also where our categorical terminal object true affinity enters. It's a poetic limit object that closes the space of coherent interpretation. The hierarchical layering is crucial for modeling cross domain observers. one to for example infer value from physics or deploy symbolic logic to manage valuation hierarchies. It also allows us to rigorously define how continuity and therefore computation means something different depending on the specific obser focus. Why do we talk about the hierarchies? It's because one of the central aims of the extension is to show that multimatic causation is natural in an observer in the general obser framework. There's no magic because all levels observers like us have access to are embedded in a single structure. We really have a single causal chain and thread through all of those domains. We attempt a formalization across domain causation for a state D in domain I and a state P in domain J . There is a causal relationship if we can find a path of transitions in the observable area connecting state T to state P . The path is a sequence of states across domains. In categorical language, we have a process category where objects are states and morphisms are processes. A morphism from state P to state P spanning different

domains is a cross influence. So as an example, a decision you make in the mental domain either with valuation or symbolic logic can abomorphically lead to a muscle muscular contraction in the physical domain via the composition of funus between those domains. Think about it as a map connecting to another map within one giant. There's nothing a causal about it. is just one connected multi-step morphism in the whole of observer constructed In the observer theory extension, causation has two main flavors. Formal causation corresponds to a higher level pattern from a less constrained domain imposing constraints on a lower level. For instance, the blueprint of an intention shaping fiscal act. You could use the example of an architect's plans for a building that don't instantiate specific materials, which is the shape of the structure. That limits what those builders in the physical domain can do to put it together, but it doesn't determine every piece of that puzzle. Efficient causation is the usual physics style cause effect like you're expiring. Mathematically we saw causor morphisms by how their constraint levels compare i.e. how many rules are required to output said morphism. If a state has less constraint and goes to one with more we'll call it formal course. If it's the other way around it's efficient. This picture shows clearly that there are just two directions of the of one unified structure. Crucially, mind body effects are simply morphisms in the observable ruler involving different domains. We don't need new laws. We don't need any magic. For example, deciding to pick up a cup. A mental domain morphism gets met by functions through the symbolic and then the physical domains eventually. The laws of physics are untouched. We're just following morphisms in the rulad under a particular observer's mapping. Recently Eric V has formalized a mathematical model of the interactions of micro macros scale causation that we feel supports this view. Let me tie these ideas together. The core new quantity we have is the information content an observer gains with state X. This is explicitly tied to how an observer reduces uncertainty. It generalizes for instance and others ideas and minim. An observer will naturally prefer observations that minimize a new information because they are cheaper computationally and enable more of their bounded resources to have availability to search more potentially increasing global and local integrated information. The sum of integrated information tells us how rich the experience is. This makes observer dependent novelty precise for any class of observer. What is new to one observer may be old to another because each has a different sampling function even if they share the same observable rad. For in through intuition you can think of identical twins. They have the same observer really add and exactly the same bodies but their divergent cause histories mean their sampling in front of them works differently. Thus this framework lets us begin to quantify novelty and surprise in a

way that depends on the individual observer filters. Finally we tackle the question of closure. Is there an infinite hall of mirrors of observers observing observers? We answer by introducing a special object called true infinity which plays the role of a terminal object in our textbook. Concrete TI has the property that every observer maps uniquely to it. In everyday language, you can think of this as the ultimate perspective. Our mind of God is not reachable or observable by any finite observer. But here it's the source of all rules that generate the ruby. It provides a big end of the line. So we're not stuck with an endless hierarchy that's computationally intractable. I at some point everything just makes sense. As we summarize in the paper, TI acts like an undershirt viewpoint or a unification or integration of all underlying information. This ensures the framework is mathematically complete. Every diagram has a limit and there's no paradox of what observes the observe. True infinity is essentially the background frame or ground truth that all observers implicitly assume but can't prove or fully attain. Does it feel like conceptually? Here it's a placeholder for ultimate computational completion. Imagine that every computational process and every observer is eventually seen from some advantage. True infinity is the vantage where everything comes together. You could call it the first mover or even a godlike viewpoint. But in this formalism is strictly a mathematical object. Because of it, our model doesn't have two sense. There is an absolute frame hidden in the background, an unachievable fixed ground against which everything else is defined. It shuts down the infinite regress problem. Nothing is left angly and philosophy. There's an ultimate to the observed matter. So it serves as both poetic closure and a precise mathematical capstone of a framework that aligns with the Garden RC formalism on the construction of the Rubad. Simply put, closure in the formal model enables the Rubad to generate geometry and then through a long chain generate physics. Without it, we don't have a structure we can do things on. It is no longer an infinite group. is an surprise the observ author's view treated as stere by formalizing observers of a broad computational universe and by introducing observer dependent measures and information metrics we can start to reconcile subject and object In our view, it doesn't introduce any contradictions with known physics. Instead, explains why macros world looks from our point. I believe this new tool to think about how consciousness and business complain sandbox move forward. This framework and guided segments in neuroscience or in the design of AI about how build reality. As just one example, if you increase observers computational boundedness or increase it, you could use say height speed, we could expand it observable. We might see phenomenology without particles. The key takeaway is that is that observation is fundamental computation. By

modeling themselves formally we can better understand physics, cognition and their intersection. It still blows my mind 100 years after Q there is still no general formal observable shortly I'm publishing a new paper that extends the extends the ideas of this paper of this model to theology ethics telos and human superructures met through Last I want to thank James Steven for all that help enjoy the All right. Sorry about the audio issues on that last one. Thank you for that very interesting talk. All right. One second before the next one. My name is All right, next up is exerceptive bias belief updating in the global model Jim Freda. This is a 51minute pre-recorded video. My name is Jim Freda and I am very very pleased to be here for the 2025 applied active inference symposium. I want to thank the organizers and for being so open in their acceptance policy and letting someone like me present. I am a deep tissue structural body worker, a manual therapist, an embodiment teacher and a somatic historian and social theorist. My interest is not in uh computation, computational neuroscience, mathematics, um computer science and um physics. But I have a deep interest in a model of life and especially in the perceptual mechanics of a model of life from a somatic perspective. My interest in active inference derives from my focus on what is called the deep core myofascial synergy which is the anatomy the synergistic and very uh new anatomy of our axial line our postural core. Sherington describes this system as proprioception or self-awareness. And when I came across Sherington's u marvelous text, the integrative action of the nervous system and his neural architecture of the animal as a whole. It really blew me away. It really changed everything about how I was working with people, working with myself, and understanding my own self-image and the world. When I came across Carl Friston's version, updated version of the model, I was really blown away, very excited, and I just love the work that the active inference community is doing. I do not have computational, neuroscientific, uh, computer scientific kinds of critiques and suggestions to offer. But my field is anatomy and from the perspective of anatomy and especially the self-organizing, selfaware, selfbalancing, self organizing and self-directed generative anatomy of the human body. I do have a lot to say. This is a new kind of anatomy. Um, for the purposes of today's talk, I'm calling it cybernetic anatomy in the cybernetic tradition or we can call it synergistic myofascial anatomy. And this is really the key to understanding the motoric element of the deep field in the model. The motoric element of the physical homeostatic function of active inference itself. We're going to talk about a few major points here. I'm going to go over in addition to some guided body meditations that I hope that you will find novel and um enlightening. We're going to spend some time reviewing Sherington's original model, the

neural architecture of the animal as a whole. I have a big section on that. There's an awful lot of text. I'm going to summarize it, but I put all the text there so that you can go back and do your research because a lot of the belief updating that we're going to be doing in this talk today involves the major contours the global outline of the superordinate model which is three receptor fields exteroception, interoception and proprioception. Now the core node proprioception I argue has been under theorized and proprioception is the least well understood of the three different perceptual systems. The model we understand contains two blankets. But I want to give the inner blanket the outer surface of the inner self in the nucleus. What would be the very importantly mechanistic nuclear membrane? The lamina membrane of the outer surface of the inner component of the cell as an intuitive uh intuitive example of of a model of the model. uh that is uh by analogy uh we can also find in our cells in our body. This motoric element has been under theorized and it's actually missing from the model. Proprioception lacks a motoric element. The deep field lacks motoric anatomy. we're going to find is exteroceptive bias. And we're going to talk about that in terms of the blanket politics in the model that if these two perceptual fields have agency and um self have identity which it's clear that they do and theoretically which they must then we can talk about their relationship in terms of blanket politics. And I'm going to call up call forth both Carl Friston's disconnection hypothesis and Gregory Bateson's double bind as a way to discuss that. I'd like to invite you into a guided meditation on on the grip on how to get a grip on the reach. The reach, the arm, the forlim of any animal is something that we study to understand its unique adaptive uh potential in its environment. The human upper extremity, the human form, the arm and the hand of course are are very very special and allow us a tremendous amount of control and sensitivity uh within our environment. And this has been studied hundreds and thousands of times by motor control theorists and neurologists and and uh biomechanists and it's something that we should really be very familiar with. Out here in the hand is the exteroceptive grip. What I want us to focus on today is the proprioceptive prior which is located in the armpit. Now that armpit is formed by a single large bone called a scapula. Now this scapula you'll notice is shaped exactly like a grip. And your hand fits around it perfectly. And indeed there are five intrinsic muscles arrayed over this that literally grab onto it above and below with their tendinous insertions wrapping around the the the shoulder joint the ro forming the rotator cuff. And this is the system that anchors the arm to the body, but also that manages that originary kind of joint, the shoulder joint that manages it and maintains centration and balance and stability in that very insecure joint. So, we're going to reach out with our hand and if

there's someone there, grasp their hand gently. If there isn't someone there that if you have a cell phone or a coffee cup, grasp that and feel how your little finger is like a hook that secures that object in your hand. How the thumb coming over the top is a kind of a clamp and how you get a grip in the exterepive system reaching out into the world. But even though this is the applied active inference conference, I'm going to suggest that we come back, not reach outward, but reach inward into the armpit, not the shoulder. Now, this is again an extraceptive knuckle, a superficial structure. Um, and a rather insignificant motorically speaking and perceptually speaking structure that is not well innervated. We're going to come into this very sensitive armpit. The armpit is a product of the scapula's attachment on the back of the ribs. those proprioceptive movements, postural movements are long, slow, continuous and mild. They really are the expression of active inference in the mortal substrate. So we have this grip in the hand with a parallel or a mirror synergy an inverse homology in the scapula in the armpit and see if you can feel that and just as an exercise let's organize our arm with our shoulder. Let's use the neck and the shoulder to lift the arm which is very common and results in all kinds of pathologies like impingement syndrome and frozen shoulder. Instead, let's come down deep within the armpit and bring that elbow in contact with the ribs and come up into the concavity, the sensitive pit of the the gradient of the armpit. and feel a hand. Literally imagine that as you have a hand out in the world, you have a hand reaching inward in the body. A literal hand grasping that scapula. Now that hand, that sense of a grip wraps around the bony scapula with its fingertips, the tendonous fingertips of those muscles wrapping around the shoulder joint like fingertips around a doorork knob forming the rotator cuff and giving it giving that insecured glenoumeral joint the stability and centration that it needs for this amazing range of motion. that we have that comes from deep within the proprioceptive prior of the armpit of the concavity. And indeed the body has a system of major concavities on the axial line all with this kind of a function. And those five intrinsic muscles really create a synergy, a mirror synergy, an inverse homology that the brain finds very easy to work with. Now the last thing that I want to say is to come out of that meditation and to look here at this theory. These are some recent very interesting motor control studies of the reach and the grasp. And you'll see here outlined in green, I've highlighted a graphic from each study. Now, this is universal. Every single motor control study originates in the shoulder, the knuckle of the shoulder on the one hand and on the other hand in the in the grip and attempts to calculate proprioception from this very false Instead, if we reach deeper within and understand the motoric anatomy of the proprioceptive of the deep field of proprioception, I wonder how that

would change our studies and our research, our thinking about proprioceptive prior. We're going to take a few minutes here to do a deep dive and I have a number of slides coming up with a lot of content. We're going to do a deep dive into the model and we're going to go back to the origin, the 1906 origin, Charles Sherrington's neural architecture of the animal as a whole. And it's Charles Sherrington. Charles Sherrington was a great scholar, a tremendous scientist and a man of letters. And when he named something, he really thought about it and his names have weight. And we've gone and really should have renamed or reinterpreted/redefined the neural architecture of the animal as a whole without considering the original model itself. So, we're going to spend some time going into that. Exteroception, interoception, and proprioception were all words that Sherrington created with a great deal of intentionality and rationale. Now it's a perceptual model but it's a spatial model which is great for somatic thinkers like myself because kinesthesia the muscle sense somatics is a spatial form of awareness where we're not looking to ask why or how but we're asking where where do we go in our experience with our perceptual faculties in order to encounter those affordances in the environment and within ourselves that possess the information theoretic content that we need. This is the nature of living systems and working with complex self-organizing and self-aware living systems. Now the model has two great separate fields. There's an outside and an inside. The debate really is what's the inside. And what modern scholarship has done is taken interoception, exteroception, interoception and made the very simplistic and naive assumption that well that's the outside and the inside must mean inside. And this isn't really true. Sherrington referred to the deep field repeatedly and proprioception is the deep field or the core in the model. Now, um I'm going to go ahead and read this third paragraph. Multicellular animals regarded broadly throughout a vast range of animal forms are cellular masses which present to the environment a surface sheet of cells and underneath that surface sheet a bulk made up of cells more or less screened from the environment by the surface sheet overlying them. Many of the agencies which the environment by which the environment acts on the organism do not penetrate to the mass of cells forming the organism's deeper part. The various stimuli expend themselves as stimuli at the surface sheet and do not penetrate into the depth of the organism. So Sherrington is really trying to talk about a part of the body that is insulated most fully from the outside world, the deep field, the core proprioception. And he identified that as a mechanical system, a muscular system, the postural system, compressions and strains produced by muscles. Not only that, but not only that, but this inner receptor field perceptual system is not other aware. It is most self-aware. Let's take a look at Sherrington's

arguments around interoception and the surface field again. The surface field is broadly subdivisible. Its subdivisions are too. So the external field is cutaneous. But the inner surface then would be elementary. Anything that passes through the boundary sheet of an organism has to be processed for example by the cytoplasm in the cell by the elementary canal by the visca by the gut what Sherrington called the kitchen. This is not the deep field. This is still something that although shielded from the outside still is basically working with processing sensing stimuli that come from the outside. It's important to recognize, I think, and to take a moment with this bottom paragraph to just go ahead and look that external is special and important in a way in that, and these words I think are really, really nice. It's facing outward on the general environment and it receives freely and it receives and has received for countless ages the full stream I love this language the full stream of all the varied agencies forever pouring on it freely from the outside world so there's a bit of a basis with exteroceptive bias in terms of the perceptual fields exteroception has to always be operative from the moment moment of birth onward. And this is one good explanation for bias in the system. But the real bias that we're facing is an accentuation of of proprioception and a diminishment of proprioception. So let's look at Sherrington. The stimuli effective on the receptors of the deep field differ fundamentally from those on either subdivision of the surface field. They are adapted for excitation by changes going forward in the or organism itself. Now in interoception in the gut the excitations are coming from um the the red pepper and the garlic that you might have had with your spaghetti. But in the deep field it really is the contractions and relaxations of muscles of the postural system hugging the core, hugging the spine itself. which are the basis the priors the postural basis for the rest of the actions of the body. Since in the deep field the stimuli to the receptors are delivered by the organism itself the deep receptors may be termed proprioceptors meaning self-awareness and the deep field of field of proprioception. Now this is a postural and a tonic system and we've discussed that but it's tightly interwoven not with interoception but with exteroception maintains tonic reactions in the skeletal musculature. Sherrington describes these wonderfully steady, long, continued, untiring and for the most part mild contractions and this is known as their tone. Tone. Sherrington spends a long time talking about tone. The muscle sense was a major topic of 19th century um explorations into perception but also into the physics of life. And I love what Sherrington uh says and he says that um tone serves as an adjuvant to so-called to the muscle sense. in the literature since Sherrington these terms have gotten twisted around quite a bit and there's a real consensus in the literature. Um these are a number of quotes here from different

serious studies and scholars making the point for example that proprioception predominates as the most misused term within the sensory motor system. Uh jumping down um proprioception is one of the least understood senses yet fundamental for control of movement and foot to the bottom. Some major textbooks claiming to cover neuroscience barely mention it or mention it only in passing. But the truth is that if you do a search online, you're not going to get the proprioception that Sharington just described, you're going to get a definition of it as positional awareness. Typically, say in the elbow or in the extremities. That's a very different creature. Positional awareness is a very different creature than self awareness and then what of interoception? Now interoception has been redefined as comprehensive of all inner sense which then includes proprioception or proprioception as positional awareness proprioceptors might be considered interoceptors. Sarius says in a in a major study um interoception then refers to the perception of the physiological condition. Now what's happened is a very simple reinterpretation and none of these studies engage sharing more than for a few lines or a few paragraphs at the most. But interoception has simply been redefined to mean inner sensation. And we now understand from Sharington that was not his intent. That does not make sense. The cyto cytoplasm is not constitutive of the inner self of the cell. That would be the nucleus. interoception is not a source of self-awareness. It's not realistic to expect from it a selfidentity. Exteroception however is. Of course, our sense of our outer sense of identity is quite rich and complete. But that inner sense of identity that comes from the functional unity, the bodywide axial core also functions at the level of of of a self identity and has the neural the neural mass to support that kind of a hypothesis in the cerebellum, the brain stem and the spinal cord. Now, I just want to take a moment here and and I'm sorry if that was rough, but I want to take a little bit of a moment, a few minutes of your time to explore this in the body. And we're just going to look at those three fields. And let's start on the outside just to rehearse the the marov blanket, the outer surface of the surface sheet. with the five senses, with the extremities, with the skin, with our distance receptors. And we can look around and we can hear the cars far off in the distance or children playing. We can touch the things in our environment and touch ourselves on the outside. We have so many ways of relating to the outer world that it can be overwhelming and and allconsuming. The importance of going deep within is so important. So we're going to start that process at the inner surface of the outer boundary. Naturally, we're going to start that process. We can start in the mouth and in the nose. Inhaling, embibing, coming down the esophagus, feeling the blood, feeling the heart, feeling the lungs, and all of the awareness that exists in the gut. That's interoception. Now

that doesn't go all the way to the core, but it does go all the way to the other end of our tubular vertebrae tors torso with its cordate body plant to the excretory system. What comes in and comes out is a matter of interoception but is part of the surface field. Next, we're going to start again here in the tongue. And we're going to identify that muscular system of the tongue running back in the back of the throat, in the back of the chest, behind the heart, in the front of the spine, behind the lungs, then expanding suddenly here into the whole diaphragm and coming down through the gut. Not in the viscera, behind the viscera, in the motoric core, the postural core, the sox muscles running down through the pelvic basin and the groin through the medial thighs and the adductors down the legs around the medial ankle and the medial arch to the big toe and the surface of the feet. a bodywide myofascial motoric muscular continuity. This deep inner field proprioception, the postural system, the deep core doesn't contain these distance receptors. It doesn't have visual and verbal and auditory awareness. It has haptic and muscular awareness which nevertheless is tremendously rich. And this is something that we're going to encounter all day every day. It is a daily part of our life. But if you go into a yoga studio, then that's that's what you're going to experience is that system as you move in those long slow continuous tonic movements. I have been I have been arguing that the core node has been under theorized in the model and that the most important aspect of that core node is the motoric anatomy of the deep field in the body. Um the deep field is the postural system and is the source of physical homeostasis. And if we do not have a clear anatomical understanding of that system, then I I think it's reasonable to say it's under theorized. it's only a recent discovery that the nucleus has a nucleoskeleton and the importance of the mechanotransductive elements in the nuclear envelope. it's also a recent discovery of the the the the discovery of the myofascial synergistic nature of the body is a recent discovery and we can see in these other pictures here the core we can see this one circular this cutaway diagram of the abdomen. Now we can see the outer blanket of the abdominal muscles of the core. The traditional core that we think of on the outside of the body is complemented by an inner core that surrounds and supports and protects and expresses its neural content. The spinal cord this inner core system is the myofascial synergy that I've been describing. so I want to talk about two blankets, a double blanket theory. I want to propose that there are two blankets in the model and that they have a relationship that is problematic. We can call it blanket politics. We can think about that problematic relationship between these two systems in the body in terms of for instance disconnection hypothesis because poor integration between these two major perceptual systems is going to have effects in terms of neuromodulation and sensory integration. Now

I'm not going to discuss that but I am going to suggest that Bateson's notion of a double bind is relevant here. Now let's look at a double bind. A double bind is a situation in which there's a victim who is exposed to multiple contradictory injunctions within a kind of a power dynamic a power differential where escape is barred. there's a dependency. Now the result of that is silence and immobilization and and we can see that in the body. Now I think it's going to take a lot of work to really flesh that out so to speak but the double mind we can think of that as a relationship say within a family between a parent and a child and let's say there's a problematic relationship. Now the child is sick and is has recovered, is getting better and the parent visits in the hospital and the child gets out of bed and wants to embrace their parent and share their their happiness and their their gratitude, but the parent backs away and signals, "No, we're not going to hug. We don't touch." And the child is immediately despondent and perhaps a little bit ashamed. And then the parent having given one contradictory signal gives the next and chides the child saying something like it's perfectly what's wrong with you? It's perfectly natural to love your parent. It's perfectly natural to want to express your joy. Why are you ashamed of yourself? And so then it puts the child this dependent being who can't escape in a situation where they they can't do and they can't not do they can't speak but they can't not speak a double bind and this the result then would be immobilization and silence and I think that's an interesting metaphor for this outer blanket the excerpt system with all of its power and it's enveloping presence and this more dependent inner very sensitive um perceptual system which nevertheless has its own impulses and needs and which can be at odds. They can be at odds with each other and so a lack of integration then we would call that the double bind. Not sure really how to work with that, but I find that to be productive in terms of my own working out my own inner blanket politics, so to speak. The solution, what's the solution? Well, first to identify that dynamic to be able to step outside of it, at least conceptually, but also empowerment to give that victim agency. And one of the uh problems about under theorization of the core node is that we don't have that we don't have that empowerment and agency at the level that we really need it for belief updating in the global model. Belief updating in the global model then just to be really simple about it. There's a lot of text here, but the global model is going to be the superordinate level of the model, which would be the model itself, as we've already discussed, and its three fields rather than some subsystem within it. It's going to be the the level of the individual organism, the person, the self, and their own different ways, their their own uh three different fields within the individual organism. but also the world model that we're working with. And we certainly can in terms of thinking about surprisal, we

certainly at the level of global phenomenon and the anthropocene and collapse and technological overshoot and uh political um uh strife that we're experiencing. We certainly can identify a great deal of surprisal which should be telling us to look at to revisit the global model and I think that this interpretation is a salient one and a relevant one. Now a little bit more about exteroceptive bias. Exteroceptive bias is going to be uh an innate tendency within the system. There simply are more receptors. There simply is a lot more going on exteroceptively than in any other field. Certainly in the proprio than in the proprioceptive field and the animal when it's born immediately has to look into the world around it to find uh warmth, shelter, food, kin and that need doesn't stop that continues. So there is an exteroceptive bias that is an innate tendency within the system although it is not meant to be overwhelming, not meant to be in an imbalance and repressive instead. Um and so what that what happens then is that a control bias emerges when the balancing proprioceptive priors are not present in the system. Then there's a an a feedback loop, a positive feedback loop, a vicious cycle that occurs, a control bias in which exacerbates that imbalance. Now we can see that in right hemispheric dominance and hyperintellectualization of our culture which is increasing exponentially with the exponential changes in technology and the built Self-evidencing then is one way of looking at how the body calibrates. Precision tracking and self-evidencing. Now self-evidencing, what is self-evidencing? I think that's a great question from an exteroceptive perspective. Self-evidencing is a little bit obscure because exteroception is other evidencing. Um, the five senses provide the basis, the evidence required by science and reason. This is conventional. There's really nothing wrong with it. But self-evidencing is a property of a self-aware a self directed system and that would be the core that would be proprioception. That would be the core node in the model. self-evidencing then is a natural property of the functioning of the proprioceptive core in its relationship the body's deep deeper inner hidden relationship with itself. Now there's a lot of ways to look at that, but we're going to talk about uh precision tracking and we're first we're going to talk a little bit about fascia and we're going to talk about precision tracking in the model. Now the rediscovery of fascia is a big deal. I'm a body worker so I work with fascia. That is a recent phenomenon and I would argue one of the most important fifth in anatomical science since Vesalius in 1543 that the restoration of the connective tissue matrix to the image of the body is a powerful powerful form of belief updating. Now we had removed the fascia because it was in the way. We thought, we believed it was simply a kind of white wrapping tissue that needed to be discarded so that we could see what was underneath the individual muscles with their tendinous connections to the bones. A

bunch of mechanistic little parts was revealed through the process not of peeling away a wrapping paper, but of cutting away powerful chunks of flesh. Because the muscles, this image that the muscle is connected to the bone and the bone is connected to another bone and to other muscles is artificial. The bones and muscles are invested within a connective tissue matrix that provides 30 to 70% of the force production of the system and which is itself an inner skin a layered complex inner skin. It's a it's an elastic source of force production. That's very important. But it's also a layered inner skin so that the intermuscular the intramuscular septa in the system the interstitial um tissues the interstitial space is very very sensitive. we now know. So we by restoring that more integrative and holistic self-image anatomical image of the body then that in itself is very empowering. It certainly is empowering for me as a therapist because I understand it. But also in terms of my movements, it gives me a lot of insight into the feel that I have for these deeper more subtle movements that we say experience in the yoga studio or in meditation in Taichi or rehabilitation etc. Myofascial synergies then are and this is the one that I've been talking about are the basis for um a kind of biomechanical belief updating in the model and my work at least comes from this wonderful text by Tom Meyers and where he has posited 12 different longitudinal myofascial continuities. This is not the only example. It is the preeminent example, but there are others in history and others available today. And an interesting hypothesis, a paper was published in the international journal of molecular um science arguing that the myofascial unit, not the individual muscle, is the physiological basic element for peripheral motor control. that the the basic unit of motor control is myofascial synergy. Now again, some different pictures of the deep core and then getting a little bit more deeply into my system. What I've tried to do is to unpack this axial skeleton in terms of that axial line. And what we see here are how that axial line is intersected at each major juncture. At the one end of the spine where the head meets the neck. At the other end of the spine where the legs meet where the hips meet the spine and in the feet where the the ground connects with where the in the feet where where the body connects with the ground. And each one of those junctures is represented by a circle because those are highly sensitive arrays, concavities, gradients if you will, pits in the body full of sensitive organs and tissues full complex junctures that come together with at the center a a bony prominent central bony servo. mechanism. I want to apply that these ideas to movement and we're going to talk about kinesiology. So, conventional kinesiology of the exteroceptive system is adequately represented by the very very primal primordial simple stick figure that we see on cave walls. And that with just a few simple lines can represent the expressions and the movements of the

body on the outside, the extremities. But it's interesting to note that that stick figure has nothing to say about what is obviously what is clearly the most important part of the body, its most sensitive contents. the the axial line, the core. there isn't a stick. We even a even the transmission of a car has linkages and articulations. So, getting back to that stick figure in conventional kinesiology, the the kinesiology, the movements are oppositional. flexion coming together, extension coming apart, adduction bringing the limbs to the body, abduction taking the limbs away from the body. They're And then internal rotation, external rotation, oppositional movements. Now on the other side of the fence in proprioception in the axial line in the core, in the synergistic system, these kinesi, this kinesiology is not oppositional. It's complimentary. It's interdependent and the movements of the axial structure are also um uh diverse and significant but we can describe them as as the grip. We talked a little bit about the grip that we can get in the armpit. The scapula doesn't just grab the arm. It reaches around the ribs in in a in a grip-like structure. and contributes to centration and pressurization in the torso. Centration coming together, pressurization, a little bit of expansion work together to create balance and stability in the body on a longitudinal level. We push down through the posterior line and experience lift coming up through the front. This is something that's uh experienced in yoga all the time. Gravity in fact translated through the tensegrity structure of the body does not pull living organisms down but in fact lifts them up. It provides a vertical kind of a gradient that my physics is getting way ahead of me but provides lift. And this is something that we can see commonly in life and that we can experience in ourselves as we bring our posture together. That poise is a form of lift. And then the third set I think of of complimentary uh cooperative sort of movements is some sort of a spiral or helical pattern in the body. Now we talked a little bit about homologies and how the in the exteceptive system in the in the body in general there are there is segmental homologies and that helps us to organize the body the body's movements but when we shift from between perceptual fields from an exteceptive to a proprioceptive field then we also experience what I think is an inverse homology and that helps the brain to sort of get a grip from within. I won't say any more about that. We talked a lot about the bony servo mechanisms here. Um we'll go into them a little bit more deeply. The different bony servo mechanisms on the axial line. Um there are no other alternatives, no other options. It's very clear uh what they are. we can come into each one of these sensitive areas and each one of these major bones in order to organize that area. Now here they are laid out in the skeleton and uh we can see the the circles around those bones and on the right hand side and highlighted in a kind of a green blue is is each one of

those prominent bony servo mechanisms. Each one is at the center of a sensitive kind of hidden concavity. The groin, the throat, the armpit in the back in the soles of the feet. Now, these are clearly areas in the body where we're meant to know ourselves. But what happens when we get confused? So this is a little diagram and I'm going to finish with this diagram today about how we actually what stereosceptive bias means as a control bias means that we wear the body backwards and the green areas are the axial postural areas the bony affordances largely in the front with the exception of the scapula. these are the areas if you think about your body now, the most sensitive areas of the body. But are these the areas where we experience the most problems? These are the areas where I think that we experience the most repression and sometimes even shame. And when we see collapse, we see a collapse in these areas with a tremendous amount of stress and tension and compensation required in their opposites and those are the red areas. So when we have a head forward position, when we have collapse in this anterior structure, the back of the neck is carrying this heavy imbalanced load. We get tension in the occiput in the neck and in the shoulders. When we um fail to attend to the proprioceptive prior in the armpit, we make the shoulder do all the work. And of course, that's the third most common site of musculoskeletal complaints. It is not where we're meant to control the body. We're meant to find coordinative awareness in these um concavities and their bony servo mechanisms of the low back in relationship to the groin, the pubic bone in the front. The pelvis very much is shaped like a wheelbarrow. And those two big bones in the back, they're like the two big wheels or handles of a wheelbarrow. And we try our best to force that load and we believe that we can control that load with brute strength until the front wheel of the wheelbarrow hits a pebble and the entire thing tips. So if we don't attend to the the little front wheel to the low front, then we're going to have dysfunction in the low back. The feet. What I'm going to say about the feet is that that big toe is very clearly the power structure of the entire foot. And what's the first thing we do when we put it in modern shoes? Hix VGUS. We collapse. We crush that So that's my talk for today and I'm that's an effort to unpack belief updating and exteceptive bias in a somatic way with reference to sharrington with reference to anatomy myofascial anatomy it's been a real pleasure I want to ask you if you've made it this far in the talk that you give me some comments give me some feedback hopefully some positive feedback I've worked very hard on this again I'm very excited to make this unusual somatic anatomical contribution to the field of active inference thank you very All right, the next talk is by Sid, 10 minutes long. Greetings everyone. My name is Siddhant and I'll walk you through our recent work on enhancing bisantine tolerant formation control for maritime

buoy swarms using you guessed it right active inference. What are maritime buoy swarms? As you can see on my screen maritime buoy swarms are a swarm of buoys like deployed in the ocean in order to achieve certain tasks. For example, if there is an oil spill or um like um an incident in the ocean, these buoys help in monitoring that. They also help support missions like surveillance, search and rescue and other kinds of environmental monitoring in hostile or contested waters. The threats that we focus on in this particular work rely on the configuration that you see in front of you. All of these buoys can be imagined as decentralized robots floating in the ocean with minimal motor control and they can communicate among each other and a centralized hub. But because of this they are also prone to cyber physical attacks. So we will focus on three kinds of attacks which can impact their operation and mission success. So for example, if the task is area coverage like taking down one or more of these buoys is equivalent to a bisonantine problem where we cannot trust certain buoys anymore and now we have to measure how to measure the mission success in this particular uh scenario so that the rest of the buoys can reorganize around this and in order to do do that they need to detect the cyberphysical attack at play. So the threats that we will focus on are GPS spoofing, communication jamming and actuator hijacking. Why these three? Because these three push the resilience limits of existing control architectures which may not use artificial intelligence or any active inference approach. Our goal in this work is to maintain formation integrity and task coverage while detecting and mitigating malicious agents. I'll start with our baseline approach and then explain how we integrated active inference, where it shines, where it still struggles, what alternatives we considered, and how we plan to push the results further. The original meth methodology without active inference uses a decentralized neighborto- neighbor approach in order to facilitate communication. Each buoy assessed trust using hand-crafted manual heruristics, rules that downweighted agents exhibiting inconsistent trajectories, communication dropouts or mission throughput anomalies. We combined that multi-dimensional trust score with an attack aware formation controller which essentially helped reate the neighbor influence based on trust estimates while enforcing formation geometry and coverage. This approach was practical and interpretable and it handled many different kinds of anomalies well as you can see in the graph in front of you. However, it lacked formal uncertainty quantification which is very important for this kind of a task. Strugg it struggled with subtle or staged attacks that we uh inflicted upon the system and sometimes even over penalized honest agents during complicated maneuvers. In our improved methodology like we see we see that we introduced an active inference-based trust assessor. The core idea here is to

move from heuristic rules to a probabilistic generative model and variational bayesian inference. Each buoy in this new model maintains an internal state predicting neighbor trajectories position, velocity and acceleration and even jerk to measure change in momentum. While taking into account estimated formation forces when reality deviates from predictions, we compute a mahalanobis distance across those dimensions. High prediction errors will suggest potential tampering. We then update our trust beliefs using bayesian filtering explicitly tracking both the mean trust score and its uncertainty. The formation controller uses those uncertainty aware trust estimates to moderate neighbor influence. Agent communication stays decentralized. The key change is that the trust updates now come from predictive modeling rather than fixed thresholds. And this as you can see improves the detection accuracy a bit across all three metrics. Where active inference is useful can be highlighted in the following ways. Like the graphs are pretty clear that active inference over a period of time can help in improving the results that we are seeking. But to do that it helps mostly in three areas. First is early detection. Predictive errors surface anomalies before they manifest as gross deviation. We see this particularly with GPS spoofing position when they diverge sharply from expected trajectories. Secondly, uncertainty quantification helps us in carrying variances along with the trust means so that we can protect the honest agents when the model is unsure. This reduces false positives during tight maneuvers. Thirdly, it helps with attack specificity. Different attack modes produce distinct error signatures which are important for us to quantify and measure. GPS spoofing elevates the position residues. Communication jamming shows up in missing observations. Actuator hijacking disrupts velocity and jerk. We calibrate these responses accordingly. In our V2 results, this translated into an 8.4% average improvement in detection accuracy, a 1.6% boost in resilience, and 6.6% gain in task coverage with actuator hijacking coverage jumping nearly 20%. And the results are pretty clear. We do not claim that active inference is always the better approach. We can see here that active inference does a lot better at task coverage and resilience score whereas it is nearly as performant in other metrics or sometimes less performant in average trust score over time. Where active inference still falters is that the generative model still simplifies the hydrodynamics and control inputs. When cooperative agents execute aggressive maneuvers, prediction error spike and the system can result in false alarms. Moreover, communication jamming creates data gaps that challenge the inference updates. Even with tempered thresholds, detection improvements for actuator hijacking are like are minuscule. So what do we do? We see that trust detection improvements are modest with only 3.5% because actuation dynamics are hard to model at the

granularity we need. So active inference is computationally heavier for these decentralized forms with lesser processing budgets. So we need reliable processing power in order to ensure real-time updates which may not always be a solution in a maritime environment. The alternative approaches that we have considered was a reinforcement learning based trust estimation where agents learn policies from simulated attacks. It offers adaptability but lacks the interpretability and uncertainty guarantees that we wanted. Another was a consensus based filtering with fall tolerant observers. Classic control theory techniques. These have strong stability proofs but require centralized coordination or strict communication assumptions. Another thing that we could look at is an hybrid approach which combines calman filters with heuristic gating which shows um which has the potential to show some promise yet um like it also has the potential to lack explicit uncertainty handling. What active inference gives us is a balance between theory interpretability and decentralized operation. The key finding for us is that active inference can outperform heuristic baselines across detection resilience and coverage while providing principled uncertainty estimates. GPS spoofing detection jumped over 15%. Uh our communication jamming detection um improved by 7%. And actuator hijacking coverage improved almost 20%. These are not basic numbers. These are quite significant. So how do we go about it in a practical setting? We confirm that these protective floors and attack specific calibrations are essential. Without them, active inference overreacts. Finally, the predictive modeling gives the commanders richer diagnostic signals. Not just what went wrong, but how confident the system is in that assessment. This is where active inference really shines. We can improve upon a few things as next steps. Our next steps focus on richer generative models incorporating better hydrodynamic parameters for this environment, external disturbance introduction and learned control inputs. We plan to fuse communication reliability signals more tightly with behavioral predictions to handle jamming gaps. We also want to explore hierarchical active inference where local agents share compressed summaries of their beliefs to enhance collective awareness without breaking decentralization. Finally, we plan to benchmark against adaptive consensus and reinforcement learning hybrids to ensure that we are harnessing the best aspects of each approach. In summary, active inference has pushed our maritime swarm resilience forward in a simulation environment, giving us predictive confidence and measurable improvements against bisonantine threats. The work is not done. We need to do a lot more to move this from a Python-based simulation environment to the real world. But the path is clearer. We need to sharpen the models, strengthen the communication inference loop, and validate under more diverse operational profiles. Thank you. Please feel

free to reach out and I'm happy to dive deeper into any section. And it has been an honor presenting here at the symposium and I wish you luck. All right, next talk is by Nicholas Hinrix. Hypers scanning under active inference for mental health. This is going to be an 11 minute recorded video. >> Hi, my name is Dr. Nicholas Henri. I'm a neuroscientist working at the Max Planck Institute for Cognitive and Brain Sciences. And hi, my name is Dr. Nicholas Henri. I'm a neuroscientist working at the Max Planck Institute for Human Cognitive and Brain Sciences and at the Okinawa Institute of Science and Technology. I study how people come to feel. Today, I want to talk about a problem that's obvious to anyone who has ever been in a therapy room. Mental health interventions depend on attunement. When therapist and client are not aligned, therapy stalls or even fails. Yet, we have almost no quantitative tools for monitoring attunement in real time. My colleagues and I propose a way forward. We build on active inference models of social interaction and link them to a geometric hypers scanning observable that is suitable for clinical deployment. We will also sketch a prospective digital twin for psychotherapy that uses real-time monitoring of neural geometry to mark and eventually predict when crucial behavioral events such as ruptures, co-regulation and reatunement occur. The common scientific approach to social cognition is to treat one person as an observer trying to infer the hidden mental state of another. This third person stance assumes social understanding happens through detached one-way observation. In contrast, second person neuroscience emphasizes that meaningful interaction is embodied and reciprocal. When two people engage, their brains form a coupled generative system. Computationally, each partner maintains a generative model of their own as well as of the other person's states. predictions are continuously updated and belief updating becomes mutual. This dissolves the regress of I model you modeling me modeling you into a shared dynamic narrative and this framework the dyad not the isolated individual is the basic unit of A effect fits naturally into this picture. Emotional balance can be formalized as inference over the coherence of a self model. Two quantities drive it. The magnitude of self-model prediction error. So how surprised you are about yourself and the self-relevance of that error. So how much the situation touches your identity. We include an effective memory term to account for emotional inertia and allow different temporal orientations. prospective, retrospective, fast, slow, intense or smoother, which we call temporal aim. The position parameters alpha and beta, which you'll see in our upcoming paper, tune the influence of surprise and memory. In a dyadic interaction, prediction errors become entangled because each agent recursively models the other. Positive balance signals smooth coordination whereas negative balance signals breakdowns within active inference. A

effect acts as a soft prior that biases policy selection towards trajectories that minimize expected surprise. Therapists sometimes use their intuition to sense when the alliance with a client is strong or weak. In social neuroscience, hyperscanning allows us to monitor brain activity simultaneously in multiple people. Most hyperscanning studies compute correlation or phase locking metrics which describe neurosynchrony but cannot tell us when and how the network reorganizes. These metrics treat interbrain coupling as somewhat static and misdynamic topological reconfigurations associated with rupture and repair. Network geometry offers a more mechanistic lens. Discrete geometric notions like the form and Ricky curvature quantify how information expands or contracts across a network. In a weighted network, positive curvature marks edges embedded in dense redundant clusters while negative curvature marks bridging edges that connect different modules. This edge centered measure captures local flow in ways that nodebased synchrony can't. Our method which we've dubbed geometric hyperscanning computes the formal rickety curvature on time resolved interbrain networks derived from dual EEG or fMRI recordings. We treat each cortical region as a node and functional connectivity as weighted edges for each edge connecting nodes I and J. The formal Ricky curvature is computed over the weight of that edge multiplied by the sum of one divided by the weight of node I and 1 divided by the weight of node J. Then you subtract the correction term which is the sum over all edges that share a node with the edge under consideration of the ratio of that edge's weight to the square root of the product of that edge's weight and its neighbors weight. This context the function w assigns weights to nodes and edges. To track how the overall geometry evolves, we compute the entropy of the curvature distribution within a sliding window. Curvature entropy at time t is calculated by taking the negative of the sum over all edges present at that time of the probability of each edge multiplied by the natural log of that probability. Here the probability of an edge is the curvature of that edge divided by the sum of curvatures across all edges. This resulting scalar curvature entropy measures the volatility of interbrain topology. Abrupt peaks or discontinuities in curvature entropy indicate topological phase transitions in the shared generative manifold. When the entropy spikes, it often corresponds to effective rupture. When it declines, repair and reatunement are underway. How do we use these insights in therapy? Each partner in therapy, so the therapist and the client is modeled as an agent whose self model contains a generative model of the other. Prediction errors in this recursive model manifest as topological changes in their combined neural activity. By computing former Ricky curvature on the weighted interbrain network this third brain between therapist

and client and tracking the entropy of its distribution. We obtain a realtime readout of when the interaction is stable, rupturing or repairing. Here's where the idea of a digital twin becomes powerful. A mental health digital twin is a virtual representation of an individual's mental states that is continually updated with data collected over the person's interaction. Such a twin can guide professionals in diagnosing and treating patients using mechanistic models, machine learning tools. The digital twin concept extends beyond simple simulations. It spans the neurophysiological cycle of an individual and updated by real world data using physical and statistical models to answer what if questions in mental health. Digital twins could forecast deterioration, predict treatment outcomes, and recommend tailored interventions.

Implementing this framework requires robust data in careful modeling. Hypers scanning signals must be pre-processed to construct reliable time resolve functional connectivity matrices. The curvature calculation depends on appropriate waiting schemes. Positive curvature highlights redundant local clusters while negative curvature highlights bridges. Entropy should be computed within windows suitable for the modality's temporal resolution. shorter windows for EG or longer ones for fMRI. In real clinical settings, the digital twin must integrate behavioral cues, gaze, procity, gestures, and self-report data with the neural one. Each of these streams can inform the posterior over latent states, ensuring the twin is grounded in multimodal evidence. Ethically, digital twins raise issues of privacy and agency. Despite these challenges, the potential benefits are considerable. By embedding geometric hypers scanning within an active inference framework, we can develop social markers, realtime signatures of interpersonal coordination analogous to biomarkers. in therapy. These could alert practitioners to ruptures they might otherwise miss, guide decisions about when to pause or reframe an intervention, and help quantify progress over sessions. In research, curvature-based metrics allows allow us to test hypotheses about how belief updating and effective revelation unfold moment by moment. In technology, digital twins could inform adaptive interfaces and social robots that adjust their behavior based on the geometry of human interaction. Mental health interventions thrive on attunement. Yet quantifying attunement has remained elusive. We propose a unified framework that models therapist client interactions as coupled generative systems, formalizes affect as inference over selfmodel coherence and measures interbrain dynamics using discrete geometry. The formal ricky curvature and its entropy reveal dynamic phase transitions, rupture, cogulation and reatunement which standard synchrony measures miss. Building on this insight, we sketch a digital twin for psychotherapy, a virtual entity that integrates real time neural, behavioral, and emotional data to timestamp and

anticipate shifts in relational state. Such a twin has a potential to assist therapists in guiding sessions towards the optimal outcome, namely healing. We invite collaborations to test this approach on real clinical data and further refine the models for broader use. Thank you. Okay, we're back. This is going to be the session that would have been immediately following Carl Fristen's keynote address earlier today due to some technical issues. The group with Ian Tennant recorded this during the time. So, here it is. It is a 45minute Yeah, it seems >> great.

>> Okay, good. Hello and welcome and thanks for joining. This presentation for the fifth applied active inference symposium is titled an applied active inference group practice for transdisciplinary collaboration. My name is Ian Tenant. I'm based at Anglia Ruskin University. So I'll start with explaining um why I'm interested in a particular type of group practice or communication style for transdisciplinary work and then think about how we're going to go about this. Today we'll do a um a live demonstration of this type of um group practice and then I'll talk about the details of what it involves before we do the live demo. So at Angler Russ University a year or so ago, a new planetary health group was set up and um the idea was it brings together a new group of peers and um colleagues who are interested in planetary health with different backgrounds. So some from engineering, some from economics, health, different types of health and um law. So we're in the stage now where we're still kind of storming, forming and norming. So understanding each other's interests and trying to figure out a shared way of looking at global planetary health issues and the idea of a transdiciplinary mindset um appealed to me but I wasn't sure exactly what it means. So I looked at the literature for some definitions and found a nice paper in 1998 titled complexity and human hast uh human health the case for a transdisciplinary paradigm. And this table sets out a a spectrum of ways in which we can look at um big problems. So in this case health problems we can draw different boundaries around the problems. So um if we considered something like a uh a health problem like kidney disease, a single discipline might be a kidney specialist thinking about the best medical uh medicine or therapeutic to um treat the disease. A multiddisciplinary uh approach might be um uh a kidney specialist teaming up with a engineer to develop a new device. uh transdisciplinary um according to this definition goes a step further and they widen the scope of the problem to look at the um situation of kidney disease from multiple levels a kind of look at health as a dynamic system operating um even beyond the individual. So um staying with the example of kidney disease here we have a picture of uh Dr. William Kul who was the first um person to develop an artificial kidney. These days we call it diialysis and he teamed up with engineers to build the first um uh kidney diialysis

machine. These days we have um transdisciplinary projects looking at things related to um uh the issue of something like endstage kidney disease. So with kidney disease um some people can develop it after having diabetes for a long time which damages the kidney. So if we were going to go to the root cause of uh draw expand the boundary of our problem and say why for example are there so many people in Europe or in or the UK developing diabetes which then reaches this stage of endstage kidney disease. we might take a transdisciplinary mindset and this project here called cities changing diabetes from 2014. It um brought together people from lots of different disciplines including urban planners and uh um maybe environmental toxicologists to look at what is it in the environment that might be um causally related to people developing diabetes and the problems that they may um come up with maybe involve redesigning cities to make them healthier places for to be in. So often these are um it's a blending of disciplines for a long-term collaboration and it may end up in resulting with new frameworks or theories to work with. However, um when new people with special um interests and working models come together, there's often some challenges during the early stages. And I'm going to use the Jiari window model here to um explain what I mean by this. So with the Jiari window model um it's it's it's called a window because it looks like four planes panes of glass. And we have in the top left corner things that are known to um the self one group of people but they're not known to other groups of people. So we could say with people with some uh specialism or one discipline, let's say a uh kidney specialist, they're things that are known to the specialist in kidneys but not known to an engineer. Then the other corner we've got um things that are not known to the self but are known to the other group. And then we've got in the bottom right corner things that are not known to either of the two groups. And the aim really um of transdisciplinary research or teams will work better if we we um expand the window where things are known to self and the other. And hopefully people familiar with um the active inference framework will recognize this as a form of reducing uncertainty or reducing surprise or or minimizing error. So um the free uh active inference um involves reducing the discrepancy between what we um our internal model and what we sense from the world and we can do this either through updating our model of the world or trying to change the world change what we sense from the world. um a slide here from one of Carl Fristen's papers where he um uses uh an example of song birds singing and turn taking. So the birds generate a shared model or or a shared generative model of their song by taking turns to listen and to sing. And this is referred to as precision engineer engineered message passing. And we all kind of do this sort of thing. Um, every day we take turns to listen and speak. But there are

different um styles or ways we can do this. The archetypal way of um thinking of people coming together to share stories is around a campfire. um but we do it uh in lots of different ways. The last month there was the sixth international workshop on active inference. So a conference is a style of um sharing generative models and um it may involve people presenting for long periods of time and then questions and answers. and we're seeking to kind of understand each other and reduce our uncertainty about each other's models. In the therapeutic world, there's lots of um approaches to this. On the right, a picture of uh a counseling session. So, in some forms of counseling, the counselor will not um necessarily uh try to um coach or guide the person in counseling. they'll literally listen and reflect um and allow the person to share. But it's a very um structured approach. Uh another style of group chat maybe around uh less structured is around uh a social event around a table where people will take turns but there um there's no structure to say who has to speak first or who gets to speak or um um who should be listening. So um it's a big spectrum of styles for sharing, listening and um speaking and listening. And they range from having a lot of structure to little structure. The approach we're going to use today, the how we're going to go about this is um inspired some by some therapeutic traditions. And one of these is um has been developed by John Vavi who's appeared on the inference uh institutes u several several times and he um calls this style dialogos and he says that it encourages a reciprocal flow state which is similar to um jazz or sparring. So um this type of style has been developed by um the circling institute and for those of you who wish to find out more uh can visit the circling institute website but um it shares similarities to lots of therapeutic styles of communication or nonviolent communication and um my partner has um um trained in this style of communication and has encouraged me to um adopt it and I'll just share a very brief example of how it's helped in our relationship. So the idea is that um if someone is speaking to you and um you rather than if it if something arises that maybe you disagree with or there's some uncertainty about you seek to understand the other person before you add your own opinion. So, this simple example was a day when we'd had a lovely day out and at the end of the day, my partner said she wasn't feeling very loved. And the part of me that might have been sensitive or um wanting to defend myself might have said um of course you're loved. We've done all these lovely things today. And so on. But because she had um helped me understand the ideas of reflective practice, um I sought to understand what she meant. So I said, "Okay, um uh in what way are you not feeling loved? Um please help me understand what you're feeling." And this led to a really interesting conversation and our relationship deepened as a result of it. So today we're going to use um a

structured form of communication and we're joined by Susan, Andrew, and Leonardo to demonstrate this. And there are um certain stages to it, but the general principles are that when you're listening to someone speaking, then you uh seek to understand what they're sharing through reflecting what you've heard back to them. And if there's an error, it gives the person who spoke a chance to clarify or they may want to add further detail. So the mindset we have to approach is to be comfort to be comfortable with the error and um if you are sharing then be um open to revealing your whole experience. So um that could be more the effective side of the experience. So uh your emotional veilance as well as just the facts. So um I propose that this type of approach um can help with generative model alignment in transis transdisciplinary research. So um through this turn taking in small groups we can continually update our beliefs about others intentions and perspectives. It can help with precision calibration. So uh the practice can highlight which signals deserve more attention and which need revisiting. It can help with epistemic action and shared anticipations over time. So hopefully the transdisciplinary group will develop a um kind of shared um expectations or uh beliefs about how they can approach the transdisciplinary problem. So a quick um bit of detail about what is involved. So we imagine we're sitting in a virtual circle. If this was face to face, you could sit in a circle and it's turn taking between listening speaking. So the first two people um person A and B. Person A will speak to the topic and person B will listen and then aim to reflect back what they've heard. And then um when person A has finished and um yeah person P person B reflects and then when they've reached the position where person A feels understood person B turns to the third person and they share what they feel was surprising um or confusing or jarring about what they heard. And then after then they add their own opinion about the topic. So their fresh opinion about the topic. It doesn't have to be related to what the previous person said. Then this continues around the circle. So person the third person then takes the role of the seeker. So seeking to understand and they reflect what person two has said and that continues. So, we're going to demonstrate this live now. And um the topic we're going to look at is um about misinformation. So, a recent um piece of research says that of all the kind of um potential threats around the world that people are worried about such as natural disasters, war um um and so on. disinformation is amongst the top. So people are worried about um false information and the threats of that. So what um I'd like the group to have in mind is how might our understanding of active inference help us notice and respond differently to the types of unhelpful information in society. That could be AI generated or it could be through social media or it could just be everyday life. So, I'm going to stop sharing the screen and we'll return

back to our small circle and um welcome back Susan, Andrew, and Leonardo. So, you're welcome to unmute now. And maybe before we start the group practice, could you just introduce yourselves and say a few words about your um interest in active inference? So we'll start with Andrew. That sounds good. So um can you hear me? Okay. >> Great. Thanks. Um so yeah, thanks Ian for for inviting me to this session. Um I have been involved with the realm of active inference for roughly three or so years. Um sort of my background has been a a very broad combination of the social sciences, economics, but actually kind of a move towards complexity science, uh neuroimaging, data analysis. Uh I work full-time as a data analyst. So I'm just I'm a very curious person. I'm very interested in many things and uh so I I interned and then uh became a member of the scientific advisory board with the the institute and some people recognized me for teaching uh the the proverbial 2022 textbook uh by uh par and pulo and frist. So thanks >> thank you Andrew uh Susan. Hi. Yeah, I'm not a neuroscientist or a data analyst. Uh I teach uh biology at uh University of California and I'm interested in active inference for the roles that it can help um us find better ways to help students learn and for the difference between how they perceive they're doing in a class and what actions they're taking and how they actually are doing. And I'm very curious about ways to break through that sort of cycle when people seem to have a large gap between where they think they want to be and the things that they're doing to get there. And so um I'm very curious about ways to get through that um log jam. >> Thank you Susan and Leonardo. So my name is Leonardo. I'm a PhD student in therapy research. My focus is on seat music uh for dementia carers and my interest in active inference um initially came came through my uh studies in neuroscience and then it I find it as an interesting framework for understanding social dynamics and social neuroscience like musical interactions. >> Thank you. Wonderful. So we have um people in the background of social science, biology and music. So um we're going to begin the group practice now. And um the topic is unhelpful information that people might be presented with in various different forms. and we're speaking to how our personal understanding of active inference may help deal with misinformation or unhelpful information. So, is there someone who'd like to speak to that first? Andrew. Yeah, I don't mind popping in. Sure. Thank you. So, yeah. Shall I just kind of go for it or >> you go first and then I'll um take the role of reflecting you afterwards and then we'll then I'll pass it on to the next person. >> Sounds great. Um so, it's interesting how you introduce the word unhelpful information uh in addition to misinformation. And I think um you know if I'm thinking of whether it be active inference or even some sort of more colloquial sense of understanding this like unhelpful information to me um because active

inference in one sense is very much related to sort of the standpoint of the observer and we heard some of that during Kristen's talk. Um, so there might be information uh that is told to me that I might perceive as unhelpful uh if I'm doing some particular project or something uh that that at least seemingly has nothing to do with music. Uh then if I'm told a lot about music in this moment, I will perceive it as more, you know, distracting. It's not particularly helpful despite that information about the music perhaps being well evidenced and researched and maybe kind of comes out of some clinical study or otherwise. So in which case so it's it's not misinformation but it is unhelpful information and I think that that kind of distinction uh in the the broader context of thinking about um global issues or or uh disasters or or you know kind of um political uh contention um I think that that distinction is very relevant right because someone who if active inference entails the idea of us being self-eviding, right? Then if I'm told repeated uh incidents incidences of misinformation that at least reconfirm my model. So say I'm listening to a particular politician, they're proposing some grand scheme for how to resolve all of our issues. Um I put in caricature and that uh policy proposal doesn't necessarily bear out in policy research. um then it is a it it could be a kind of misinformation, but it's very helpful to me because it makes me feel good, right? Reconfirms a lot of the beliefs I came to to listen to the politicians talk with, right? So, um it's I'll maybe pause there, but I just thought it was a very interesting way to go with that. >> Thank you. So, I'll aim to reflect and repeat back what I've heard. So, um you found it interesting um the the putting the word unhelpful and misinformation, unhelpful information and misinformation next to each other and you um say that um these two things are not equal. So, information might not be viewed by the observer to be helpful. So it might be viewed as unhelpful information but it may um actually be true in some sense. It may actually reflect something um accurate. So it may not be misinformation in the sense of false information. Um and you use the example of um maybe something a repeated message by a um politician for example that may confirm reaffirm someone's belief. So even if it's misinformation, it may be used as it may be viewed by the receiver as helpful even though um it's not necessarily accurate. So you were um aiming to kind of show some examples of when information might be viewed as helpful or unhelpful or accurate or unaccurate. Um and that those two things don't always they're not always equivalent to each other. Anything else you'd like to add or correct about my summary? I I think you did a pretty good job of the summary. Uh yeah, I think I think that um I think it's useful to include the idea of what does it mean for information to be helpful or unhelpful and then kind of juxtapose that with what misinformation is cuz I think that kind of um by

understanding the idea of information being helpful or not to someone, it actually can kind of help us potentially better understand the dynamics of the spread of misinformation, right? that to kind of it's just reiterating what you said uh regarding re reiterating what I said uh that is uh in our in our turn taking style here that uh yeah to for for information to to make me feel good for me to self-evidence for me to oh I don't have to change my model of the world very much because these people are telling me what I think I already knew right uh it's quite comfortable it's very low cognitive cost for me to process that information and update my model. Um, and perhaps benown to us, unbeknownst to us, it nonetheless can be misinformation. So, >> thank you. So, um, you were um, saying that you think it can be useful to include the notion of helpfulness or unhelpfulness um, when talking about misinformation. Anything else to add before we move around the circle? >> I I think we should keep turn taking >> and taking. Okay. So, I'm going to turn to Susan next. >> And what I do now is briefly summarize what I found interesting about Andrew's um sharing before I share myself. So, yeah. Um what I really liked about Andrew, what Andrew shared is that um he gave an example of when something can be um viewed as unhelpful but actually be accurate and then he gave a an example of the opposite case. So um for me that opened up something around transdisciplinary work when new teams come together and they um you may have an expert who's got a um special interest and they are hearing lots of information from someone else. So it could be they're hearing some information from a finance team for example um or a engineer but they don't view that information as helpful. Their interest might be on um let's say the psychology of the issue. So they may discard it as not helpful information but actually for the project as a whole it may be useful. So for me that presents a real challenge. How do you help each member of a transdisciplinary group see the value of information if it has got value? That's all I want to share. Okay. Um, so reflecting back, I think Ian was thinking about I think Ian, you were thinking about situations where you hear or one hears information and one isn't necessarily sure in a group why that information is being presented. Is is that correct? >> Yes. Yep. >> And then you have to try to figure out what to do about that presentation. And um did you come up with any have you thought of ideas for what ways to manage a situation where information is coming up and you're not sure how to how to use it? >> I guess um what's coming to mind is there's a proactive and a reactive way. So the proactive way might be a group member might sort of put their hand up and say, "Excuse me, I'm not um sure about the significance of that information." um or I'm not sure about a terminology, maybe I just don't understand the vocabulary being used. That requires the person to feel confident about their

uncertainty. Um another way I can imagine is to um maybe the facilitator or the organizer supporting or scaffolding that uncertainty in some way. So being prepared that people will not know the significance of each other's information and putting in place some kind of process either background information or some kind of um glossary or translation tool to help members to um realize the significance for new groups. So you're saying basically that if in a group where information is presented uh someone's struggling to understand whether it's useful or not, they could ask the person themselves and say, "I'm not sure why you're saying that or or maybe I just don't understand it." Or um which puts pressure on them in a sort of social context to say that they don't understand which may be uncomfortable. And then on the other side, it could be the role of the facilitator to realize that there are gaps in communication and step in and try to fill them. Is that sort of what you're suggesting? >> That's it. Yes. Yeah. Thank you. >> so I I don't think I um I don't think I have any of anything else to add at this point. So, um, Susan, you're welcome to turn virtually towards Leonardo and share your thoughts on this topic. Yeah, I think this topic is really interesting because what it it seems often in transdisciplinary groups, it's very hard to understand whether you don't understand what they're saying or you don't understand why what they're saying is relevant or they're what they're saying isn't relevant. And so you're always in this sort of internal quandry and trying to think how do I present this in some way that you know yes I may have to show my ignorance but on the in the goal of moving the discussion forward how do I react to this and so I think that that people um I think facilitators are often hired for reasons that they can try to bridge those gaps and help people understand them. But I think sometimes facilitators either don't know the topics well enough to be facilitating those topics and that can create more trouble. So um I found some of these discussions to be kind of difficult for the reasons that uh both Andrew and Ian have mentioned. And I wonder Leonardo um what are your thoughts on this? And is anything I've said resonating? >> Um thank you Susan. This is a very interesting topic. I'll just start by uh making sure I understand what you've uh explained to me. So you're saying that um you can see the importance of the of understanding unhelpful information in transdisciplinary groups. Um because we as a listener we might uh develop internal troubles uh because we might be afraid of looking ignorant if we don't understand something. We have to balance it with the uh desire to progress the conversation. And so um am I understanding? Is this relevant? Well, maybe a facilitator can help to understand to make sure that the language is a bit more uh commonly understood. But the problem here is that uh they might not be an expert and might actually make the discussion uh even muddier. Is that's um the main topic or

is there something else you you'd like to mention? >> No, I I think that's the main topic. I think the other piece is trying to figure out whether the what the person said really was uh on par for the discussion or was sort of something that was leading the whole group off in the wrong direction. And I really don't know how you deal with that. >> I see. So that is a interesting problem to tackle. in in problem solving situations. Yeah. So now do I turn to someone or >> Yes, Leonardo. So >> So um Susan, do you have anything else to add before? >> I do not. >> So now we would at the bare minimum we would close the circle. So, Leonardo, you would turn towards Andrew and share a bit of a summary of what you've heard before and then add your own new insight. >> Fantastic. So the discussion Andrew so far um has been about around uh really unhelpful information whether we can uh in transdisciplinary groups um able to see what is really significant to the overall topic conversation with the group or what can we ignore and how can we address this problem of whether something is unhelpful given even that we're not an expert in what the speaker is saying or even the facilitator is not an expert on what's being discussed. Um I found the role of the facilitator actually very important in the what I've been hearing in this discussion. It seems like it's a good solution could be sending some premeating readings that aren't too long. Maybe all individual all the individuals should send out at least um a few bullet points relating to the discussion before. Um then so in the situation of misinformation and unhelpful um the difference in that I mean that I feel like if if you're saying something as an expert um it's maybe important to say that this is a true thing this is evidence backed and maybe coming back in your explanations that um well this relates in this way to the overall topic and I mentioned on the the fact of saying that what the statement you've expressed is true um to emphasize the truthfulness of your statement because if if it's um and false it it won't bring to the conversation. So it's just saying well I'm not trying to invent something. So yes, I think that was um to emphasize the truthfulness of Usain and the relevance to the overall topic every time we express a statement our premeating readings. So what do you think of that Andrew? >> Sure. Thanks Leonardo. Um so to kind of recap my understanding of everything um so we have we're looking at the notion of unhelpful information in the context of transdisciplinary work. Um how do we kind of see the overall significance of particular pieces of information to the whole group? What can we ignore or perhaps what should we realize we shouldn't ignore? Um and you laid emphasis upon the role of the facilitator. Uh seems to be very important and they could perhaps help with organizing a bed through uh sending out premeating readings or or bullet points in advance of the discussion uh to maybe establish some kind of initial

common ground between everyone and maybe what they're thinking about. But then that said, um, being able to relate different pieces of information to various topics or in the context of transdisciplinary research, uh, relating like the same topic to potentially different disciplines, I could also see coming in there and, um, so being able to keep those connections. Um, have I done a fair enough job in kind of kind of recapping there a bit? >> Definitely. Definitely. Um, and there was just one point I wasn't too sure how to address it in the but it would be that to emphasize the truthfulness of the statements. So I guess as an expert you you want to you suppose you're saying something truthful but you know I don't know if that's um it's just an idea. I'm not sure if it's um practical but um in the in the um definitely the relevance to the overall topic is the most important as you've uh repeated Andrew. I say Ian, do I should I continue? >> So you can you can maybe um so Leonardo wanted to add an extra detail. So if you're able to can you reflect that back to him? >> Yeah. Uh I mean to to emphasize the truthfulness of the information you're sharing. I mean, I find that quite significant as far as I understand bringing that up because if you know, if you're working with a team of people and you introduce, you know, say a new concept that's otherwise foreign to them, um, then if they're not able to sort of relate it back to their own uh, discipline and, you know, you could sort of inject the the active inference framework here a little bit and say like if you say something that simply does not sort of correspond respond in some meaningful way at least at a baseline at a micro scale as a starting point with their their prior beliefs about you know whether it be the world at large or at least like how their discipline works or their general topics and and and folks that they focus on then yeah I think I think well in that case you'll really want to like bridge that gap for them uh in a way that you can you can justify by saying that there's some truthfulness here. So, um, so like social science, uh, music and clinical research, um, you know, if you and I could establish some kind of gap or even if I was sort of like proverbially selling active inference to you, saying like, oh, you should use active inference, here's why. Here are various, you know, papers that have been published on like dynamics and clinical therapy that relate back to the active inference framework and how it can be useful. So being able to just kind of provide that sort of additional information that can help establish that connection potentially in a way that exceeds just me saying it to you, right? It's like no, it's, you know, not to appeal to authority here, but here are some published papers, you know, that that have much more detail if you want to pursue that further if it piques your interest. So that's one way to deal with >> Great. Yeah. Yeah. Yeah. Referring to the published evidence, it's a great way. Thank you. Okay, thank you. So, I think we'll close the the

demonstration and close the circle there. What can be done with this type of um process is that you can keep going round the circle again. So, you can see what else evolves with another another round. But given the the time constraints today, um we'll draw towards the end. I would like to hear um any views that you have and it can be kind of popcorn style so any any person can speak first um about how you found this process. So you all kindly gave up some time before this session to have a practice with me and we practiced with the simple question of please tell me about your name and we went through some uh a round of reflection there. So you you've experienced it twice now. Do you have any thoughts about um how this process of turn taking and reflection um could be useful or not or where it could not be useful? >> I can go first. Yeah, I I think it is I think it is helpful because you uh get a chance to sort of hear how other people perceived what you said which helps you clarify your own thinking. Uh you think oh is that how I put that? Okay. And you get so so you have that role of self-reflection on your own thought but then you also hear how the um how the thoughts of others dovetail and interact. And I felt like we were moving towards a conclusion. If we had another couple of rounds, we might have come up with a sort of, you know, tiny version of how you should start organizing a certain kind of discussion. So, I thought it was helpful. >> Thank you. Anything else to add, Andrew or Leonardo? I don't know why I saying I agree with Susan that it is helpful and we were getting closer to a a conclusion for sure. It still feels a bit uncomfortable to make sure you're Oh, am I fully understanding? Am I having am I repeating well, but um I think it's a good process. Yeah. >> Yeah. I uh I mean I think I think it's great and uh just so that I can add a little bit more. Hopefully not too much. I'm trying not to go into like teaching mode, but you brought up the the bird song example in your introductory slides to this. And what I love about that example is through the turn taking between the the song birds, it's not just that they're kind of mutually uh reducing prediction error, but if you actually look, and this is where it gets a little mathy, I don't want to get too abstract, but if you actually look at sort of the state space that they collaboratively explore, um it starts to expand. So they start to synchronize but explore new places in the state space that neither one of them have individually explored before. So through this kind of collective emergent behavior they traverse new territory. And so if you bring that back to us talking, it's kind of like, well, we all sort of have our own mutive explored spaces around these topics, you know, our own degree of interest, but by going around the circle, I I agree with with Leonardo and Susan that it did seem like we're kind of reaching a new sort of conclusion about these things. And in some way it's kind of like where it's through this turn taking that we can kind of reach new places and

conversation and our thinking that wasn't may or may not uh I won't be overly optimistic but may or may not have been immediately available to us before doing this. So basically collaborative exploration leads to new insights that weren't available before potentially. So yeah I think it's great. >> Very nice. >> Thank you. Yeah, that was really um useful to hear that view from an expert in the maths behind active inference. So thank you three for helping uh to rescue this session with the technical difficulties and I hope it will be made available um to the symposium and I'm also interested in um working up this style of uh communication and group practice and translating it into the language of active inference for other people to use. So, thank you for joining. >> You're welcome. All right, next pre-recorded video is going to be Sanjie Nam joi fundamentals of active inference a self-guided textbook for learning and applying active inference from first principles. This is going to be a 20inut video. Hello, my name is Sanjie Namjoshi and I'm a machine learning engineer working at verses. If you've ever tried to learn active inference, you probably know what it feels like to get lost in a maze of subdisciplines. Neuroscience, information theory, Beijian statistics, physics, and more. Every paper seems to assume you already know three other fields with different mathematical notations. equations seem to materialize out of nowhere with few good resources explaining where they came from or how they're even useful in the first place. Today, I would like to talk to you about my my progress toward providing greater accessibility, visibility, and knowledge of active inference and the free energy principle. I'm excited to announce that my textbook fundamentals of active inference is complete and will be released in March 2026 from the MIT press. My aim for this textbook is to provide a unified, systematic and rigorous foundation for students and researchers entering the field to reduce the steep learning curve. However, an important motivating question is why should you learn active inference in 2025? To answer this question, I think it is helpful to turn the clock back 2006 and draw a comparison between the state of deep learning in its early days and active inference as we understand it today. Prior to the mid-2000s, neural networks were just one of many possible models rather than the dominant choice in academia and industry. This all changed in 2006 when Hinton and his colleagues released the deep belief network paper which is generally understood as the start of deep learning as we understand it today. After some various hardware innovations and the release of the well-known imageet library, we started to see more coverage and success of AI in the news. But in 2012, deep learning truly provided its value with AlexNet. And for the first time, it achieved better than human performance on image detection tasks. But this was just the start. What followed was a proliferation of deep

learning across industry and research. I've added some well-known milestones just to highlight the explosion of progress in deep learning in the last decade. But there is so much more that deserves mention and attention here. And since I presented about my textbook at the 2023 active inference symposium, the state of AI has dramatically changed once again. Today, large language models dominate the AI landscape and have managed to overshadow many other fields and alternative approaches. New and more powerful language models are released every quarter, though the rate of advancement over previous models is beginning to slow. So what about the state of active inference as a field? From my perspective, active inference lies in the same position as deep learning did in 2006, influential and on the brink of exploding in popularity and relevance. This paper here um from which this figure comes from is from contributors at the active inference institute and shows the current growth of publications in the active inference community. In the last few years, active inference as a field has seen a number of important milestones. Here I show just a few that broaden the scope and attention to the field. We have the first international workshop on active inference in 2020. We have the first applied active inference symposium and the foundation of the active inference institute. We have the release of the par pizzazzulo and fristen 2022 book as well as the pmdp library. And from 2023 to present there are so many pieces and changes here. Uh some of which include new Julia packages RX infer um and many many more papers in robotics and other industries as the international workshop of active inference has begun to grow and grow and grow. I see active inference as perfectly poised for greater visibility and attention, especially given the current interest in generative AI and deep reinforcement learning. Working alongside the active inference institute and other organizations, my aim here is to provide some of the fundamental materials to capture the attention and notice of machine learning researchers and students to bridge this gap to bring active inference into its renaissance. So why learn active inference in 2025? Well, active inference is highly interpretable. The very nature of Bejjian networks means that you can follow the chain of reasoning through a particular problem to see why a certain um outcome was chosen over others for a particular model, why a particular policy was chosen. For example, active inference comes equipped with a principled modeling system, a way to model any system of interest using an elegant and all-encompassing encompassing formalism based in information theory. Active inference also has a more robust interpretation of agency than current LLMs as we talk about agency in a lot of industry contexts today and includes an aspect of control of the external environment. Active inference is fundamentally compatible with the notion of world models that capture the cause effect

understanding of the external world and it's believed these world models are very important for preventing hallucinations that we see in large language models. Active inference can also be used to solve everyday practical problems in research and in industry. And it may also expand the field of complex systems modeling which is central to fields such as physics, climate science, economics, political theory, psychology, sociology, and more. Active inference also provides a convenient way to unify many disparate fields that are exploring the same ideas in different languages or scientific conventions. And although it's been yet to be fully demonstrated, active inference does have the potential to be more cost and sample efficient at scale. And more research needs to go into this area, which is why we need a new generation of active inference researchers coming to contribute to the field. Due to the current state of active inference and its potential as a premier AI research framework, my aim is to expand the reach of active inference beyond its current confines. To this end, I've aimed to make my textbook as accessible for the widest audience possible with the least amount of required prerequisites. The major focus is writing this book for an engineering audience. Both machine learning engineers and traditional engineering disciplines. Neuroscience is largely out of scope for this book and there are many other resources available if you're interested in that particular aspect of the free energy principle and active inference. Many of the recommended and even optional prerequisites that you see here are typically known by undergraduate students in science and engineering and certainly by graduate students in these fields. Nevertheless, the book is also written to be readable for those that desire to focus on everything. The completionist that wants to know all the math, the concepts, and the code. It's also suited for people who are more interested in the theory. uh maybe just interested in the math and concepts to get a general sense of things so they can dive into the literature further. And my hope is that ultimately those who are intuitionists who want to skim over the math and just get a sense of the ideas should still be able to read this book. And ultimately I hope to convince you by reading this textbook that active inference may be challenging but it is not impossible to learn by any means. Part one of the book called fundamentals will set the stage by describing the statistical setting with an agent interacting with its environment. We start as simple as possible with a single state and a single data point and then we begin to expand from there. Connections are made in chapter 3 to other classical parameter estimation techniques in statistics as well as a complete investigation into the basics of variational bayesian inference which happens in chapter 4 where variational free energy is also introduced. The field of predictive coding is also explored in part one which we can see as a kind of precursor one of

the precursors to the modern formulation of active inference. And this will also allow us to introduce the notion of prediction errors which are ubiquitous in the active inference and also the neuroscience literature. Part two, the active inference core now introduces the core active inference methodologies. Chapters 6, seven and 8 focusing on continuous state spaces and chapters 9 and 10 uh focusing on discrete state spaces. Like part one, we start simple in the simplest possible model we can imagine and then begin to grow and build up the model to the full vanilla active inference agent step by step. Part three, applications and extensions cover various developments that have uh occurred in the last uh 8 to 7 8 to 10 years uh in the field which include um modifications and extensions to base active inference and also a number of other topics included here like structure learning, factor graphs, robotic applications, different types of free energies, hybrid discrete and continuous models, the free energy principle and so on. There are also extensive appendices covering the history of the field and its connection to other fields, a deep overview of all notation in the book, a uh exhaustive summary of the different forms of variational free energy you might find in the literature, and a least you need to know for all background mathematical topics in active inference. Um, you can see here that there's a lot of coverage that gives you the basic details. So even if you're not an expert in all of these different background topics, you have enough of a sense of how to actually understand why they're relevant in the context of active inference and understand when they show up in various papers of interest. So now I'd like to give you a preview of what's contained in the textbook itself. Type setting and style is extremely important to successful learning. So I've spent a lot of time attempting to make my work clear and readable. Margins are used for footnotes and supplemental figures uh that can aid reading comprehension. There's a large focus on building intuitive understanding of concepts. For example, all algorithms are explained from scratch. And in this book we uh start from a description of the agent environment modeling problem. Describe the univariate case extend the multivariate case. Then we introduce variational inference. Add dynamics. Add generalized coordinates if applicable. Uh we look at hierarchical models and we also look at action building on it step by step and then looking also at learning and attention as well to see how they fit into the broader framework. Figures are designed to clearly walk the reader through the text and give detailed visualizations of important concepts. There are also concept maps provided for most chapters which lay out the fundamental vocabulary that will be covered in the chapter and show how they're all interrelated to one another. And this should hopefully form a useful guide as the different sections are also numbered and shown in the concept map so you can understand

where that topic comes from and how it relates to something else that you may have read earlier in the text. The early part of the book focuses on basic concepts such as hidden state estimation that is estimating the conditional distribution of a latent variable given observed data. And the aim here is to explain the modeling paradigm in the context of an agent attempting to infer the states of the environment. This is an interaction between a generative model and a generative process. And this is a perspective that differs from the way that Bayesian inference is normally taught. In chapter 3 and also in the code notebooks that I'll talk about in a moment, we go through this perspective taking something as simple as linear regression and putting it into the context of an agent environment interaction to really make those concepts clear and understandable. Typically, introductory Bayesian inference books focus on parameter estimation or what we call learning in this book. And in part one, I introduced the expectation maximization algorithm to make it clear what the relationship is between classical parameter learning and estimation techniques that you might see in a Bayesian context in certain textbooks and this notion of hidden state inference. We deal with latent variable models to understand how they connect together within one algorithm which sets the stage for the active inference models that come later. Variational inference is also explained in detail. The book features a catalog of all the different forms of variational free energy or expected free energy in the literature and how they can be derived from one another and how the different forms interrelate. The book also covers raw and Ballard style predictive coding terms and ideas such as prediction error minimization as well as a clear explanation of fundamental concepts such as surprisal and variational free energy. I'm also making extensive use of Bayesian networks in this book so that all models can be seen graphically and we also build up starting simple to more and more complex models throughout the book. Factor graphs which are an alternative way of representing Bayesian networks that have a number of other interesting and very useful properties are explained in chapter 12 where I go through all the different models we've seen so far in the book and show how they can be represented as a factor graph to build that intuition and understanding of how these kinds of models work. There are more than 230 custom figures that have been created for the book, most of which are full color. And the figures are extremely detailed to give the reader a deep understanding of the content that I provide in the book. And many of the models presented in the book are also shown in pseudo code which would aid in implementation. My aim for readers is to go through these algorithms and be able to implement them in code yourselves. Although code will also be provided for those who also want to learn through reading the notebooks and running pre-

existing code. Finally, each chapter is filled with numerous examples and simulations. These examples correspond to the Jupyter notebooks that I will be talking about in a moment and try to show the application of a particular concept in a simulated environment. So now let's shift our attention to some of the supplemental materials that are going to be provided. A few months after publication, um Jupyter notebooks will be uh released on GitHub and be fully reproducible using Docker or other version um version handling tools. The notebooks will recapitulate every single figure that's shown in the book, all simulations in the book and all background material with explanations for all this all these uh different items. And it will also come with a library of modular functions that can be used to build custom active inference agents in environments fit for learning purposes. These modular pieces can essentially be mixed and matched inside different environments or agents. And that's how these Jupyter notebooks often proceed. They'll explain the basic concept first and then use those components or tools to build up an agent that does a particular task. Uh that's corresponds to whatever is being uh presented in that particular chapter. There's a direct correspondence between the equations that you see in the text and the uh notebook u Jupyter notebook code as well. So here's an example showing um variational free energy um in one particular simplified form that you can see here and I've tried to match the kinds of style of variables on the left here uh showing you how the matrix multiplication and addition and these different components fit together. So as close as I possibly can I've tried to match them so it's very easy to see when you read through the notebooks exactly what that equation is telling you. And this is just another example here showing that um there are detailed walkthroughs of different kinds of things. Here I'm showing you colored noise and how you can smooth out a noisy signal by passing over a Gaussian filter and get smooth or colored noise which is very useful in active inference simulations and is present in a lot of raw data uh that sensors might pick up. There are also going to be uh full video presentations for each chapter. Um this is going to be a project that I'll be working on in the uh late future after the book is released. Um but this uh the idea would be to eventually have an active inference course and you can see some of the details here of uh some slides that I've created uh along the route to this this goal. A good majority of proofs had to be cut from the book due to space concerns. um and I will collect all of those as well as other proofs um in a document on archive that will um contain all proofs for every equation in the active inference literature. So the aim is if you ever see an equation you don't know where it comes from, you should be able to look it up in this guide and be able to find how it was derived and what assumptions were made in that derivation. So finally, let's

talk about a few of the planned future resources I'd like to work on to further support the educational tooling around the book. There are two potential side projects I'm interested in working on with collabor collaborators. The first is providing a software suite for continuous state space active inference and a basian mechanics simulation suite. Um these will both be very useful additions to the field and um allow people from other disciplines to begin experimenting with active inference code and fill the gap right now in the software ecosystem. I'm also very much interested in interactive learning. The aim would be to have concepts explained in the textbook with various simulations um interspersed in the forms of plots uh plot demos and other visualizations maybe sliders that you can move around. You can tweak initial conditions and parameters and and see how the resulting plots change. And this would support um everything from bejjian mechanics to continuous and discrete space state space active inference. As you can see, a lot of this work will um be very involved and take a lot of time. So, I'm very open to uh any volunteers that would like to help me with these projects. So, please contact me. I have my contact information shown at the end if you're interested in contributing to these projects um as I continue to expand this ecosystem. I'm also working on a second book um also to be published by MIT Press called Fundamentals of Bejjian Mechanics. If you're not familiar with Bejjian mechanics, it's an entirely separate field um related to active inference and the free energy principle that talks about the separation between different kinds of stochastic processes uh separated by what's called a marov blanket rendering the two systems conditionally independent. And this forms the background formalism and structure around which active inference models can be derived. So it's kind of looking at the bigger picture, looking at more thermodynamics and living systems and the mathematics of different kinds of interacting stoastic processes. This book which is at the intersection of information theory and physics um is currently under um a work in progress and will be released probably sometime in late 2027. Um and I'll be giving more updates on that book as I proceed. So just as another reminder again, my textbook will be released in March next year and I'm very excited and looking forward to educating and collaborating with a whole new generation of active inference researchers. I would like to thank the uh active inference institute for hosting um presentation and uh code during the time when the book was being written. Um I got a lot of very useful feedback from everybody that joined in those sessions. Um, and I really appreciate all of your hard work. You've all been acknowledged uh in my um book's preface as well for all that for all the contributions there. And um I'd also like to thank my my employer versus for their continued support of my book as well as the active inference

community in general with the international workshops on active inference. And this is my contact information. Please feel free to contact me at any time and email is the easiest way, but I'm also sometimes on the active inference institute discord as well if you'd like to see me on there. All right, that's the end of this stream. Next stream tomorrow, part two. Bye.